Delayed blow-up by transport noise for the 3D Navier–Stokes equation with Navier-slip boundary conditions

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Abstract

We study the vorticity formulation of the 3D Navier–Stokes equation driven by transport noise in a periodic channel with Navier-slip boundary conditions. We consider both non-degenerate transport noise and degenerate tangential transport noise. For any prescribed $T > 0$ and $\epsilon > 0$, we prove that, by choosing the noise intensity sufficiently large and concentrating the noise on sufficiently high modes, the solution exists up to T with probability at least $1 - \epsilon$.

A main contribution of this work is to identify and analyze the interaction between the enhanced dissipation induced by transport noise and physical boundary effects. The no-flux condition breaks the isotropy of the noise and changes the scaling limit of the Itô–Stratonovich corrector. In the non-degenerate case, a boundary feedback term appears in the limiting effective operator; in the degenerate case, the limiting operator is a nonlocal anisotropic tangential dissipation. The proof is based on a combination of a boundary correction operator, a Meyers-type estimate, a boundary-adapted scaling-limit analysis of the Itô–Stratonovich corrector, and resolvent estimates for the deterministic limiting equations.

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1 Introduction

In this paper, we study the vorticity formulation of the 3D Navier–Stokes (NS, for short) equation driven by transport noise in the periodic channel $D := \{(x, y)|x \in \mathbb{T}^2, y \in (0, 1)\}$ with Navier-slip boundary conditions

$$\begin{cases} \partial_t w - \nu\Delta w + (u \cdot \nabla)w - (w \cdot \nabla)u = \sqrt{2\kappa}\Pi(\dot{W}(t, x, y) \circ \nabla)w, \\ u = \mathrm{curl}^{-1} w, \\ (\partial_y w_3 - \alpha w_3)|_{y=0} = (w_h - \alpha u_h^{\perp})|_{y=0} = 0, \\ (\partial_y w_3 + \alpha w_3)|_{y=1} = (w_h + \alpha u_h^{\perp})|_{y=1} = 0, \\ w|_{t=0} = w_{\mathrm{in}}. \end{cases} \tag{1.1}$$

Here, $\nu > 0$ is the viscosity, $\alpha > 0$ is the friction parameter of the boundary, curl^{-1} is the Biot–Savart operator, and w, u denote the vorticity and the corresponding velocity field, respectively, with w_h, u_h being the tangential components and $u_h^{\perp} := (-u_2, u_1)$. The driving noise $W(t, x, y)$ is a divergence-free Gaussian random field, whose precise definition will be given below. The parameter $\kappa > 0$ characterizes the noise intensity, the symbol $\circ$ denotes the Stratonovich integral, and $\Pi : L^2 \to H_{\mathrm{curl}}$ is a projection operator, where

$$H_{\mathrm{curl}} := \left\{ w \in L^2 \middle| \operatorname{div} w = 0, \int_{\mathbb{T}^2} w_3(x, y)\mathrm{d}x = 0 \text{ a.e. } y \in (0, 1) \right\} \tag{1.2}$$

is the curl-admissible class endowed with the usual L^2-norm $\|\cdot\|$.

1.1 The noise structure

We consider two types of Gaussian random fields W. The first is the non-degenerate case, in which each component of W is not identically zero and W depends on both the tangential and vertical variables (x, y). The second is the degenerate case, in which the vertical component vanishes and W depends only on the tangential variables x, namely

$$W := (W_1(t, x), W_2(t, x), 0).$$

In this case, the stochastic transport reduces to

$$(\dot{W} \circ \nabla)w = (\dot{W}_h \circ \nabla_x)w,$$

and therefore acts only in the tangential directions. Such a stochastic transport is naturally relevant in situations where the dominant motion is tangential, for instance in oceanic and atmospheric circulations.

The non-degenerate transport noise

Let

$$k := (k_h, m/2) = (k_1, k_2, m/2), \qquad k_h \in \mathbb{Z}^2,\ m \in \mathbb{Z}$$

be a wave number. For $k \neq 0$, define $a_{k,1}, a_{k,2}$ as unit vectors in $\mathbb{R}^3$ satisfying

$$a_{k,1} \cdot a_{k,2} = 0, \qquad a_{k,j} \cdot k = 0, \qquad a_{k,j} = a_{-k,j}, \qquad j = 1, 2. \tag{1.3}$$

In particular, for $m = 0$, we define

$$a_{k,1} := (a_{k,1,h}, 0), \qquad a_{k,2} := (0, 0, 1),$$

where $a_{k,1,h} \cdot k_h = 0$ and $a_{k,1,h} = a_{-k,1,h}$. By definition, one has

$$\textstyle\sum_{j=1,2} a_{k,j} \otimes a_{k,j} = I - \dfrac{k \otimes k}{|k|^2}, \qquad k \neq 0, \tag{1.4}$$

where $k \otimes k := kk^T$ is the usual tensor product. Introduce the solenoidal Fourier sine-cosine modes

$$\sigma_{k,j}(x, y) := c_k e^{2\pi i k_h \cdot x} \begin{pmatrix} a_{k,j,1} \cos m\pi y \\ a_{k,j,2} \cos m\pi y \\ i a_{k,j,3} \sin m\pi y \end{pmatrix} \tag{1.5}$$

with the normalization coefficient $c_k = \sqrt{2}$ if $m \neq 0$ and $c_k = 1$ if $m = 0$. Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space with a filtration $\mathbb{F} := \{\mathscr{F}_t\}_{t \geq 0}$ satisfying the usual conditions. Suppose that $\{B_\cdot^{k,j}\}_{k,j}$ is a sequence of complex Brownian motions defined on Ω, which satisfies

$$\overline{B_t^{-k,j}} = B_t^{k,j}, \qquad [\![B_\cdot^{k,j}, B_\cdot^{k',j'}]\!]_t = 2t\delta_{k+k'=0}\delta_{j=j'}. \tag{1.6}$$

Here, $[\![\cdot, \cdot]\!]$ denotes the quadratic covariation. We define the non-degenerate Gaussian random field by

$$W(t, x, y) := \textstyle\sum_{k \neq 0} \sum_{j=1,2} \theta_k \sigma_{k,j} B_t^{k,j}, \tag{1.7}$$

where $\{\theta_k\}_{k \neq 0}$ is a sequence of real numbers satisfying

$$\|\theta_\cdot\|_{\ell^2} = 1 \qquad \text{and} \qquad \theta_k = \theta_{k'}, \qquad \text{if } |k| = |k'|. \tag{1.8}$$

Under the above notations, the transport noise takes the form

$$\sqrt{2\kappa}\Pi(\dot{W} \circ \nabla)w = \sqrt{2\kappa} \textstyle\sum_{k \neq 0} \sum_{j=1,2} \theta_k \Pi(\sigma_{k,j} \cdot \nabla)w \circ \dot{B}_t^{k,j}.$$

The degenerate tangential transport noise

Next, we introduce the degenerate tangential transport noise. Let $\mathbb{Z}_0^2 := \mathbb{Z}^2 - \{0\}$. Define the solenoidal tangential Fourier modes

$$\sigma_k(x) := \begin{cases} \frac{k^\perp}{|k|} e^{2\pi i k \cdot x}, & k \in \mathbb{Z}_{0,+}^2 := \{k \in \mathbb{Z}_0^2 | k_2 > 0 \text{ or } k_2 = 0, k_1 > 0\}, \\ -\frac{k^\perp}{|k|} e^{2\pi i k \cdot x}, & k \in \mathbb{Z}_{0,-}^2 := \mathbb{Z}_0^2 - \mathbb{Z}_{0,+}^2. \end{cases} \tag{1.9}$$

Fix a filtered probability space $(\Omega, \mathscr{F}, \mathbb{F}, \mathbb{P})$ with $\mathbb{F} := \{\mathscr{F}_t\}_{t \geq 0}$ satisfying the usual conditions. Suppose that $\{B_\cdot^k\}_k$ is a sequence of complex Brownian motions defined on Ω, which satisfies

$$\overline{B_t^{-k}} = B_t^k, \qquad [\![B_\cdot^k, B_\cdot^{k'}]\!]_t = 2t\delta_{k+k'=0}.$$

The degenerate random field W is defined by

$$W(t, x, y) := (W_h(t, x), 0), \qquad W_h(t, x) := \textstyle\sum_{k \neq 0} \theta_k \sigma_k(x) B_t^k, \tag{1.10}$$

where $\{\theta_k\}_{k \neq 0}$ satisfies the normalization and radial symmetry conditions (1.8). Therefore, the transport noise reduces to

$$\sqrt{2\kappa}\Pi(\dot{W}_h \circ \nabla_x)w = \sqrt{2\kappa} \textstyle\sum_{k \neq 0} \theta_k \Pi(\sigma_k \cdot \nabla_x)w \circ \dot{B}_t^k. \tag{1.11}$$

1.2 The main result and literature review

To present our main result, we first introduce the notion of solutions used in this paper.

Definition 1.1. Let $\tau : \Omega \to [0,\infty)$ and $w : \Omega \times [0,\tau) \times D \to \mathbb{R}^3$ be a stopping time and a progressively measurable process, respectively. We call (w,τ) a solution of (1.1), if the following conditions are satisfied.

1. $w(\cdot \wedge \tau) \in L^\infty(0,\infty; H_{\mathrm{curl}}) \cap L^2_{\mathrm{loc}}(0,\infty; H^1 \cap H_{\mathrm{curl}})$ $\mathbb{P}$-almost surely.
2. The boundary conditions for w_h are satisfied in the sense of traces, namely,
$$(w_h - \alpha u_h^\perp)|_{y=0} = (w_h + \alpha u_h^\perp)|_{y=1} = 0, \qquad u := \mathrm{curl}^{-1}\, w$$
in $L^2(0,\tau; H^{1/2}(\mathbb{T}^2))$ $\mathbb{P}$-almost surely. The Robin boundary conditions for w_3 are encoded by the boundary term in the weak formulation (1.13).
3. For any test function
$$\varphi := (\varphi_h, \varphi_3) \in C^\infty \cap H_{\mathrm{curl}} \qquad \text{with} \qquad \varphi_h \in C_c^\infty(D), \tag{1.12}$$
the following identity holds $\mathbb{P}$-almost surely for any $t \geq 0$:
$$\begin{aligned}\langle w(t\wedge\tau),\varphi\rangle + \int_0^{t\wedge\tau} \big(\nu\langle\nabla w, \nabla\varphi\rangle + \nu\alpha\langle w_3,\varphi_3\rangle_{L^2(\partial D)} + \langle (u\cdot\nabla)w - (w\cdot\nabla u),\varphi\rangle\big)\,\mathrm{d}s \\ = \langle w_{\mathrm{in}},\varphi\rangle + \sqrt{2\kappa}\sum_{k\neq 0}\sum_{j=1,2}\int_0^{t\wedge\tau} \theta_k\langle \Pi(\sigma_{k,j}\cdot\nabla)w,\varphi\rangle \circ \mathrm{d}B_s^{k,j}.\end{aligned} \tag{1.13}$$

For the degenerate case, the solution is defined as above, with the noise term in (1.13) replaced by the right-hand side of (1.11).

Define the anisotropic Sobolev space

$$\mathcal{X}^m := \Big\{ f \in L^2 \Big| \|f\|_{m,0}^2 := \textstyle\sum_{|l|\leq m} \|\partial_x^l f\|^2 < \infty \Big\}, \qquad m \in \mathbb{N}. \tag{1.14}$$

The main result is presented below.

Theorem 1.2. *Let $m > 2$ be an integer, and $\alpha, \nu > 0$. Suppose that W is either the non-degenerate random field defined in (1.7), or the degenerate random field defined in (1.10). Then, for any $\epsilon, T > 0$ and $\mathscr{F}_0$-measurable initial data*

$$w_{\mathrm{in}} \in L^\infty(\Omega; H_{\mathrm{curl}} \cap H^1 \cap \mathcal{X}^m),$$

there exist $\kappa > 0$ independent of T and $\theta \in \ell^2$ supported in sufficiently high modes and satisfying (1.8) such that the stochastic NS equation (1.1) has a solution (w,τ), which exists up to time T with high probability:

$$\mathbb{P}\{\tau \geq T\} \geq 1 - \epsilon.$$

To the best of our knowledge, this is the first result that addresses the interaction between the enhanced dissipation induced by transport noise and physical boundary effects in nonlinear fluid equations. The key new feature is that, due to the presence of the Navier boundary conditions, the stabilizing mechanism generated by transport noise does not merely add bulk dissipation; it also interacts with the boundary vorticity generation mechanism. Consequently, the structure of the effective dissipation operator arising in the scaling limit differs from its boundaryless counterpart; see Propositions 3.2 and 3.3. Understanding this interaction is one of the central themes of the present work.

The stabilizing effect of transport noise has attracted considerable attention in recent years. A central mechanism behind this phenomenon is that, under a suitable high-frequency

scaling, the Itô–Stratonovich corrector converges to an effective dissipation operator. In many boundaryless settings, this limiting operator acts as an additional viscosity and thus produces enhanced dissipation. This mechanism was first identified by Galeati [Gal20] for the transport equation with high-frequency transport noise, where the solution was shown to converge to a solution of a deterministic parabolic equation. It was later applied by Flandoli, Galeati, and Luo [FGL21] to nonlinear PDEs with possible finite-time blow-up, leading to delayed blow-up and global existence with high probability.

For fluid equations, this mechanism was first applied by Flandoli and Luo [FL21] to the vorticity formulation of NS equation on $\mathbb{T}^3$. After their work, a natural question is how this mechanism can be extended to fluid models with more physically relevant structures. There are at least two possible directions. The first one is to refine the noise structure itself, for instance by formulating the transport noise at the velocity level, or by considering the Lie-transport noise in the vorticity equation, where the associated stochastic stretching effect is also taken into account. The second direction, as pursued in the present paper, is to keep the high-frequency transport-noise mechanism and investigate how it is modified by physical boundaries.

Recent developments along the first direction include several works. By applying stochastic maximal regularity combined with Meyers-type estimates, Agresti established anomalous dissipation for the velocity-NS equation on $\mathbb{T}^2$ and a delayed blow-up result for a hyperviscous velocity-NS equation on $\mathbb{T}^3$; see [Agr25, Agr26b]. More recently, delayed blow-up for the classical velocity-NS equation on $\mathbb{T}^3$ with high-mode transport noise was obtained by Agresti in [Agr26a] via quantitative homogenization techniques combined with localized stochastic maximal regularity. Another related development is due to Agresti, Butori, and Luongo, who proved a delayed blow-up result for a logarithmically hyperviscous vorticity-NS equation on $\mathbb{T}^3$ driven by high-mode Lie-transport noise; see [ABL26]. Jiao and Romito introduced a class of pseudo-transport noises and proved delayed blow-up results for the velocity formulations of the Euler and NS equations on $\mathbb{T}^3$; see [JR26].

Let us also mention a related research topic on Kraichnan-type noise. In the Kraichnan model, a passive scalar is transported by a Gaussian velocity field which is white-in-time and correlated-in-space. This model provides a classical paradigm for turbulent mixing and anomalous dissipation. Recent mathematical developments include anomalous dissipation and regularization for fluid equations; see, e.g., [CM23, Row24, BGM25, GL25, GGM26] and the references therein.

However, much less is known in the presence of physical boundaries. Existing results in this direction mainly concern linear passive scalar or heat-type equations, where transport noise still generates enhanced dissipation despite boundary effects; see [FGL22, FL22]. To the best of our knowledge, delayed blow-up or global existence results by transport noise for nonlinear fluid equations with physical boundary conditions have not been available so far.

The presence of a physical boundary creates several difficulties which are absent in the periodic setting. First, the vorticity formulation is coupled with nonhomogeneous mixed Dirichlet–Robin boundary conditions, which are nonlocal through the Biot–Savart law. Thus, the vorticity Laplacian is not directly self-adjoint, and the basic energy estimate has to be recovered through a boundary correction. Second, the projection Π onto the curl-admissible class H_{curl} is more delicate than the usual Leray projection on the torus. More precisely, the gradient part in the Helmholtz decomposition (A.1) solves an elliptic boundary value problem, and its boundary condition produces boundary layer terms in the scaling limit of the Itô–Stratonovich corrector. Third, the random field W has to satisfy the no-flux condition at the boundary, and is therefore not isotropic as in the periodic case. This boundary-induced anisotropy affects the limiting operator in different ways depending on the structure of the noise. In the non-degenerate case, a nontrivial boundary contribution survives in the scaling limit

and produces an additional boundary feedback term; see (3.64). In the degenerate case, the effective operator becomes a nonlocal anisotropic tangential dissipation; see (3.75). Together with the nonhomogeneous vorticity boundary conditions, these boundary effects also make the limiting deterministic equation harder to analyze. In particular, the enhanced dissipation cannot be read off directly from energy estimates, and we instead prove resolvent estimates for the corresponding effective dissipation operators and convert them into exponential decay by a semigroup stability criterion.

1.3 The scheme of the proof

We reformulate equation (1.1) into the following Itô form

$$\begin{cases} \partial_t w - \nu\Delta w + (u\cdot\nabla)w - (w\cdot\nabla)u = S_\theta w + \sqrt{2\kappa}\Pi(\dot{W}\cdot\nabla)w, \\ u = \operatorname{curl}^{-1} w, \\ (\partial_y w_3 - \alpha w_3)|_{y=0} = (w_h - \alpha u_h^\perp)|_{y=0} = 0, \\ (\partial_y w_3 + \alpha w_3)|_{y=1} = (w_h + \alpha u_h^\perp)|_{y=1} = 0, \\ w|_{t=0} = w_{\mathrm{in}}, \end{cases} \tag{1.15}$$

where

$$S_\theta w := \begin{cases} 2\kappa \sum_{k\neq 0}\sum_{j=1,2} \theta_k^2 \Pi(\sigma_{k,j}\cdot\nabla)\Pi(\sigma_{-k,j}\cdot\nabla)w, & \text{in the non-degenerate case,} \\ 2\kappa \sum_{k\neq 0} \theta_k^2 \Pi(\sigma_k\cdot\nabla_x)\Pi(\sigma_{-k}\cdot\nabla_x)w, & \text{in the degenerate case} \end{cases} \tag{1.16}$$

is the Itô–Stratonovich corrector. It is important to note that S_θ does not yield a net dissipation at the stochastic level: in the energy estimate, its contribution is compensated by the quadratic variation of the transport noise. The actual stabilization appears only after passing to the high-frequency scaling limit, where S_θ converges to an effective deterministic dissipation operator.

We now describe the main ideas of the proof. The argument consists of three steps, each of which is designed to overcome one of the boundary difficulties discussed above.

The first step is a uniform-in-θ estimate for the stochastic linear Stokes equation (2.1). This requires two ingredients. The first is a boundary correction. Due to the nonhomogeneous vorticity boundary conditions, the Laplacian Δ is not self-adjoint at the vorticity level. We therefore introduce a boundary correction operator $\mathfrak{J}$ and define $\tilde{w} := \mathfrak{J}w$ as the corrected vorticity. We show that $\mathfrak{J}$ can be decomposed into an isomorphism plus a compact perturbation. This, together with a Fredholm argument, ensures the invertibility of $\mathfrak{J}$. Thus, one may equivalently work with the corrected unknown $\tilde{w}$, which satisfies homogeneous boundary conditions and admits a uniform energy estimate.

The second ingredient is a Meyers-type estimate in the periodic channel D. Here, a new difficulty appears because of the anisotropy induced by the physical boundary. In the periodic setting, the Itô–Stratonovich corrector S_θ can be naturally decomposed into a principal Laplacian Δ plus a perturbative remainder. Then, one may absorb the remainder by a continuity argument combined with the maximal regularity of the Laplacian Δ; see [Agr25]. However, in our case, this method does not yield a useful decomposition for the Itô–Stratonovich corrector $\mathfrak{J}S_\theta\mathfrak{J}^{-1}$ in the corrected vorticity formulation. Indeed, the leading part of the corrector behaves like a second-order operator with coefficients depending on y coupled with the nonlocal operators $\mathfrak{J}, \mathfrak{J}^{-1}$, rather than a constant multiple of the Laplacian Δ. We overcome this difficulty by using a different splitting, in which the corrector is combined with the corrected Laplacian $\mathfrak{J}\Delta\mathfrak{J}^{-1}$. Using the correction operator $\mathfrak{J}$, we show that the remainder in the splitting can be decomposed into an absorbable second order term plus some

lower-order terms. This, combined with the continuity method, ensures the uniform-in-θ Meyers estimate; see Lemma 2.3 for the details.

In the second step, our goal is to pass to the high-frequency scaling limit of the stochastic NS equation (3.1) with cut-off. The tightness of the cut-off solutions follows directly from the uniform-in-θ estimate obtained in the first step, so the main task is to identify the limit of the Itô–Stratonovich corrector S_θ. Since the random field W has to satisfy the no-flux condition, the transport vector fields σ entering the definition (1.16) of S_θ are tangent to the boundary. Consequently, the corrector does not by itself select any normal boundary condition. On the other hand, in the non-degenerate case, the bulk scaling limit of S_θ is a full Laplace operator Δ, which has to be realized as a mixed Dirichlet–Robin Laplacian. This creates an unavoidable mismatch in boundary behavior between the corrector S_θ and its bulk scaling limit Δ. The additional boundary feedback term in the limiting effective operator compensates precisely for this mismatch. By contrast, in the degenerate case, the limiting operator is a nonlocal anisotropic tangential dissipation. Since it involves no normal derivatives, the above boundary mismatch is absent, and no boundary feedback term appears.

At a more technical level, the boundary feedback term arises as follows. The projections involved in S_θ lead to elliptic boundary value problems, which become singularly perturbed in the high-frequency scaling limit. Consequently, the corrector can be decomposed into bulk and boundary-layer contributions. The boundary-layer parts are controlled and do not contribute to the limiting operator. The feedback term is produced instead by a vertically oscillatory component in the bulk part, generated by the anisotropy imposed by the no-flux condition on the noise field. In the scaling limit, this component contributes no interior term but produces a boundary Dirac mass; see Proposition 3.2 for the details.

The third step is the analysis of the limiting deterministic equations (4.1) and (4.54). A direct energy estimate for the corrected unknown is not sufficient for our purpose, since the commutator between the boundary correction operator $\mathfrak{J}$ and the linear part may produce growth terms, which would lead to a smallness restriction on the friction parameter α. To obtain a result for arbitrary $\alpha > 0$, we instead use a resolvent approach. In the non-degenerate case, this requires studying the nonlocal resolvent equation (4.4). A key observation is that the equation for the third component is closed, while the tangential components solve a local resolvent equation with a forcing term depending on the third component. We exploit the vertical symmetry of the closed resolvent equation for w_3, and decompose the right-hand side f_3 into odd and even parts. This reduces the construction of w_3 to a superposition of two elementary one-dimensional boundary value problems, whose explicit estimates yield the required bound for w_3. Based on this estimate, we then prove the tangential resolvent bound by a contradiction argument. Finally, the resolvent estimate is converted into exponential decay of the linear semigroup by an exponential stability criterion; see Theorem 4.2. In the degenerate case, since no nonlocal boundary feedback term appears, the resolvent estimate and the corresponding linear decay can be obtained more directly by contradiction.

Combining the three steps, choosing κ sufficiently large, and taking the noise coefficients θ supported on sufficiently high modes, the solution of the stochastic NS equation (1.1) remains close to that of the controlled deterministic limiting equations with high probability. This completes the proof of Theorem 1.2.

1.4 Further comments

We close the introduction with three comments. First, Theorem 1.2 is stated on an arbitrary but fixed interval $[0, T]$, rather than on $[0, \infty)$. The obstruction is the lack of a small-data infinite-lifespan theory for the stochastic NS equation (1.1). In the periodic case, since the Itô–Stratonovich corrector is exactly cancelled by the quadratic variation, and no boundary contribution appears, the L^2 estimate is essentially pathwise. Thus, the small-data global

theory is close to the deterministic one. In our situation, however, as the boundary correction operator $\mathfrak{J}$ does not commute with the stochastic transport terms, the resulting commutators make the L^2-theory genuinely stochastic.

Existing small-data theories do not seem to cover this regime. The results of Kukavica–Xu [KX24] and Aydin–Kukavica–Xu [AKX25] yield infinite-lifespan solutions with high probability, but the noise under consideration is of order 0 and does not involve the gradient of the unknown. On the other hand, Agresti–Veraar [AV24] allow transport-type noise, but the solution exists up to any fixed time T with high probability. None of these results provides an infinite-lifespan theory applicable to our setting.

Second, the Navier boundary conditions are not imposed merely for technical convenience. They provide a physically meaningful wall law allowing tangential slip with friction, and are widely used in fluid-mechanical and engineering models; see [NEB+05, LBS07]. Moreover, from a mesoscopic point of view, these conditions arise naturally from Maxwell-type kinetic boundary conditions through hydrodynamic limits or Knudsen-layer expansions; see [Cor89, JM17]. Thus, they provide a natural and physically relevant setting in which to study how the enhanced dissipation induced by transport noise interacts with physical boundary effects.

At the same time, the present method is specific to the Navier boundary conditions and does not directly extend to the no-slip case. The reason is that at the level of estimates, transport noise behaves like a second-order operator. In the case of Navier boundary conditions, the boundary correction operator $\mathfrak{J}$ is given by $\mathfrak{J} = I - \mathcal{B}$, where $\mathcal{B}$ is an operator of order -1. Therefore, the commutators between $\mathcal{B}$ and the stochastic transport terms are lower-order contributions. This is crucial for the uniform-in-θ energy estimate. For the no-slip boundary conditions, however, such a property does not hold for the boundary correction operator. The corresponding commutator with the transport noise would then remain a leading-order term, and the cancellation between the Itô–Stratonovich corrector and the quadratic variation would no longer close the energy estimate uniformly in the noise coefficients. For this reason, the no-slip case would require a substantially different approach.

Third, the modes in (1.5) are chosen mainly for notational convenience: this choice keeps the notation close to the standard three-dimensional Fourier notation and is useful in the scaling-limit computations. To give a formulation that is independent of the underlying basis, we present the covariance kernel of the non-degenerate random field W:

$$Q(x,y;x',y') = 2 \sum_{k=(k_h,m/2)\neq 0} \theta_k^2 c_k^2 e^{2\pi i k_h\cdot(x-x')} \mathfrak{M}_m(y) \left(I_3 - \frac{k\otimes k}{|k|^2}\right) \mathfrak{M}_{-m}(y'),$$

where $\mathfrak{M}_m(y) := \operatorname{diag}(\cos(m\pi y), \cos(m\pi y), i\sin(m\pi y))$.

Organization of the paper

The paper is organized as follows. Section 2 is devoted to the uniform-in-θ estimates for the stochastic linear Stokes equation. In Section 3, we study the scaling limit of the stochastic NS equation with cut-off. Then, in Section 4, we prove small-data global well-posedness for the limiting NS-type equations. Finally, in Section 5, we combine all the results obtained in the previous sections to complete the proof of Theorem 1.2. The appendices collect the auxiliary results on the curl-admissible class, the Biot–Savart operator, the boundary correction operator, and the linear estimate used in Section 4.

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Notation

We shall use the following standard notations.

- For $q \in [1,\infty]$, L^q is the usual Lebesgue space endowed with the norm $\|\cdot\|_q$. For simplicity, we write $\|\cdot\| := \|\cdot\|_2$. For $s \geq 0$, $H^{s,q}$ denotes the Bessel potential space with the conventions $H^{0,q} := L^q$ and $H^{s,2} := H^s$. It is known that $H^{m,q} = W^{m,q}$ for all $m \in \mathbb{N}$ and $q \in (1,\infty)$, where $W^{m,q}$ denotes the classical Sobolev space. For $p \in [1,\infty]$, $B^s_{q,p}$ denotes the Besov space.
- For $q \in (1,\infty)$ and integer $m \geq 1$, we define
$$H^{-m,q} := (H^{m,q'})^*, \qquad H_0^{-m,q} := (H_0^{m,q'})^*, \tag{1.17}$$
where $\frac{1}{q} + \frac{1}{q'} = 1$. We shall simply write H^{-m} and H_0^{-m}, if $q = 2$. Define the spaces
$$X^{1,q} := \{f = (f_1, f_2, f_3) | f_1, f_2 \in H_0^{1,q}, f_3 \in H^{1,q}\} \tag{1.18}$$
and
$$X^{-1,q} := (X^{1,q'})^* = \{f = (f_1, f_2, f_3) | f_1, f_2 \in H_0^{-1,q}, f_3 \in H^{-1,q}\} \tag{1.19}$$
endowed with the natural product norms, respectively.
- H_{curl} is the curl-admissible class defined in (1.2), $\widetilde{H}_{\mathrm{curl}}$ is the corrected vorticity class defined in (B.1), and $\mathcal{X}_m$ is the anisotropic Sobolev space given in (1.14).
- By e_j, we denote the unit vector in the Euclidean space with the j-th component being 1. For a matrix $\mathcal{M}$, $\mathrm{Row}_j\,\mathcal{M}$ denotes its j-th row. For a three-dimensional vector w, we denote by w_h its horizontal components, and by w_3 its vertical component. Define $\nabla_x := (\partial_{x_1}, \partial_{x_2})$, $\Delta_x := \partial_{x_1}^2 + \partial_{x_2}^2$, and $\mathrm{div}_h\, w_h := \partial_{x_1} w_1 + \partial_{x_2} w_2$.
- By $\mathbb{I}_\cdot$, we denote the usual indicator function. We use the notations $\lesssim$, $\lesssim_\alpha$, $\lesssim_\nu$, etc., to indicate inequalities that hold up to an inessential multiplicative constant, such as C, C_α, C_ν, and so on. By $A \sim_\alpha B$, we denote the relation $B \lesssim_\alpha A \lesssim_\alpha B$.
- For any Banach spaces X and Y, $\mathcal{L}(X,Y)$ is the space of all bounded linear operators from X to Y, and $\mathcal{L}(X) := \mathcal{L}(X,X)$. For any two operators $\mathcal{T}_1, \mathcal{T}_2$, we define the commutator $[\mathcal{T}_1, \mathcal{T}_2] := \mathcal{T}_1\mathcal{T}_2 - \mathcal{T}_2\mathcal{T}_1$.
- For any stochastic processes $X(t)$ and $Y(t)$, $[\![X,Y]\!]_t$ is the quadratic covariation of X, Y, and $[\![X]\!]_t := [\![X,X]\!]_t$.

2 Uniform-in-θ estimate for the linear Stokes equation with transport noise

In this section, we establish estimates for the linear Stokes equation (2.1) with transport noise, which are uniform with respect to the noise coefficients. These estimates will be used in Section 3 to prove the tightness of the cut-off solutions under the high-frequency scaling. We consider

$$\begin{cases} \partial_t w - \nu\Delta w = \mathrm{curl}\, f + S_\theta w + \sqrt{2\kappa}\Pi(\dot{W}\cdot\nabla)w, \qquad u = \mathrm{curl}^{-1} w, \\ (\partial_y w_3 - \alpha w_3)|_{y=0} = (w_h - \alpha u_h^\perp)|_{y=0} = 0, \\ (\partial_y w_3 + \alpha w_3)|_{y=1} = (w_h + \alpha u_h^\perp)|_{y=1} = 0, \\ w|_{t=0} = w_{\mathrm{in}}, \end{cases} \tag{2.1}$$

where W is given by either (1.7) or (1.10), and S_θ is defined accordingly in (1.16). The main difficulty is that the vorticity boundary conditions are nonhomogeneous and nonlocal through the Biot–Savart law. We remove this difficulty by introducing a boundary correction

operator, which gives a uniform energy estimate. Then, by establishing a Meyers-type estimate, we upgrade this energy estimate to a uniform estimate in the space $H_t^{s,p}H_{x,y}^{1-2s,q}$, with p, q sufficiently close to 2 and $s \in (0, \frac{1}{2})$.

2.1 Uniform energy estimate

The main result of this subsection is stated as follows.

Proposition 2.1. *Suppose that w_{in} is $\mathscr{F}_0$-measurable, f is progressively measurable, and*

$$w_{\mathrm{in}} \in L^2(\Omega; H_{\mathrm{curl}}), \qquad f \in L^2_{\mathrm{loc}}(\Omega \times (0,\infty); L^2).$$

Then, there exists a unique global solution w of (2.1). Moreover, for any $T > 0$,

$$\mathbb{E}\sup_{t\le T}\|w(t)\|^2 + \mathbb{E}\int_0^T \|\nabla w\|^2 \mathrm{d}t \lesssim_{\alpha,\nu,\kappa,T} \mathbb{E}\left(\|w_{\mathrm{in}}\|^2 + \int_0^T \|f\|^2 \mathrm{d}t\right). \tag{2.2}$$

Due to the nonhomogeneous Dirichlet boundary conditions satisfied by the horizontal components w_h, the Laplacian is no longer self-adjoint, and hence direct energy estimates are not available. To overcome this difficulty, we introduce the boundary correction operator

$$\mathfrak{J} := I - \mathcal{B}, \tag{2.3}$$

where $\mathcal{B}$ is a boundary map defined by

$$\mathcal{B} : H_{\mathrm{curl}} \cap H^{s,q} \to H^{s+1,q}, \qquad w \mapsto \begin{pmatrix} \alpha(1-2y)\left[(\mathrm{curl}^{-1} w)_h\right]^{\perp} \\ 0 \end{pmatrix}. \tag{2.4}$$

Let

$$\tilde{w} := \mathfrak{J} w$$

be the corrected vorticity. We collect basic properties of $\mathfrak{J}$ in Appendix B.

Proof of Proposition 2.1. We prove the estimate in the non-degenerate case. The degenerate case follows from the same argument, with minor modifications indicated at the end of the proof.

Step I. Applying $\mathfrak{J}$ to both sides of (2.1) and using Proposition B.2, one has

$$\begin{cases} \partial_t \tilde{w} - \nu\Delta\tilde{w} = \mathrm{curl}\, f + S_\theta w + h + \sqrt{2\kappa}\sum_{k\neq 0}\sum_{j=1,2}\theta_k(\Pi(\sigma_{k,j}\cdot\nabla)w + g_{k,j})\dot{B}_t^{k,j}, \\ (\partial_y\tilde{w}_3 - \alpha\tilde{w}_3)|_{y=0} = (\partial_y\tilde{w}_3 + \alpha\tilde{w}_3)|_{y=1} = \tilde{w}_h|_{y=0,1} = 0, \\ \tilde{w}|_{t=0} = \tilde{w}_{\mathrm{in}}, \end{cases} \tag{2.5}$$

where

$$h := -\nu[\mathcal{B}, \partial_y^2]w - \mathcal{B}S_\theta w - \mathcal{B}\,\mathrm{curl}\, f, \qquad g_{k,j} := -\mathcal{B}\Pi(\sigma_{k,j}\cdot\nabla)w.$$

We turn to the estimate of $h, g_{k,j}$. Throughout the paper, all L^2-pairings involving complex Fourier modes are understood as the real parts of the corresponding complex L^2-inner products. First, by Proposition B.2, it follows that

$$\nu\|[\mathcal{B}, \partial_y^2]w\| \lesssim_{\alpha,\nu} \|w\|. \tag{2.6}$$

We estimate the term $\mathcal{B}S_\theta w$ as follows. Let $\psi \in H^1$ be a test function. Since $\sigma_{k,j}$ is divergence-free and $\sigma_{k,j,3}|_{y=0,1} = 0$, by integrating by parts and using (1.8) and Corollary

A.2, one gets

$$\begin{aligned}\langle S_\theta w,\psi\rangle &= 2\kappa\sum_{k\neq 0}\sum_{j=1,2}\theta_k^2\langle\Pi(\sigma_{k,j}\cdot\nabla)\Pi(\sigma_{-k,j}\cdot\nabla)w,\psi\rangle\\ &= -2\kappa\sum_{k\neq 0}\sum_{j=1,2}\theta_k^2\langle\Pi(\sigma_{-k,j}\cdot\nabla)w,(\sigma_{-k,j}\cdot\nabla)\Pi\psi\rangle\\ &\lesssim_\kappa \|\nabla w\|\|\nabla\Pi\psi\| \lesssim_\kappa \|\nabla w\|\|\psi\|_{H^1},\end{aligned}$$

which, together with Corollary A.5, gives

$$\|\mathcal{B}S_\theta w\| \lesssim_\alpha \|S_\theta w\|_{H^{-1}} \lesssim_{\alpha,\kappa} \|\nabla w\|. \tag{2.7}$$

Next, we turn to the estimate of $\mathcal{B}\operatorname{curl} f$. By the definition (A.19) of curl^{-1}, one has

$$\operatorname{curl}^{-1}\operatorname{curl} f = \mathrm{P}f - \left(\int_D (\mathrm{P}f)_1 \mathrm{d}x\mathrm{d}y, \int_D (\mathrm{P}f)_2 \mathrm{d}x\mathrm{d}y, 0\right),$$

where P is the usual Leray projection. This implies

$$\|\mathcal{B}\operatorname{curl} f\| \lesssim_\alpha \|\operatorname{curl}^{-1}\operatorname{curl} f\| \lesssim_\alpha \|\mathrm{P}f\| \lesssim_\alpha \|f\|, \tag{2.8}$$

which, combined with (2.6) and (2.7), yields

$$\|h\| \lesssim_{\alpha,\nu,\kappa} \|w\|_{H^1} + \|f\|. \tag{2.9}$$

Finally, we estimate $g_{k,j}$. Notice that

$$\int_D (\sigma_{k,j}\cdot\nabla)w_3 \mathrm{d}x\mathrm{d}y = 0,$$

by Corollaries A.2 and A.5, one has

$$\|g_{k,j}\| \lesssim_\alpha \|\Pi(\sigma_{k,j}\cdot\nabla)w\|_{H^{-1}} \lesssim_\alpha \|(\sigma_{k,j}\cdot\nabla)w\|_{H^{-1}} \lesssim_\alpha \|w\|. \tag{2.10}$$

Step II. Applying the Itô formula and using (1.6), it follows that

$$\begin{aligned}\mathrm{d}\|\tilde{w}\|^2 &= 2\langle\tilde{w},\nu\Delta\tilde{w}+\operatorname{curl} f+S_\theta w+h\rangle\mathrm{d}t\\ &\quad+2\sqrt{2\kappa}\sum_{k\neq 0}\sum_{j=1,2}\theta_k\langle\tilde{w},\Pi(\sigma_{k,j}\cdot\nabla)w+g_{k,j}\rangle\mathrm{d}B_t^{k,j}\\ &\quad+4\kappa\sum_{k\neq 0}\sum_{j=1,2}\theta_k^2\|\Pi(\sigma_{k,j}\cdot\nabla)w+g_{k,j}\|^2\mathrm{d}t.\end{aligned} \tag{2.11}$$

The quadratic variation term is estimated by combining it with the Itô–Stratonovich corrector and exploiting their cancellation property. Indeed, notice that

$$\begin{aligned}\mathcal{I}_1 :&= 2\langle\tilde{w},S_\theta w\rangle + 4\kappa\sum_{k\neq 0}\sum_{j=1,2}\theta_k^2\|\Pi(\sigma_{k,j}\cdot\nabla)w+g_{k,j}\|^2\\ &= 2\langle w,S_\theta w\rangle - 2\langle\mathcal{B}w,S_\theta w\rangle + 4\kappa\sum_{k\neq 0}\sum_{j=1,2}\theta_k^2\|\Pi(\sigma_{k,j}\cdot\nabla)w+g_{k,j}\|^2\\ &= 4\kappa\sum_{k\neq 0}\sum_{j=1,2}\theta_k^2\left(\|\Pi(\sigma_{k,j}\cdot\nabla)w+g_{k,j}\|^2 - \|\Pi(\sigma_{k,j}\cdot\nabla)w\|^2\right) - 2\langle\mathcal{B}w,S_\theta w\rangle,\end{aligned}$$

where, by using (2.7), (2.10), and Corollary A.5, one has

$$\begin{aligned}\|\Pi(\sigma_{k,j}\cdot\nabla)w+g_{k,j}\|^2 - \|\Pi(\sigma_{k,j}\cdot\nabla)w\|^2 &= 2\langle\Pi(\sigma_{k,j}\cdot\nabla)w,g_{k,j}\rangle + \|g_{k,j}\|^2\\ &\lesssim_\alpha \|w\|\|w\|_{H^1},\end{aligned}$$

and

$$-2\langle\mathcal{B}w,S_\theta w\rangle \lesssim \|\mathcal{B}w\|_{H^1}\|S_\theta w\|_{H^{-1}} \lesssim_{\alpha,\kappa} \|w\|\|\nabla w\|.$$

Therefore, it follows that

$$\mathcal{I}_1 \lesssim_{\alpha,\kappa} \|w\|\|w\|_{H^1}. \tag{2.12}$$

Now, by first integrating from 0 to T and taking expectations on both sides of (2.11), and then using (2.9), (2.12), and Proposition B.1, one gets

$$\begin{aligned}\mathbb{E}\|\tilde{w}(T)\|^2-\mathbb{E}\|\tilde{w}_0\|^2+2\nu\mathbb{E}\int_0^T\left(\|\nabla\tilde{w}\|^2+\alpha\|\tilde{w}_3\|^2_{L^2(\partial D)}\right)\mathrm{d}t&\\ \lesssim_{\alpha,\nu,\kappa}\mathbb{E}\int_0^T\left(\|\nabla\tilde{w}\|\|f\|+\|\tilde{w}\|(\|w\|_{H^1}+\|f\|)+\|w\|\|w\|_{H^1}\right)\mathrm{d}t&\\ \lesssim_{\alpha,\nu,\kappa}\mathbb{E}\int_0^T\left(\|\tilde{w}\|_{H^1}\|f\|+\|\tilde{w}\|\|\tilde{w}\|_{H^1}\right)\mathrm{d}t.&\end{aligned}$$

On the other hand, since the forcing terms on the right-hand side of $(2.1)_1$ belong to H_{curl}, by taking integration with respect to $x\in\mathbb{T}^2$ on both sides of the third component, the horizontal average $\int_{\mathbb{T}^2}w_3\mathrm{d}x$ solves the homogeneous 1D heat equation with Robin boundary conditions and zero initial data:

$$\int_{\mathbb{T}^2}w_{\mathrm{in},3}\mathrm{d}x=0,\qquad\text{a.e. } y\in(0,1).$$

Therefore, one has

$$\int_{\mathbb{T}^2}\tilde{w}_3(t,x,y)\mathrm{d}x=\int_{\mathbb{T}^2}w_3(t,x,y)\mathrm{d}x=0,\qquad\text{for any } t\geq 0 \text{ and a.e. } y\in(0,1),$$

which, together with the homogeneous Dirichlet boundary conditions for the horizontal components $\tilde{w}_h$, ensures the Poincaré inequality

$$\|\tilde{w}\|_{H^{1,q}}\lesssim\|\nabla\tilde{w}\|_q,\qquad q\in(1,\infty).\tag{2.13}$$

Combining this with the Gronwall inequality, one gets

$$\mathbb{E}\|\tilde{w}(T)\|^2+\nu\mathbb{E}\int_0^T\left(\|\nabla\tilde{w}\|^2+\alpha\|\tilde{w}_3\|^2_{L^2(\partial D)}\right)\mathrm{d}t\lesssim_{\alpha,\nu,\kappa,T}\mathbb{E}\|\tilde{w}_0\|^2+\mathbb{E}\int_0^T\|f\|^2\mathrm{d}t.\tag{2.14}$$

Step Ⅲ. By taking supremum in $t\in[0,T]$ on both sides of (2.11), and using (1.8), (2.9), (2.10), (2.12), Proposition B.1, and the Burkholder–Davis–Gundy inequality, it follows that

$$\begin{aligned}\mathbb{E}\sup_{t\leq T}\|\tilde{w}(t)\|^2-\mathbb{E}\|\tilde{w}_{\mathrm{in}}\|^2&\lesssim_{\alpha,\nu,\kappa}\mathbb{E}\int_0^T\left(\|\tilde{w}\|_{H^1}\|f\|+\|\tilde{w}\|\|\tilde{w}\|_{H^1}\right)\mathrm{d}t\\ &\quad+\mathbb{E}\sup_{t\leq T}\left|\int_0^t\textstyle\sum_{k\neq0}\sum_{j=1,2}\theta_k\langle\tilde{w},\Pi(\sigma_{k,j}\cdot\nabla)w+g_{k,j}\rangle\mathrm{d}B_t^{k,j}\right|\\ &\lesssim_{\alpha,\nu,\kappa}\mathbb{E}\int_0^T\left(\|\tilde{w}\|_{H^1}\|f\|+\|\tilde{w}\|^2_{H^1}\right)\mathrm{d}t+\mathbb{E}\left(\int_0^T\|\tilde{w}\|^2\|\tilde{w}\|^2_{H^1}\mathrm{d}t\right)^{\frac{1}{2}},\end{aligned}$$

where, by using the Cauchy–Schwarz inequality, the Poincaré inequality (2.13), and (2.14), one has

$$\mathbb{E}\int_0^T\left(\|\tilde{w}\|_{H^1}\|f\|+\|\tilde{w}\|^2_{H^1}\right)\mathrm{d}t\lesssim_{\alpha,\nu,\kappa,T}\mathbb{E}\|\tilde{w}_0\|^2+\mathbb{E}\int_0^T\|f\|^2\mathrm{d}t,$$

and

$$\mathbb{E}\left(\int_0^T\|\tilde{w}\|^2\|\tilde{w}\|^2_{H^1}\mathrm{d}t\right)^{\frac{1}{2}}\leq\epsilon\mathbb{E}\sup_{t\leq T}\|\tilde{w}(t)\|^2+C_{\epsilon,\alpha,\nu,\kappa,T}\left(\mathbb{E}\|\tilde{w}_0\|^2+\mathbb{E}\int_0^T\|f\|^2\mathrm{d}t\right)$$

Consequently, by choosing $\epsilon > 0$ sufficiently small, one gets

$$\mathbb{E}\sup_{t\le T}\|\tilde{w}(t)\|^2 \lesssim_{\alpha,\nu,\kappa,T} \mathbb{E}\|\tilde{w}_0\|^2 + \mathbb{E}\int_0^T \|f\|^2 \mathrm{d}t.$$

Together with (2.14) and Proposition B.1, this implies (2.2).

Step IV. The existence of global solutions to (2.1) follows from standard iteration arguments applied to the equivalent formulation (2.5); we therefore omit the details. The degenerate case is treated in the same way. Indeed, the noise coefficients $\theta_\cdot$ satisfy (1.8), and each solenoidal tangential Fourier mode σ_k, viewed as a three-dimensional vector field with zero vertical component, satisfies the divergence-free and no-flux conditions. □

2.2 Improved regularity via a Meyers-type estimate

In this subsection, our goal is to improve the uniform energy estimate (2.2) to the fractional space-time regularity $H^{s,p}(0,T;H^{1-2s,q})$ with p, q close to 2 and $s \in [0, 1/2)$.

Proposition 2.2. *Suppose that w_{in} is $\mathscr{F}_0$-measurable, f is progressively measurable, and*

$$w_{\mathrm{in}} \in L^p(\Omega; B^{1-2/p}_{q,p} \cap H_{\mathrm{curl}}), \qquad f \in L^p(\Omega\times(0,T); L^q).$$

Then, there exists $p_0 > 2$ depending on α, ν, κ such that for any $2 \le q \le p < p_0$ and $s \in [0, 1/2)$, the unique global solution w of (2.1) satisfies

$$\|w\|_{L^p(\Omega;H^{s,p}(0,T;H^{1-2s,q}))} \lesssim_{s,p,q,\alpha,\nu,\kappa,T} \|f\|_{L^p(\Omega\times(0,T);L^q)} + \|w_{\mathrm{in}}\|_{L^p(\Omega;B^{1-2/p}_{q,p})}. \tag{2.15}$$

As before, we focus on the non-degenerate case and indicate the minor modifications needed for the degenerate case at the end of this subsection. The key ingredient is the following Meyers-type estimate for the associated model problem

$$\begin{cases} \partial_t\tilde{w} - \nu\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w} + \lambda\tilde{w} = \mathfrak{J}F + \mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w} \\ \qquad\qquad + \sqrt{2\kappa}\sum_{k\neq 0}\sum_{j=1,2}\big(\theta_k\Pi(\sigma_{k,j}\cdot\nabla)\tilde{w} + G_{k,j}\big)\dot{B}^{k,j}_t, \\ (\partial_y\tilde{w}_3 - \alpha\tilde{w}_3)|_{y=0} = (\partial_y\tilde{w}_3 + \alpha\tilde{w}_3)|_{y=1} = \tilde{w}_h|_{y=0,1} = 0, \\ \tilde{w}|_{t=0} = \tilde{w}_{\mathrm{in}}, \end{cases} \tag{2.16}$$

where $\lambda > 0$ is a parameter to be chosen below, S_θ is defined in $(1.16)_1$, $\mathfrak{J}$ is given in (2.3), the coefficients $\theta_\cdot$ satisfy (1.8), and

$$F \in L^p(\Omega\times\mathbb{R}_+; X^{-1,q}), \qquad G := \{G_{k,j}\}_{k,j} \in L^p(\Omega\times\mathbb{R}_+; L^q(\ell^2)),$$

with $X^{-1,q}$ defined in (1.19).

Lemma 2.3. *Let $\tilde{w}$ be a solution of (2.16) satisfying the path regularity*

$$\tilde{w} \in L^\infty(\mathbb{R}_+; \widetilde{H}_{\mathrm{curl}}), \qquad \mathbb{P}\text{-almost surely}, \tag{2.17}$$

where $\widetilde{H}_{\mathrm{curl}}$ is the corrected vorticity class defined in (B.1). Then, there exists $p_0 > 2$ and $\lambda_0 > 0$ depending on α, ν, κ such that for any $2 \le q \le p < p_0$ and $\lambda \ge \lambda_0$,

$$\begin{aligned}\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;H^{1,q})} + \lambda\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q)} &\lesssim_{p,q,\alpha,\nu,\kappa} \|\mathfrak{J}F\|^2_{L^p(\Omega\times\mathbb{R}_+;X^{-1,q})} \\ &\quad + \|G\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))} + \|\tilde{w}_{\mathrm{in}}\|^2_{L^p(\Omega;B^{1-2/p}_{q,p})}. \end{aligned}\tag{2.18}$$

Proof. **Step I.** In this step, we consider the following heat-type equation

$$\partial_t\tilde{w} - (\nu + 9\kappa)\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w} + \lambda\tilde{w} = \mathfrak{J}F \tag{2.19}$$

with zero initial data and boundary conditions $(2.16)_2$. Define

$$\|\tilde{w}\|_{X^{1,q}_{\alpha,\nu+9\kappa}} := \Big((\nu+9\kappa)\|\nabla\tilde{w}\|_q^2 + (\nu+9\kappa)\alpha\|\tilde{w}_3\|^2_{L^q(\partial D)}\Big)^{\frac{1}{2}}, \tag{2.20}$$

which is an equivalent norm of the space $X^{1,q}$ defined in (1.18), due to the Poincaré-type inequality

$$\|\tilde{w}\|_q \lesssim \|\nabla\tilde{w}\|_q + \|w_3\|_{L^q(\partial D)}.$$

Since equation (2.19) can be reformulated into the following form

$$\partial_t\tilde{w} - (\nu+9\kappa)\Delta\tilde{w} + \lambda\tilde{w} = (\nu+9\kappa)[\partial_y^2, \mathcal{B}]\mathfrak{J}^{-1}\tilde{w} + \mathfrak{J}F, \tag{2.21}$$

by applying the maximal regularity of heat equations, it follows

$$\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;X^{1,q}_{\alpha,\nu+9\kappa})} + \lambda\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;L^q)} \lesssim_{p,q,\alpha,\nu,\kappa} \|(\nu+9\kappa)[\partial_y^2,\mathcal{B}]\mathfrak{J}^{-1}\tilde{w} + \mathfrak{J}F\|^2_{L^p(\mathbb{R}_+;\mathfrak{X}^{-1,q}_{\alpha,\nu+9\kappa,\lambda})},$$

where we used the notation

$$\|F\|_{\mathfrak{X}^{-1,q}_{\alpha,\nu+9\kappa,\lambda}} := \sup_{\psi\in X^{1,q'}} \frac{|\langle F,\psi\rangle|}{\Big(\|\psi\|^2_{X^{1,q'}_{\alpha,\nu+9\kappa}} + \lambda\|\psi\|^2_{q'}\Big)^{\frac{1}{2}}}.$$

Combining the above estimate with Proposition B.2, there is a constant $\lambda_0(p,q,\alpha,\nu,\kappa) > 0$ such that if $\lambda \geq \lambda_0(p,q,\alpha,\nu,\kappa)$, one has

$$\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;X^{1,q}_{\alpha,\nu+9\kappa})} + \frac{\lambda}{2}\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;L^q)} \lesssim_{p,q,\alpha,\nu,\kappa} \|\mathfrak{J}F\|^2_{L^p(\mathbb{R}_+;\mathfrak{X}^{-1,q}_{\alpha,\nu+9\kappa,\lambda})}.$$

Now we introduce another norm

$$\|F\|_{\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda}} = \sup_{\psi\in X^{1,q'}\cap\widetilde{H}_{\mathrm{curl}}} \frac{|\langle F,\psi\rangle|}{\Big(\|\psi\|^2_{X^{1,q'}_{\alpha,\nu+9\kappa}} + \lambda\|\psi\|^2_{q'}\Big)^{\frac{1}{2}}}, \tag{2.22}$$

where the test functions are chosen from $X^{1,q'}\cap\widetilde{H}_{\mathrm{curl}}$, and $\widetilde{H}_{\mathrm{curl}}$ is the corrected vorticity class defined in (B.1). Let $\tilde{\Pi} : L^2 \to \widetilde{H}_{\mathrm{curl}}$ be the projection operator onto $\widetilde{H}_{\mathrm{curl}}$. For any $\psi \in X^{1,q'}$, by applying Propositions B.4 and B.5, one has $\tilde{\Pi}\psi \in X^{1,q'}\cap\widetilde{H}_{\mathrm{curl}}$. Together with $\tilde{\Pi}\mathfrak{J} = \mathfrak{J}$, this implies

$$\begin{aligned}\langle\mathfrak{J}F,\psi\rangle = \langle\mathfrak{J}F,\tilde{\Pi}\psi\rangle &\leq \|\mathfrak{J}F\|_{\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda}}\Big(\|\tilde{\Pi}\psi\|^2_{X^{1,q'}_{\alpha,\nu+9\kappa}} + \lambda\|\tilde{\Pi}\psi\|^2_{q'}\Big)^{\frac{1}{2}}\\ &\leq \max\Big(\|\tilde{\Pi}\|_{\mathcal{L}(X^{1,q'}_{\alpha,\nu+9\kappa})}, \|\tilde{\Pi}\|_{\mathcal{L}(L^{q'})}\Big)\|\mathfrak{J}F\|_{\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda}}\Big(\|\psi\|^2_{X^{1,q'}_{\alpha,\nu+9\kappa}} + \lambda\|\psi\|^2_{q'}\Big)^{\frac{1}{2}}.\end{aligned}$$

Consequently, it follows

$$\|\mathfrak{J}F\|_{\mathfrak{X}^{-1,q}_{\alpha,\nu+9\kappa,\lambda}} \lesssim_{q,\alpha,\nu,\kappa} \|\mathfrak{J}F\|_{\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda}}, \tag{2.23}$$

and thus

$$\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;X^{1,q}_{\alpha,\nu+9\kappa})} + \frac{\lambda}{2}\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;L^q)} \lesssim_{p,q,\alpha,\nu,\kappa} \|\mathfrak{J}F\|^2_{L^p(\mathbb{R}_+;\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda})}. \tag{2.24}$$

Fix $\alpha,\nu,\kappa > 0$ and $\bar{p} \in (2,3]$. Let $C_{p,q}$ be the optimal constant such that the above estimate holds, and choose

$$\lambda_0 := \sup_{2\leq q\leq p\leq\bar{p}} \lambda_0(p,q,\alpha,\nu,\kappa), \qquad \lambda \geq \lambda_0. \tag{2.25}$$

For $p = q = 2$, by applying the standard energy method combined with Proposition B.2 and (2.17), and enlarging λ_0 if necessary, it follows

$$\|\tilde{w}\|^2_{L^2(\mathbb{R}_+;X^{1,2}_{\alpha,\nu+9\kappa})} + \frac{\lambda}{2}\|\tilde{w}\|^2_{L^2(\mathbb{R}_+;L^2)} \le \|\mathfrak{J}F\|^2_{L^2(\mathbb{R}_+;\tilde{\mathfrak{X}}^{-1,2}_{\alpha,\nu+9\kappa,\lambda})}.$$

Therefore, one gets $C_{2,2} \le 1$. Together with the direct interpolation argument for the optimal constants in [Agr25], see (A.10)–(A.12) therein, this yields

$$\limsup_{p\to 2} C_p \le C_{2,2} \le 1, \qquad C_p := \sup_{2\le q\le p} C_{p,q}. \tag{2.26}$$

Step II. In this step, we consider the following equation

$$\partial_t \tilde{w} - \nu\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w} + \lambda\tilde{w} = \mathfrak{J}F + \mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w} \tag{2.27}$$

with zero initial data and boundary conditions $(2.16)_2$. Notice that the above equation can be reformulated as

$$\partial_t \tilde{w} - (\nu+9\kappa)\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w} + \lambda\tilde{w} = \mathfrak{J}F + \mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w} - 9\kappa\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w}, \tag{2.28}$$

so $\tilde{w}$ satisfies (2.19) with the forcing term replaced by $\mathfrak{J}F + \mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w} - 9\kappa\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w}$, and estimate (2.24) is applicable.

We turn to the L^2-based estimate of $\mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w} - 9\kappa\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w}$. Let $\tilde{\psi} \in X^{1,2} \cap \widetilde{H}_{\mathrm{curl}}$, and write $\psi := \mathfrak{J}^{-1}\tilde{\psi}, w := \mathfrak{J}^{-1}\tilde{w}$. By Proposition B.1, one has $\psi, w \in H^1 \cap H_{\mathrm{curl}}$. Then,

$$\begin{aligned}
\langle \mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w} - 9\kappa\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w}, \tilde{\psi}\rangle &= \langle \mathfrak{J}S_\theta w, \mathfrak{J}\psi\rangle - 9\kappa\langle \mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w}, \tilde{\psi}\rangle \\
&= \langle S_\theta w, \psi\rangle - 9\kappa\langle \Delta\tilde{w}, \tilde{\psi}\rangle - \langle \mathcal{B}S_\theta w, \psi\rangle - \langle \mathfrak{J}S_\theta w, \mathcal{B}\psi\rangle + 9\kappa\langle [\mathcal{B}, \partial_y^2]\mathfrak{J}^{-1}\tilde{w}, \tilde{\psi}\rangle \\
&=: \mathcal{I}_2(\tilde{w}, \tilde{\psi}) + \mathcal{I}_3(\tilde{w}, \tilde{\psi}),
\end{aligned}$$

where

$$\mathcal{I}_2(\tilde{w}, \tilde{\psi}) := \langle S_\theta w, \psi\rangle - 9\kappa\langle \Delta\tilde{w}, \tilde{\psi}\rangle + c_1\langle \tilde{w}, \tilde{\psi}\rangle,$$

and

$$\mathcal{I}_3(\tilde{w}, \tilde{\psi}) := -\langle \mathcal{B}S_\theta w, \psi\rangle - \langle \mathfrak{J}S_\theta w, \mathcal{B}\psi\rangle + 9\kappa\langle [\mathcal{B}, \partial_y^2]\mathfrak{J}^{-1}\tilde{w}, \tilde{\psi}\rangle - c_1\langle \tilde{w}, \tilde{\psi}\rangle.$$

We claim that $\mathcal{I}_2(\tilde{w}, \tilde{\psi})$ is a positive symmetric bilinear form. Indeed, the symmetric property follows from integration by parts. The positivity is established in the following way. By integrating by parts, and using the relation $w = \tilde{w} + \mathcal{B}\mathfrak{J}^{-1}\tilde{w}$ combined with Corollary A.5, Proposition B.1, and the Young inequality, one has

$$\begin{aligned}
\mathcal{I}_2(\tilde{w}, \tilde{w}) &= -2\kappa \textstyle\sum_{k\neq 0}\sum_{j=1,2} \theta_k^2 \|\Pi(\sigma_{k,j}\cdot\nabla)w\|^2 + 9\kappa(\|\nabla\tilde{w}\|^2 + \alpha\|\tilde{w}_3\|^2_{L^2(\partial D)}) + c_1\|\tilde{w}\|^2 \\
&\ge -8\kappa\|\nabla w\|^2 + 9\kappa(\|\nabla\tilde{w}\|^2 + \alpha\|\tilde{w}_3\|^2_{L^2(\partial D)}) + c_1\|\tilde{w}\|^2 \\
&\ge -8\kappa(\|\nabla\tilde{w}\| + C_\alpha\|\tilde{w}\|)^2 + 9\kappa(\|\nabla\tilde{w}\|^2 + \alpha\|\tilde{w}_3\|^2_{L^2(\partial D)}) + c_1\|\tilde{w}\|^2 \\
&\ge \frac{\kappa}{2}\|\nabla\tilde{w}\|^2 + 9\kappa\alpha\|\tilde{w}_3\|^2_{L^2(\partial D)} + (c_1 - C_{\alpha,\kappa})\|\tilde{w}\|^2.
\end{aligned}$$

Choosing $c_1 \ge C_{\alpha,\kappa}$, one gets the positivity of $\mathcal{I}_2$. Consequently, by applying the polar identity, it follows

$$|\mathcal{I}_2(\tilde{w}, \tilde{\psi})| \le \mathcal{I}_2(\tilde{w}, \tilde{w})^{\frac{1}{2}}\mathcal{I}_2(\tilde{\psi}, \tilde{\psi})^{\frac{1}{2}},$$

where, by enlarging λ_0 in (2.25) so that $\lambda \ge 2c_1\frac{\nu+9\kappa}{9\kappa}$, one has

$$\mathcal{I}_2(\tilde{w}, \tilde{w}) \le 9\kappa(\|\nabla\tilde{w}\|^2 + \alpha\|\tilde{w}_3\|^2_{L^2(\partial D)}) + c_1\|\tilde{w}\|^2 \le \frac{9\kappa}{\nu+9\kappa}\left(\|\tilde{w}\|^2_{X^{1,2}_{\alpha,\nu+9\kappa}} + \frac{\lambda}{2}\|\tilde{w}\|^2\right),$$

and thus

$$|\mathcal{I}_2(\tilde{w},\tilde{\psi})| \leq \frac{9\kappa}{\nu+9\kappa}\left(\|\tilde{w}\|^2_{X^{1,2}_{\alpha,\nu+9\kappa}} + \frac{\lambda}{2}\|\tilde{w}\|^2\right)^{\frac{1}{2}}\left(\|\tilde{\psi}\|^2_{X^{1,2}_{\alpha,\nu+9\kappa}} + \lambda\|\tilde{\psi}\|^2\right)^{\frac{1}{2}}. \tag{2.29}$$

It remains to bound $\mathcal{I}_3$. By applying Corollary A.5, Propositions B.1 and B.2, and the definition (2.22), it follows

$$\begin{aligned}|\mathcal{I}_3(\tilde{w},\tilde{\psi})| &\lesssim_{\alpha,\kappa} \|\nabla w\|\|\psi\| + \|\tilde{w}\|\|\tilde{\psi}\| \lesssim_{\alpha,\kappa} \|\tilde{w}\|_{X^{1,2}}\|\tilde{\psi}\| \\ &\lesssim_{\alpha,\nu,\kappa} \lambda^{-\frac{1}{2}}\left(\|\tilde{w}\|^2_{X^{1,2}_{\alpha,\nu+9\kappa}} + \frac{\lambda}{2}\|\tilde{w}\|^2\right)^{\frac{1}{2}}\left(\|\tilde{\psi}\|^2_{X^{1,2}_{\alpha,\nu+9\kappa}} + \lambda\|\tilde{\psi}\|^2\right)^{\frac{1}{2}}.\end{aligned}$$

Together with (2.29), this yields

$$\|\mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w} - 9\kappa\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w}\|_{\tilde{\mathfrak{X}}^{-1,2}_{\alpha,\nu+9\kappa,\lambda}} \leq C_2'\left(\|\tilde{w}\|^2_{X^{1,2}_{\alpha,\nu+9\kappa}} + \frac{\lambda}{2}\|\tilde{w}\|^2\right)^{\frac{1}{2}}, \tag{2.30}$$

where, by choosing λ_0 in (2.25) sufficiently large, one has

$$C_2' := \left(\frac{9\kappa}{\nu+9\kappa} + C_{\alpha,\nu,\kappa}\lambda^{-\frac{1}{2}}\right) < 1.$$

Next, we establish an L^q-based estimate for $\mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w} - 9\kappa\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w}$. Indeed, by applying Corollary A.5, Propositions B.1 and B.2, and integrating by parts, for any $\phi \in X^{1,q'} \cap \widetilde{H}_{\mathrm{curl}}$, it follows

$$\begin{aligned}|\langle\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w},\phi\rangle| &\lesssim_{\alpha,q} \|\tilde{w}\|_q\|\phi\|_{q'} + \|\tilde{w}_3\|_{L^q(\partial D)}\|\phi_3\|_{L^{q'}(\partial D)} + \|\nabla\tilde{w}\|_q\|\nabla\phi\|_{q'} \\ &\lesssim_{\alpha,q} \|\tilde{w}\|_{X^{1,q}}\|\phi\|_{X^{1,q'}},\end{aligned} \tag{2.31}$$

and

$$|\langle\mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w},\phi\rangle| \lesssim_{\alpha,q} \|S_\theta\mathfrak{J}^{-1}\tilde{w}\|_{H^{-1,q}}\|\phi\|_{X^{1,q'}}.$$

Moreover, for any $\varphi \in H^{1,q'}$, by integrating by parts again and applying Corollary A.2, one has

$$\begin{aligned}|\langle S_\theta\mathfrak{J}^{-1}\tilde{w},\varphi\rangle| &\lesssim_\kappa \|\Pi\|_{\mathcal{L}(L^q)}\sum_{k\neq 0}\sum_{j=1,2}\theta_k^2\|(\sigma_{-k,j}\cdot\nabla)\mathfrak{J}^{-1}\tilde{w}\|_q\|(\sigma_{-k,j}\cdot\nabla)\Pi\varphi\|_{q'} \\ &\lesssim_\kappa \|\Pi\|_{\mathcal{L}(L^q)}\|\Pi\|_{\mathcal{L}(H^{1,q'})}\|\tilde{w}\|_{X^{1,q}}\|\varphi\|_{H^{1,q'}} \lesssim_{\kappa,q} \|\tilde{w}\|_{X^{1,q}}\|\varphi\|_{H^{1,q'}},\end{aligned}$$

which implies

$$|\langle\mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w},\phi\rangle| \lesssim_{\alpha,\kappa,q} \|\tilde{w}\|_{X^{1,q}}\|\phi\|_{X^{1,q'}}.$$

Together with (2.31) and the definition (2.22), this implies

$$\|\mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w} - 9\kappa\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w}\|_{\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda}} \leq C_{\alpha,\nu,\kappa,q}\left(\|\tilde{w}\|^2_{X^{1,q}_{\alpha,\nu+9\kappa}} + \frac{\lambda}{2}\|\tilde{w}\|_q^2\right)^{\frac{1}{2}}.$$

Let C_q' be the optimal constant such that the above estimate holds. Applying the interpolation argument for the optimal constants in [Agr25] combined with (2.30), one has

$$\limsup_{q\to 2} C_q' \leq C_2' < 1. \tag{2.32}$$

Now applying the estimate (2.24) to equation (2.28) and using the Young and Minkowski inequalities, it follows

$$\begin{aligned}\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;X^{1,q}_{\alpha,\nu+9\kappa})}+\frac{\lambda}{2}\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;L^q)}&\le C_{p,q}(1+\delta^{-1})\|\mathfrak{J}F\|^2_{L^p(\mathbb{R}_+;\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda})}\\&\quad+C_{p,q}(1+\delta)\|\mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w}-9\kappa\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w}\|^2_{L^p(\mathbb{R}_+;\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda})}\\&\le C_{p,q}(1+\delta^{-1})\|\mathfrak{J}F\|^2_{L^p(\mathbb{R}_+;\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda})}\\&\quad+C_{p,q}C'^2_q(1+\delta)\left\|\left(\|\tilde{w}\|^2_{X^{1,q}_{\alpha,\nu+9\kappa}}+\frac{\lambda}{2}\|\tilde{w}\|^2_q\right)^{\frac12}\right\|^2_{L^p(\mathbb{R}_+)}\\&\le C_{p,q}(1+\delta^{-1})\|\mathfrak{J}F\|^2_{L^p(\mathbb{R}_+;\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda})}\\&\quad+C_{p,q}C'^2_q(1+\delta)\left(\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;X^{1,q}_{\alpha,\nu+9\kappa})}+\frac{\lambda}{2}\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;L^q)}\right).\end{aligned}$$

Choosing p_0 sufficiently close to 2 and δ sufficiently small, and using (2.26), (2.32), for any $2\le q\le p\le p_0$, one has

$$\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;X^{1,q}_{\alpha,\nu+9\kappa})}+\frac{\lambda}{2}\|\tilde{w}\|^2_{L^p(\mathbb{R}_+;L^q)}\lesssim_{p,q,\alpha,\nu,\kappa}\|\mathfrak{J}F\|^2_{L^p(\mathbb{R}_+;\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda})}.\tag{2.33}$$

Step III. In this step, we consider

$$\partial_t\tilde{w}-\nu\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w}+\lambda\tilde{w}=\mathfrak{J}F+\mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w}+\sqrt{2\kappa}\sum_{k\neq0}\sum_{j=1,2}G_{k,j}\dot{B}^{k,j}_t\tag{2.34}$$

with the initial data $\tilde{w}_{\rm in}$ and boundary conditions $(2.16)_2$. Let z be the solution of

$$\begin{cases}\partial_t z-\nu\Delta z+\lambda z=\nu[\partial_y^2,\mathcal{B}]\mathfrak{J}^{-1}z+\sqrt{2\kappa}\sum_{k\neq0}\sum_{j=1,2}G_{k,j}\dot{B}^{k,j}_t,\\(\partial_y z_3-\alpha z_3)|_{y=0}=(\partial_y z_3+\alpha z_3)|_{y=1}=z_h|_{y=0,1}=0,\\z|_{t=0}=\tilde{w}_{\rm in}.\end{cases}$$

It is known that the operator $-\Delta+\lambda$, endowed with the Dirichlet boundary conditions on the horizontal components and the Robin boundary conditions on the vertical component, admits a bounded H^∞-calculus on L^q with angle less than $\pi/2$, see Section 2 of [VNVW12]. Therefore, by applying Theorem 1.1 therein combined with the standard heat semigroup estimate for the deterministic part, and choosing larger λ_0 in (2.25) if necessary in order to absorb the lower order term, this yields

$$\|\nabla z\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q)}+\lambda\|z\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q)}\lesssim_{p,q,\alpha,\nu,\kappa}\|G\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}+\|\tilde{w}_{\rm in}\|^2_{L^p(\Omega;B^{1-2/p}_{q,p})}.\tag{2.35}$$

Let $\bar{w}:=\tilde{w}-z$. Notice that $\bar{w}$ satisfies equation (2.27) with the right-hand side replaced by $\mathfrak{J}F+\mathfrak{J}S_\theta\mathfrak{J}^{-1}z+\mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w}$. Then, an application of (2.33) and (2.35) combined with the estimate

$$\|\mathfrak{J}S_\theta\mathfrak{J}^{-1}z\|_{\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda}}\lesssim_{\alpha,\nu,\kappa,q}\|z\|_{X^{1,q}}$$

gives

$$\begin{aligned}\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;X^{1,q}_{\alpha,\nu+9\kappa})}+\lambda\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q)}&\lesssim_{p,q,\alpha,\nu,\kappa}\|\mathfrak{J}F\|^2_{L^p(\Omega\times\mathbb{R}_+;\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda})}\\&\quad+\|G\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}+\|\tilde{w}_{\rm in}\|^2_{L^p(\Omega;B^{1-2/p}_{q,p})}.\end{aligned}\tag{2.36}$$

Step IV. Finally, we consider equation (2.16). Define

$$B_{p,q,\theta,\kappa}(\tilde{w}) := \sqrt{2\kappa}\left\|\left(\sum_{k\neq 0}\sum_{j=1,2}\theta_k^2|\Pi(\sigma_{k,j}\cdot\nabla)\tilde{w}|^2\right)^{\frac{1}{2}}\right\|_{L^p(\Omega\times\mathbb{R}_+;L^q)}.$$

Since $q\geq 2$, by the Minkowski inequality, one has

$$\begin{aligned}B_{p,q,\theta,\kappa}(\tilde{w}) &\leq \sqrt{2\kappa}\left\|\left(\sum_{k\neq 0}\sum_{j=1,2}\theta_k^2\|\Pi(\sigma_{k,j}\cdot\nabla)\tilde{w}\|_q^2\right)^{\frac{1}{2}}\right\|_{L^p(\Omega\times\mathbb{R}_+)}\\ &\leq 2\sqrt{2\kappa}\|\Pi\|_{\mathcal{L}(L^q)}\|\nabla\tilde{w}\|_{L^p(\Omega\times\mathbb{R}_+;L^q)}. \end{aligned}\tag{2.37}$$

On the other hand, by the definitions (2.20) and (2.22), it follows

$$\|\tilde{w}\|_{X^{1,q}_{\alpha,\nu}}\leq\|\tilde{w}\|_{X^{1,q}_{\alpha,\nu+9\kappa}},\qquad \|\mathfrak{J}F\|_{\tilde{\mathfrak{X}}^{-1,q}_{\alpha,\nu+9\kappa,\lambda}}\leq\|\mathfrak{J}F\|_{\mathfrak{X}^{-1,q}_{\alpha,\nu,0}}.$$

Together with (2.36) and (2.37), this implies the existence of an optimal constant $C''_{p,q}$ such that

$$\begin{aligned}&\frac{1}{4}\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;X^{1,q}_{\alpha,\nu})}+B^2_{p,q,\theta,\kappa}(\tilde{w})+\frac{\lambda}{2}\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q)}\\ &\leq C''^2_{p,q}\left(\frac{1}{2}\|\mathfrak{J}F\|^2_{L^p(\Omega\times\mathbb{R}_+;\mathfrak{X}^{-1,q}_{\alpha,\nu,0})}+2\kappa\|G\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}+\frac{1}{2}\|\tilde{w}_{\mathrm{in}}\|^2_{L^p(\Omega;B^{1-2/p}_{q,p})}\right)\end{aligned}\tag{2.38}$$

for all solutions $\tilde{w}$ of (2.34). In particular, for $p=q=2$, by the Itô formula, one has

$$\begin{aligned}\mathbb{E}\int_0^\infty\left(\nu\|\nabla\tilde{w}\|^2+\nu\alpha\|\tilde{w}_3\|^2_{L^2(\partial D)}\right)\mathrm{d}t+B^2_{2,2,\theta,\kappa}(\tilde{w})+\lambda\mathbb{E}\int_0^\infty\|\tilde{w}\|^2\mathrm{d}t&\\ \leq\mathbb{E}\int_0^\infty|\langle\mathfrak{J}F+[S_\theta,\mathcal{B}]\mathfrak{J}^{-1}\tilde{w}+\nu[\partial_y^2,\mathcal{B}]\mathfrak{J}^{-1}\tilde{w},\tilde{w}\rangle|\mathrm{d}t&\\ +2\kappa\|G\|^2_{L^2(\Omega\times\mathbb{R}_+;L^2(\ell^2))}+\frac{1}{2}\|\tilde{w}_{\mathrm{in}}\|^2_{L^2(\Omega;L^2)},&\end{aligned}$$

where, by Corollary A.5 and Propositions B.1 and B.2, one has

$$|\langle\mathfrak{J}F+[S_\theta,\mathcal{B}]\mathfrak{J}^{-1}\tilde{w}+\nu[\partial_y^2,\mathcal{B}]\mathfrak{J}^{-1}\tilde{w},\tilde{w}\rangle|\leq\frac{3}{4}\|\tilde{w}\|^2_{X^{1,2}_{\alpha,\nu}}+\frac{1}{2}\|\mathfrak{J}F\|^2_{\mathfrak{X}^{-1,2}_{\alpha,\nu,0}}+C_{\alpha,\nu,\kappa}\|\tilde{w}\|^2.$$

Therefore, by choosing larger λ_0 in (2.25) if necessary, one gets $C''_{2,2}\leq 1$. Combining this with the interpolation argument for the optimal constants in [Agr25], one has

$$\limsup_{p\to 2}C''_p\leq C''_{2,2}\leq 1,\qquad C''_p:=\sup_{2\leq q\leq p}C''_{p,q}.$$

Now, let $\tilde{w}$ be a solution of (2.16). Notice that $\tilde{w}$ solves (2.34) with the stochastic forcing $G_{k,j}$ replaced by $\theta_k\Pi(\sigma_{k,j}\cdot\nabla)\tilde{w}+G_{k,j}$. By applying (2.38), it follows

$$\begin{aligned}&\frac{1}{4}\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;X^{1,q}_{\alpha,\nu})}+B^2_{p,q,\theta,\kappa}(\tilde{w})+\frac{\lambda}{2}\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q)}\\ &\leq C''^2_{p,q}\left(\frac{1}{2}\|\mathfrak{J}F\|^2_{L^p(\Omega\times\mathbb{R}_+;\mathfrak{X}^{-1,q}_{\alpha,\nu,0})}+\frac{1}{2}\|\tilde{w}_{\mathrm{in}}\|^2_{L^p(\Omega;B^{1-2/p}_{q,p})}\right)\\ &\quad+C''^2_{p,q}\left(\sqrt{2\kappa}\|G\|_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}+\sqrt{2\kappa}\|\{\theta_k\Pi(\sigma_{k,j}\cdot\nabla)\tilde{w}\}_{k,j}\|_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}\right)^2\\ &\leq C''^2_{p,q}\left(\frac{1}{2}\|\mathfrak{J}F\|^2_{L^p(\Omega\times\mathbb{R}_+;\mathfrak{X}^{-1,q}_{\alpha,\nu,0})}+\frac{1}{2}\|\tilde{w}_{\mathrm{in}}\|^2_{L^p(\Omega;B^{1-2/p}_{q,p})}\right)\\ &\quad+C''^2_{p,q}(1+\delta^{-1})2\kappa\|G\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}+C''^2_{p,q}(1+\delta)B^2_{p,q,\theta,\kappa}(\tilde{w}),\end{aligned}$$

which, together with (2.37), yields

$$\begin{aligned}\frac{1}{4}\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;X^{1,q}_{\alpha,\nu})}+\frac{\lambda}{2}\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q)} &\le 2\kappa C''^2_{p,q}(1+\delta^{-1})\|G\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}\\ &\quad+\frac{C''^2_{p,q}}{2}\left(\|\mathfrak{J}F\|^2_{L^p(\Omega\times\mathbb{R}_+;\mathfrak{X}^{-1,q}_{\alpha,\nu,0})}+\|\tilde{w}_{\rm in}\|^2_{L^p(\Omega;B^{1-2/p}_{q,p})}\right)\\ &\quad+8(C''^2_{p,q}(1+\delta)-1)\kappa\|\Pi\|^2_{\mathcal{L}(L^q)}\|\nabla\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q)}.\end{aligned}$$

Since $\|\Pi\|_{\mathcal{L}(L^2)}=1$ and by interpolation, $\|\Pi\|_{\mathcal{L}(L^q)}\to 1$ as $q\to 2$, one has

$$\limsup_{p\to 2}8(C''^2_p(1+\delta)-1)\kappa\|\Pi\|^2_{\mathcal{L}(L^q)}\le 8\delta\kappa.$$

By choosing p_0 sufficiently close to 2 such that for any $2\le q\le p<p_0$,

$$8(C''^2_p(1+\delta)-1)\kappa\|\Pi\|^2_{\mathcal{L}(L^q)}\le 10\delta\kappa,$$

and δ sufficiently small so that $\delta<\frac{\nu}{160\kappa}$, we obtain (2.18). □

Now we are in a position to prove Proposition 2.2.

Proof of Proposition 2.2. **Step I.** We consider a shifted version of the corrected equation (2.5)

$$\begin{cases}\partial_t\tilde{w}-\nu\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde{w}+\lambda\tilde{w}=\mathfrak{J}\operatorname{curl}f+\mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w}\\ \qquad\qquad+\sqrt{2\kappa}\sum_{k\neq 0}\sum_{j=1,2}\theta_k\mathfrak{J}\Pi(\sigma_{k,j}\cdot\nabla)\mathfrak{J}^{-1}\tilde{w}\dot{B}^{k,j}_t,\\ (\partial_y\tilde{w}_3-\alpha\tilde{w}_3)|_{y=0}=(\partial_y\tilde{w}_3+\alpha\tilde{w}_3)|_{y=1}=\tilde{w}_h|_{y=0,1}=0,\\ \tilde{w}|_{t=0}=\tilde{w}_{\rm in},\end{cases}\tag{2.39}$$

where $\lambda\ge\lambda_0$ with λ_0 given in Lemma 2.3. Since

$$\theta_k\mathfrak{J}\Pi(\sigma_{k,j}\cdot\nabla)\mathfrak{J}^{-1}\tilde{w}=\theta_k\Pi(\sigma_{k,j}\cdot\nabla)\tilde{w}+\theta_k\Pi(\sigma_{k,j}\cdot\nabla)\mathcal{B}\mathfrak{J}^{-1}\tilde{w}-\theta_k\mathcal{B}\Pi(\sigma_{k,j}\cdot\nabla)\mathfrak{J}^{-1}\tilde{w},$$

by applying Lemma 2.3, it follows that

$$\begin{aligned}\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;H^{1,q})}+\lambda\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q)} &\lesssim_{p,q,\alpha,\nu,\kappa}\|\mathfrak{J}\operatorname{curl}f\|^2_{L^p(\Omega\times\mathbb{R}_+;X^{-1,q})}\\ &\quad+\|\tilde{w}_{\rm in}\|^2_{L^p(\Omega;B^{1-2/p}_{q,p})}+\|\{\theta_k\Pi(\sigma_{k,j}\cdot\nabla)\mathcal{B}\mathfrak{J}^{-1}\tilde{w}\}_{k,j}\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}\\ &\quad+\|\{\theta_k\mathcal{B}\Pi(\sigma_{k,j}\cdot\nabla)\mathfrak{J}^{-1}\tilde{w}\}_{k,j}\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}.\end{aligned}\tag{2.40}$$

Applying an L^q-version of (2.8) and integrating by parts, one has

$$\|\mathfrak{J}\operatorname{curl}f\|_{X^{-1,q}}\le\|\operatorname{curl}f\|_{X^{-1,q}}+\|\mathcal{B}\operatorname{curl}f\|_{X^{-1,q}}\lesssim_{\alpha,q}\|f\|_q.\tag{2.41}$$

On the other hand, by Corollary A.5, Propositions B.1 and B.2, and the Minkowski inequality, it follows

$$\begin{aligned}\|\{\theta_k\Pi(\sigma_{k,j}\cdot\nabla)\mathcal{B}\mathfrak{J}^{-1}\tilde{w}\}_{k,j}\|_{L^q(\ell^2)} &\le\|\{\theta_k\|\Pi(\sigma_{k,j}\cdot\nabla)\mathcal{B}\mathfrak{J}^{-1}\tilde{w}\|_q\}_{k,j}\|_{\ell^2}\\ &\lesssim_q\|\nabla\mathcal{B}\mathfrak{J}^{-1}\tilde{w}\|_q\lesssim_{\alpha,q}\|\tilde{w}\|_q.\end{aligned}\tag{2.42}$$

Similarly,

$$\|\{\theta_k\mathcal{B}\Pi(\sigma_{k,j}\cdot\nabla)\mathfrak{J}^{-1}\tilde{w}\}_{k,j}\|_{L^q(\ell^2)}\lesssim_{\alpha,q}\|\tilde{w}\|_q.\tag{2.43}$$

Combining (2.40)–(2.43) and enlarging λ_0 if necessary, one gets

$$\|\tilde{w}\|^2_{L^p(\Omega\times\mathbb{R}_+;H^{1,q})}\lesssim_{p,q,\alpha,\nu,\kappa}\|f\|^2_{L^p(\Omega\times\mathbb{R}_+;L^q)}+\|\tilde{w}_{\rm in}\|^2_{L^p(\Omega;B^{1-2/p}_{q,p})}.\tag{2.44}$$

Step II. Next, we establish fractional regularity for the shifted equation (2.39). Applying stochastic maximal regularity of the heat semigroup to equation (2.39), see Theorem 1.2 in [VNVW12], it follows

$$\begin{aligned}\|\tilde{w}\|_{L^p(\Omega;H^{s,p}(\mathbb{R}_+;H^{1-2s,q}))} \lesssim_{s,p,q,\alpha,\nu,\kappa} \|\tilde{w}_{\mathrm{in}}\|_{L^p(\Omega;B^{1-2/p}_{q,p})} \\ + \|\mathfrak{J}\operatorname{curl} f + \mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w} + \nu[\partial_y^2,\mathcal{B}]\mathfrak{J}^{-1}\tilde{w}\|_{L^p(\Omega\times\mathbb{R}_+;X^{-1,q})} \\ + \|\{\theta_k\mathfrak{J}\Pi(\sigma_{k,j}\cdot\nabla)\mathfrak{J}^{-1}\tilde{w}\}_{k,j}\|_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}, \end{aligned} \tag{2.45}$$

where, by using (2.41), Corollary A.5, Propositions B.1 and B.2, and the Minkowski inequality, one has

$$\|\mathfrak{J}\operatorname{curl} f + \mathfrak{J}S_\theta\mathfrak{J}^{-1}\tilde{w} + \nu[\partial_y^2,\mathcal{B}]\mathfrak{J}^{-1}\tilde{w}\|_{X^{-1,q}} \lesssim_{\alpha,\nu,\kappa,q} \|f\|_q + \|\tilde{w}\|_{H^{1,q}},$$

and

$$\|\{\theta_k\mathfrak{J}\Pi(\sigma_{k,j}\cdot\nabla)\mathfrak{J}^{-1}\tilde{w}\}_{k,j}\|_{L^q(\ell^2)} \lesssim_{\alpha,q} \|\tilde{w}\|_{H^{1,q}}.$$

Plugging the above estimates into (2.45) and applying (2.44), one gets

$$\|\tilde{w}\|_{L^p(\Omega;H^{s,p}(\mathbb{R}_+;H^{1-2s,q}))} \lesssim_{s,p,q,\alpha,\nu,\kappa} \|\tilde{w}_{\mathrm{in}}\|_{L^p(\Omega;B^{1-2/p}_{q,p})} + \|f\|_{L^p(\Omega\times\mathbb{R}_+;L^q)}. \tag{2.46}$$

Step III. Finally, by applying the transform $\widehat{w}(t) := e^{-\lambda t}\tilde{w}(t)$ and (2.46), we obtain (2.15) for the corrected vorticity $\tilde{w}$. This, together with Proposition B.1, completes the proof for the non-degenerate case. The degenerate case is handled in the same way, with the following notational modifications. Throughout the proof of Proposition 2.2 and Lemma 2.3, one replaces the transport term $\sigma_{k,j}\cdot\nabla$ by $\sigma_k\cdot\nabla_x$, and S_θ is defined in $(1.16)_2$. Since σ_k, considered as a three-dimensional vector field with zero vertical component, is divergence-free and tangent to the boundary ∂D, all integrations by parts used above produce no boundary contribution. Therefore, Proposition 2.2 and Lemma 2.3 hold verbatim in the degenerate case. □

3 Scaling limit of the Navier–Stokes equation with transport noise and cut-off

In this section, we study the stochastic NS equation with cut-off

$$\begin{cases} \partial_t w - \nu\Delta w + \eta_R(\|w\|)[(u\cdot\nabla)w - (w\cdot\nabla)u] = S_\theta w + \sqrt{2\kappa}\Pi(\dot{W}\cdot\nabla)w, \\ u = \operatorname{curl}^{-1} w, \\ (\partial_y w_3 - \alpha w_3)|_{y=0} = (w_h - \alpha u_h^\perp)|_{y=0} = 0, \\ (\partial_y w_3 + \alpha w_3)|_{y=1} = (w_h + \alpha u_h^\perp)|_{y=1} = 0, \\ w|_{t=0} = w_{\mathrm{in}}, \end{cases} \tag{3.1}$$

where W is given in (1.7) or (1.10), S_θ is defined accordingly in (1.16), and $\eta_R : [0,\infty) \to [0,1]$ is a smooth function satisfying $\eta_R \equiv 1$ on $[0,R]$ and $\eta_R \equiv 0$ on $[2R,\infty)$. For $\gamma > 0$, define the noise coefficients $\theta_\cdot$ by

$$\theta_k^N := \frac{\mathcal{C}_N}{|k|^\gamma}\mathbb{I}_{\{N\le|k|\le 2N\}}, \tag{3.2}$$

where $\mathcal{C}_N$ is chosen so that $\|\theta_\cdot^N\|_{\ell^2} = 1$ for every N. We denote by w_{cut}^N the solution of (3.1) corresponding to the noise coefficient $\theta_\cdot^N$.

The main goal of this section is to investigate the asymptotic behavior of w_{cut}^N as $N\to\infty$. The analysis is carried out in three steps. First, using the Meyers-type estimate obtained in Proposition 2.2, we prove the tightness of the laws $\{\mathcal{D}(w_{\mathrm{cut}}^N)\}_{N\ge1}$ in a suitable space-time topology. Second, under this topology, we identify the limit of the Itô–Stratonovich corrector S_θ. Finally, we show that $\{w_{\mathrm{cut}}^N\}_{N\ge1}$ converges to a process $w_{\mathrm{cut}}^{\mathrm{det}}$, which solves a limiting deterministic equation.

3.1 Tightness of the cut-off solutions

The main result of this subsection is stated below.

Proposition 3.1. *Let $\alpha, \nu, \kappa, R > 0$, and*

$$w_{\text{in}} \in L^\infty(\Omega; H_{\text{curl}} \cap H^1).$$

Then, for each $N \geq 1$, there exists a unique global solution w^N_{cut} of the cut-off equation (3.1). Moreover, there exists δ_0 depending on α, ν, κ such that for any $0 < \delta < \delta_0$ and $T > 0$, the laws $\{\mathcal{D}(w^N_{\text{cut}})\}_{N\geq 1}$ are tight in

$$\chi_{\delta,T} := C([0,T]; H^\delta) \cap L^2(0,T; H^{1-\delta}).$$

Proof. We only give the proof of the non-degenerate case, since the degenerate case follows from the same argument by replacing $\sigma_{k,j} \cdot \nabla$ by $\sigma_k \cdot \nabla_x$.

Step I. In this step, we derive some uniform-in-N a priori estimates. Utilizing the identity

$$(u \cdot \nabla)w - (w \cdot \nabla)u = \operatorname{curl}((u \cdot \nabla)u),$$

the cut-off equation (3.1) can be formulated in the form

$$\partial_t w - \nu \Delta w = \operatorname{curl} f_R(w) + S_\theta w + \sqrt{2\kappa}\Pi(\dot{W} \cdot \nabla)w, \tag{3.3}$$

where

$$f_R(w) := -\eta_R(\|w\|)(u \cdot \nabla)u, \qquad u = \operatorname{curl}^{-1} w.$$

By applying Proposition 2.2 with $q = 2$, $p \in [2, p_0)$, and $s \in [0, 1/2)$, one has

$$\|w^N_{\text{cut}}\|_{L^p(\Omega; H^{s,p}(0,T;H^{1-2s}))} \lesssim_{s,p,\alpha,\nu,\kappa,T} \|f_R(w^N_{\text{cut}})\|_{L^p(\Omega\times(0,T);L^2)} + \|w_{\text{in}}\|_{B^{1-2/p}_{2,p}}, \tag{3.4}$$

where, by the Hölder inequality, the Sobolev embedding $H^1 \hookrightarrow L^6$, Corollary A.5, and the Gagliardo–Nirenberg interpolation inequality

$$\|v\|_3 \lesssim \|v\|^{\frac{1}{2}} \|v\|^{\frac{1}{2}}_{H^1},$$

it follows

$$\|(u \cdot \nabla)u\| \lesssim \|u\|_6 \|\nabla u\|_3 \lesssim \|u\|^{\frac{3}{2}}_{H^1} \|\nabla u\|^{\frac{1}{2}}_{H^1} \lesssim \|w\|^{\frac{3}{2}} \|w\|^{\frac{1}{2}}_{H^1}.$$

This implies

$$\|f_R(w^N_{\text{cut}})\| \lesssim_R 1 + \|\nabla w^N_{\text{cut}}\|^{\frac{1}{2}},$$

and thus

$$\|f_R(w^N_{\text{cut}})\|_{L^p(\Omega\times(0,T);L^2)} \leq C_{T,R,\epsilon} + \epsilon \|\nabla w^N_{\text{cut}}\|_{L^p(\Omega\times(0,T);L^2)}. \tag{3.5}$$

Plugging the above estimate into (3.4) and choosing $s = 0$ and ϵ sufficiently small, one has

$$\|w^N_{\text{cut}}\|_{L^p(\Omega\times(0,T);H^1)} \lesssim_{p,\alpha,\nu,\kappa,T,R} 1 + \|w_{\text{in}}\|_{B^{1-2/p}_{2,p}},$$

which, together with (3.4), (3.5), and the embedding $H^1 \hookrightarrow B^{1-2/p}_{2,p}$, yields

$$\sup_{N\geq 1} \|w^N_{\text{cut}}\|_{L^p(\Omega; H^{s,p}(0,T;H^{1-2s}))} \lesssim_{s,p,\alpha,\nu,\kappa,T,R} 1 + \|w_{\text{in}}\|_{H^1}. \tag{3.6}$$

Step II. The unique solvability of (3.1) follows from a standard Galerkin-type argument combined with the a priori estimate (3.6). For simplicity, we give a sketch and omit the details. Indeed, the Galerkin approximation is a finite-dimensional SDE with locally Lipschitz

coefficients. Moreover, the cut-off makes the nonlinear drift globally controlled, and the estimate (3.6) is uniform in the Galerkin dimension. Passing to the limit ensures a global solution. The uniqueness holds, because of the local Lipschitz property of $w \mapsto f_R(w)$. Therefore, for each $N \geq 1$, there exists a unique global solution w^N_{cut}.

Step III. It remains to prove the tightness of the laws $\{\mathcal{D}(w^N_{\mathrm{cut}})\}_{N\geq 1}$. Now choose $p \in (2, p_0)$ and $\frac{1}{p} < s < \frac{1}{2}$. Set

$$\delta_0 := 1 - 2s.$$

Fix arbitrary positive $\delta < \delta_0$. By applying Theorems 2.1 and 2.2 in [FG95] combined with the embedding

$$H^{s,p}(0,T;H^{1-2s}) \hookrightarrow W^{s,p}(0,T;H^{1-2s}),$$

where $W^{s,p}$ denotes the fractional Slobodeckij–Sobolev space, one has

$$H^{s,p}(0,T;H^{1-2s}) \hookrightarrow\hookrightarrow C([0,T];H^{\delta})$$

and

$$L^p(0,T;H^1) \cap H^{s,p}(0,T;H^{1-2s}) \hookrightarrow\hookrightarrow L^p(0,T;H^{1-\delta}) \hookrightarrow L^2(0,T;H^{1-\delta}).$$

Together with (3.6), this ensures the tightness of $\{\mathcal{D}(w^N_{\mathrm{cut}})\}_{N\geq 1}$ in $\chi_{\delta,T}$. □

3.2 Scaling limit of the Itô–Stratonovich corrector

Next, we justify the scaling limit of the Itô–Stratonovich corrector under the convergence topology specified in Proposition 3.1.

3.2.1 The non-degenerate case

We first address the non-degenerate case. For $s > 1/2$, define

$$\mathfrak{T}_h : (L^2(\partial D))^2 \to (H^{-s}(D))^2, \qquad g \mapsto \mathfrak{T}_h g := (g|_{y=0})\delta_{y=0} - (g|_{y=1})\delta_{y=1},$$

where $\delta_{y=\cdot}$ denotes the Dirac mass at $\{y = \cdot\}$. Equivalently, $\mathfrak{T}_h g$ is given by

$$\langle \mathfrak{T}_h g, \vartheta \rangle = \int_{\mathbb{T}^2} g \cdot \vartheta|_{y=0} \mathrm{d}x - \int_{\mathbb{T}^2} g \cdot \vartheta|_{y=1} \mathrm{d}x. \tag{3.7}$$

Due to the trace theorem, the above operator $\mathfrak{T}_h$ is well-defined and bounded. Based on $\mathfrak{T}_h$, we introduce the operator $\mathfrak{T} : (L^2(\partial D))^2 \to (H^{-s}(D))^3$ by

$$\langle \mathfrak{T} g, \phi \rangle := \langle \mathfrak{T}_h g, \phi_h \rangle, \qquad \forall \phi \in (H^s(D))^3. \tag{3.8}$$

The main result is stated as follows.

Proposition 3.2. *Let $\{\theta^N_\cdot\}_{N\geq 1}$ be defined in (3.2) and $s \in (1/2, 1]$. For any $\varphi \in H_{\mathrm{curl}} \cap C^\infty$ satisfying $\varphi_h \in C^\infty_c(D)$,*

$$\lim_{N\to\infty} \left\| S_{\theta^N}\varphi - \frac{4\kappa}{5}\Delta\varphi - \mathfrak{T}\left(\frac{2\kappa}{15}\nabla_x \varphi_3|_{\partial D}\right)\right\|_{H^{-s}} = 0, \tag{3.9}$$

where S_θ is defined in $(1.16)_1$, and $\mathfrak{T}$ is given in (3.8).

Proof. **Step I.** We first consider the case of a single mode. Let

$$l := (l_h, n/2) = (l_1, l_2, n/2), \qquad l_1, l_2, n \in \mathbb{Z}$$

be a wave number, and

$$\varphi := e^{2\pi i l_h \cdot x} \begin{pmatrix} ib_1 \sin n\pi y \\ ib_2 \sin n\pi y \\ b_3 \cos n\pi y \end{pmatrix} \tag{3.10}$$

with $b := (b_1, b_2, b_3)$ satisfying $b \cdot l = 0$. By the Euler formula, one has

$$\varphi = \frac{1}{2}(e^{2\pi i l^+ \cdot \mathfrak{z}} b^+ - e^{2\pi i l^- \cdot \mathfrak{z}} b^-),$$

where

$$\mathfrak{z} := (x, y), \qquad l^+ := l, \qquad b^+ = b, \qquad l^- := (l_h, -n/2), \qquad b^- := (b_h, -b_3).$$

Similarly, from (1.5), it follows that

$$\sigma_{k,j} = \frac{c_k}{2}(e^{2\pi i k^+ \cdot \mathfrak{z}} a^+_{k,j} + e^{2\pi i k^- \cdot \mathfrak{z}} a^-_{k,j}), \tag{3.11}$$

which implies

$$(\sigma_{-k,j} \cdot \nabla)\varphi = \frac{c_k}{2}\pi i \sum_{\mathfrak{r},\mathfrak{s}=\pm} e^{2\pi i (l^{\mathfrak{r}} - k^{\mathfrak{s}})\cdot \mathfrak{z}} (a^{\mathfrak{s}}_{-k,j} \cdot l^{\mathfrak{r}}) \mathfrak{d}_{\mathfrak{r}} b^{\mathfrak{r}} \tag{3.12}$$

with $\mathfrak{d}_+ := 1$ and $\mathfrak{d}_- := -1$. Consequently, by Proposition A.1, one has

$$\Pi(\sigma_{-k,j} \cdot \nabla)\varphi = (\sigma_{-k,j} \cdot \nabla)\varphi - \nabla \psi_{k,j}, \tag{3.13}$$

where $\psi_{k,j}$ solves

$$\begin{cases} \Delta \psi_{k,j} = -c_k \pi^2 \sum_{\mathfrak{r},\mathfrak{s}=\pm} e^{2\pi i (l^{\mathfrak{r}} - k^{\mathfrak{s}})\cdot \mathfrak{z}} (a^{\mathfrak{s}}_{-k,j} \cdot l^{\mathfrak{r}})(\mathfrak{d}_{\mathfrak{r}} b^{\mathfrak{r}} \cdot (l^{\mathfrak{r}} - k^{\mathfrak{s}})), \\ \psi_{k,j}|_{y=0,1} = 0. \end{cases}$$

Plugging the ansatz $\psi_{k,j} = \phi_{k,j}(y) e^{2\pi i (l_h - k_h)\cdot x}$ into the above equation and applying the principle of superposition, one has the decomposition $\phi_{k,j} = \sum_{\mathfrak{r},\mathfrak{s}=\pm} \phi^{\mathfrak{r},\mathfrak{s}}_{k,j}$ with $\phi^{\mathfrak{r},\mathfrak{s}}_{k,j}$ satisfying

$$\begin{cases} (\partial_y^2 - 4\pi^2 |l_h - k_h|^2)\phi^{\mathfrak{r},\mathfrak{s}}_{k,j} = -c_k \pi^2 e^{\pi i (\mathfrak{r} n - \mathfrak{s} m) y} (a^{\mathfrak{s}}_{-k,j} \cdot l^{\mathfrak{r}})(\mathfrak{d}_{\mathfrak{r}} b^{\mathfrak{r}} \cdot (l^{\mathfrak{r}} - k^{\mathfrak{s}})) =: F^{\mathfrak{r},\mathfrak{s}}_{k,j}, \\ \phi^{\mathfrak{r},\mathfrak{s}}_{k,j}|_{y=0,1} = 0. \end{cases} \tag{3.14}$$

Since θ^N_k is supported on high modes $\{N \le |k| \le 2N\}$, to study the scaling limit of S_{θ^N}, one has to understand the asymptotic behavior of $\phi^{\mathfrak{r},\mathfrak{s}}_{k,j}$ as $|k| \to \infty$. Notice that the above equation with large $|k|$ is a singular-limit problem. It is natural to introduce the decomposition

$$\phi^{\mathfrak{r},\mathfrak{s}}_{k,j} = \phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j} + \phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}. \tag{3.15}$$

Here, the bulk part

$$\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j} := -\frac{1}{4\pi^2 |l^{\mathfrak{r}} - k^{\mathfrak{s}}|^2} F^{\mathfrak{r},\mathfrak{s}}_{k,j} \tag{3.16}$$

satisfies

$$(\partial_y^2 - 4\pi^2 |l_h - k_h|^2)\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j} = F^{\mathfrak{r},\mathfrak{s}}_{k,j},$$

and the boundary layer term $\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}$ is introduced to compensate the mismatch of the boundary conditions of $\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}, \phi^{\mathfrak{r},\mathfrak{s}}_{k,j}$. Notice that $\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}$ satisfies the following equation

$$\begin{cases} (\partial_y^2 - 4\pi^2 |l_h - k_h|^2)\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j} = 0, \\ \phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}|_{y=0,1} = -\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}|_{y=0,1}, \end{cases}$$

which can be solved explicitly by the formula

$$\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j} = \begin{cases} -\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}(0)\dfrac{\sinh \lambda_k (1-y)}{\sinh \lambda_k} - \phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}(1)\dfrac{\sinh \lambda_k y}{\sinh \lambda_k} & \text{if } \lambda_k \neq 0, \\ -y\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}(1) - (1-y)\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}(0) & \text{if } \lambda_k = 0 \end{cases} \tag{3.17}$$

with $\lambda_k := 2\pi|l_h - k_h|$.

To summarize, from (3.13), it follows that

$$\Pi(\sigma_{-k,j}\cdot\nabla)\varphi = (\sigma_{-k,j}\cdot\nabla)\varphi - \nabla\left[e^{2\pi i(l_h-k_h)\cdot x}\sum_{\mathfrak{r},\mathfrak{s}=\pm}\left(\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}+\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}\right)\right].$$

This gives the decomposition

$$S_{\theta^N}\varphi = S_{\theta^N,\mathrm{bulk}}\varphi - S_{\theta^N,\mathrm{bd}}\varphi, \tag{3.18}$$

where

$$\begin{aligned} S_{\theta^N,\mathrm{bulk}}\varphi :=\ & 2\kappa\sum_{k\neq 0}\sum_{j=1,2}(\theta^N_k)^2\Pi(\sigma_{k,j}\cdot\nabla)(\sigma_{-k,j}\cdot\nabla)\varphi \\ & - 2\kappa\sum_{k\neq 0}\sum_{j=1,2}(\theta^N_k)^2\Pi(\sigma_{k,j}\cdot\nabla)\nabla\left(e^{2\pi i(l_h-k_h)\cdot x}\sum_{\mathfrak{r},\mathfrak{s}=\pm}\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}\right), \end{aligned}$$

and

$$S_{\theta^N,\mathrm{bd}}\varphi = 2\kappa\sum_{k\neq 0}\sum_{j=1,2}(\theta^N_k)^2\Pi(\sigma_{k,j}\cdot\nabla)\nabla\left(e^{2\pi i(l_h-k_h)\cdot x}\sum_{\mathfrak{r},\mathfrak{s}=\pm}\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}\right). \tag{3.19}$$

Step II. In this step, we estimate $S_{\theta^N,\mathrm{bd}}\varphi$. For $\lambda_k \neq 0$, by the solution formulas (3.16), (3.17), and the estimate

$$\left|\frac{\sinh(\lambda_k(1-y))}{\sinh\lambda_k}\right| \lesssim e^{-\lambda_k y}, \qquad \left|\frac{\sinh(\lambda_k y)}{\sinh\lambda_k}\right| \lesssim e^{-\lambda_k(1-y)}, \tag{3.20}$$

it follows that

$$\begin{aligned} \|\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}\| &\lesssim \|\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}\|_\infty\left(\|e^{-\lambda_k y}\| + \|e^{-\lambda_k(1-y)}\|\right) \\ &\lesssim \frac{1}{\sqrt{\lambda_k}|l^{\mathfrak{r}}-k^{\mathfrak{s}}|^2}\|F^{\mathfrak{r},\mathfrak{s}}_{k,j}\|_\infty \lesssim \frac{1}{\sqrt{|l_h-k_h|}|l^{\mathfrak{r}}-k^{\mathfrak{s}}|}, \end{aligned} \tag{3.21}$$

and

$$\|\partial_y\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}\| \lesssim \frac{\sqrt{|l_h-k_h|}}{|l^{\mathfrak{r}}-k^{\mathfrak{s}}|}, \qquad \|\partial_y^2\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}\| \lesssim \frac{|l_h-k_h|^{\frac{3}{2}}}{|l^{\mathfrak{r}}-k^{\mathfrak{s}}|}. \tag{3.22}$$

If $\lambda_k = 0$, then $\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}$ is affine in y and $l_h - k_h = 0$, which implies

$$(\sigma_{k,j}\cdot\nabla)\nabla\left(e^{2\pi i(l_h-k_h)\cdot x}\sum_{\mathfrak{r},\mathfrak{s}=\pm}\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}\right) = 0,$$

and therefore its contribution to $S_{\theta^N,\mathrm{bd}}\varphi$ vanishes. On the other hand, since

$$\sum_{N\leq|k|\leq 2N}|k|^{-2\gamma} \sim N^{3-2\gamma},$$

it follows that the constant $\mathcal{C}_N$ in (3.2) satisfies $\mathcal{C}_N \sim N^{\gamma-3/2}$, which yields

$$\theta^N_k \sim N^{-\frac{3}{2}}, \qquad N\leq|k|\leq 2N. \tag{3.23}$$

Combining (3.19)–(3.23), one gets

$$\begin{aligned} \|S_{\theta^N,\mathrm{bd}}\varphi\| &\lesssim \kappa\sum_{N\leq|k|\leq 2N}\sum_{j=1,2}N^{-3}\left\|\nabla^2\left(e^{2\pi i(l_h-k_h)\cdot x}\sum_{\mathfrak{r},\mathfrak{s}=\pm}\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}\right)\right\| \\ &\lesssim \kappa N^{-3}\sum_{N\leq|k|\leq 2N}\frac{|l_h-k_h|^{\frac{3}{2}}}{|l^{\mathfrak{r}}-k^{\mathfrak{s}}|} \lesssim \kappa N^{\frac{1}{2}}. \end{aligned} \tag{3.24}$$

Next, we establish an estimate in H^{-1} for $S_{\theta^N,\mathrm{bd}}\varphi$. Let $\vartheta \in H^1$ be a test function. By integrating by parts and using (3.21) and (3.22), one has

$$\begin{aligned}
&\left\langle \Pi(\sigma_{k,j}\cdot\nabla)\nabla\left(e^{2\pi i(l_h-k_h)\cdot x}\textstyle\sum_{\mathfrak{r},\mathfrak{s}=\pm}\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}\right),\vartheta\right\rangle \\
&\qquad = \left\langle \nabla\left(e^{2\pi i(l_h-k_h)\cdot x}\textstyle\sum_{\mathfrak{r},\mathfrak{s}=\pm}\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}\right),(\sigma_{-k,j}\cdot\nabla)\Pi\vartheta\right\rangle \\
&\qquad \lesssim \frac{\sqrt{|l_h-k_h|}}{|l^{\mathfrak{r}}-k^{\mathfrak{s}}|}\|\vartheta\|_{H^1},
\end{aligned}$$

which implies

$$\left\|\Pi(\sigma_{k,j}\cdot\nabla)\nabla\left(e^{2\pi i(l_h-k_h)\cdot x}\textstyle\sum_{\mathfrak{r},\mathfrak{s}=\pm}\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bd},k,j}\right)\right\|_{H^{-1}} \lesssim \frac{\sqrt{|l_h-k_h|}}{|l^{\mathfrak{r}}-k^{\mathfrak{s}}|},$$

and thus

$$\|S_{\theta^N,\mathrm{bd}}\varphi\|_{H^{-1}} \lesssim \kappa\textstyle\sum_{k\neq 0}(\theta^N_k)^2\dfrac{\sqrt{|l_h-k_h|}}{|l^{\mathfrak{r}}-k^{\mathfrak{s}}|} \lesssim \kappa N^{-\frac{1}{2}}. \tag{3.25}$$

Together with interpolation and (3.24), this implies

$$\lim_{N\to\infty}\left\|S_{\theta^N,\mathrm{bd}}\varphi\right\|_{H^{-s}} = 0, \qquad s\in(1/2,1]. \tag{3.26}$$

Step III. We turn to the bulk part $S_{\theta,\mathrm{bulk}}$. For simplicity, in the computation below, we shall take the normalization coefficient $c_k=\sqrt{2}$ in (1.5) for all k. This does not change the scaling limit. Indeed, it only modifies the modes on the layer $\{m=0\}$, which contains only $O(N^2)$ lattice points in the shell $\{N\le|k|\le 2N\}$. Moreover, on this layer, one has

$$\sigma_{k,j}\cdot\nabla = c_k e^{2\pi i k_h\cdot x}a_{k,j,h}\cdot\nabla_x, \qquad a_{k,j,h}\cdot k_h = 0,$$

which, together with (3.14) and (3.16), implies

$$\begin{aligned}
&(\sigma_{k,j}\cdot\nabla)\nabla\big(e^{2\pi i(l_h-k_h)\cdot x}\textstyle\sum_{\mathfrak{r},\mathfrak{s}=\pm}\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}\big) \\
&\qquad = -2\pi^2 c_k e^{2\pi i l_h\cdot x}(a_{k,j,h}\cdot l_h)(2(l_h-k_h),\mathfrak{r}n)\left(\textstyle\sum_{\mathfrak{r},\mathfrak{s}=\pm}\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}\right),
\end{aligned}$$

and

$$(\sigma_{k,j}\cdot\nabla)(\sigma_{-k,j}\cdot\nabla)\varphi = c_k^2(a_{k,j,h}\cdot\nabla_x)^2\varphi.$$

Combining this with (3.16) and (3.23), it follows

$$\begin{aligned}
&\|\textstyle\sum_{k\neq0,m=0}\sum_{j=1,2}(\theta^N_k)^2\Pi(\sigma_{k,j}\cdot\nabla)\nabla\big(e^{2\pi i(l_h-k_h)\cdot x}\sum_{\mathfrak{r},\mathfrak{s}=\pm}\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}\big)\| \\
&\qquad\qquad \lesssim \textstyle\sum_{N\le|k|\le2N,m=0}(\theta^N_k)^2 \lesssim N^{-1},
\end{aligned}$$

and

$$\|\textstyle\sum_{k\neq0,m=0}\sum_{j=1,2}(\theta^N_k)^2\Pi(\sigma_{k,j}\cdot\nabla)(\sigma_{-k,j}\cdot\nabla)\varphi\| \lesssim \sum_{N\le|k|\le2N,m=0}(\theta^N_k)^2 \lesssim N^{-1}.$$

Therefore, modifying the normalization coefficient c_k on $\{m=0\}$ does not change the scaling limit of $S_{\theta^N,\mathrm{bulk}}\varphi$. Since the estimates (3.24) and (3.25) are uniform with respect to c_k, the modification does not affect the vanishing of $S_{\theta^N,\mathrm{bd}}\varphi$, and thus does not change the scaling limit of $S_{\theta^N}\varphi$.

Now let

$$\mathcal{A}^{\mathfrak{r},\mathfrak{s}}_{k,j} := -\sqrt{2}\pi^2(a^{\mathfrak{s}}_{-k,j}\cdot l^{\mathfrak{r}})\mathfrak{d}_{\mathfrak{r}}b^{\mathfrak{r}}. \tag{3.27}$$

Under the above notation, from (3.12), (3.14), and (3.16), it follows

$$F^{\mathfrak{r},\mathfrak{s}}_{k,j} = e^{\pi i(\mathfrak{r}n-\mathfrak{s}m)y}\mathcal{A}^{\mathfrak{r},\mathfrak{s}}_{k,j}\cdot(l^{\mathfrak{r}}-k^{\mathfrak{s}}), \qquad \phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j} = -\frac{e^{\pi i(\mathfrak{r}n-\mathfrak{s}m)y}\mathcal{A}^{\mathfrak{r},\mathfrak{s}}_{k,j}\cdot(l^{\mathfrak{r}}-k^{\mathfrak{s}})}{4\pi^2|l^{\mathfrak{r}}-k^{\mathfrak{s}}|^2},$$

and

$$(\sigma_{-k,j}\cdot\nabla)\varphi = \frac{1}{2\pi i}\sum_{\mathfrak{r},\mathfrak{s}=\pm} e^{2\pi i(l^{\mathfrak{r}}-k^{\mathfrak{s}})\cdot\mathfrak{z}}\mathcal{A}^{\mathfrak{r},\mathfrak{s}}_{k,j}.$$

Consequently, one has

$$\begin{aligned}&2\kappa\sum_{k\neq 0}\sum_{j=1,2}(\theta^N_k)^2\Pi(\sigma_{k,j}\cdot\nabla)(\sigma_{-k,j}\cdot\nabla)\varphi\\ &\qquad = \frac{-i\kappa}{\pi}\Pi\sum_{k\neq 0,j=1,2}\sum_{\mathfrak{r},\mathfrak{s}=\pm}(\theta^N_k)^2\mathcal{A}^{\mathfrak{r},\mathfrak{s}}_{k,j}(\sigma_{k,j}\cdot\nabla)e^{2\pi i(l^{\mathfrak{r}}-k^{\mathfrak{s}})\cdot\mathfrak{z}},\end{aligned}$$

and

$$\begin{aligned}&2\kappa\sum_{k\neq 0}\sum_{j=1,2}(\theta^N_k)^2\Pi(\sigma_{k,j}\cdot\nabla)\nabla\left(e^{2\pi i(l_h-k_h)\cdot x}\sum_{\mathfrak{r},\mathfrak{s}=\pm}\phi^{\mathfrak{r},\mathfrak{s}}_{\mathrm{bulk},k,j}\right)\\ &\quad = \frac{-\kappa}{2\pi^2}\sum_{k\neq 0,j=1,2}\sum_{\mathfrak{r},\mathfrak{s}=\pm}(\theta^N_k)^2\Pi(\sigma_{k,j}\cdot\nabla)\nabla\left(e^{2\pi i(l^{\mathfrak{r}}-k^{\mathfrak{s}})\cdot\mathfrak{z}}\frac{\mathcal{A}^{\mathfrak{r},\mathfrak{s}}_{k,j}\cdot(l^{\mathfrak{r}}-k^{\mathfrak{s}})}{|l^{\mathfrak{r}}-k^{\mathfrak{s}}|^2}\right)\\ &\quad = \frac{-i\kappa}{\pi}\Pi\sum_{k\neq 0,j=1,2}\sum_{\mathfrak{r},\mathfrak{s}=\pm}(\theta^N_k)^2\left(\mathcal{P}_{l^{\mathfrak{r}}-k^{\mathfrak{s}}}\mathcal{A}^{\mathfrak{r},\mathfrak{s}}_{k,j}(\sigma_{k,j}\cdot\nabla)e^{2\pi i(l^{\mathfrak{r}}-k^{\mathfrak{s}})\cdot\mathfrak{z}}\right),\end{aligned}$$

where for any vector $\zeta\in\mathbb{R}^3$,

$$\mathcal{P}_\zeta := \frac{\zeta\otimes\zeta}{|\zeta|^2}, \qquad \mathcal{P}^\perp_\zeta := I_3 - \mathcal{P}_\zeta$$

are projection matrices in $\mathbb{R}^3$. This yields

$$S_{\theta^N,\mathrm{bulk}}\varphi = \frac{-i\kappa}{\pi}\Pi\sum_{k\neq 0,j=1,2}\sum_{\mathfrak{r},\mathfrak{s}=\pm}(\theta^N_k)^2\mathcal{P}^\perp_{l^{\mathfrak{r}}-k^{\mathfrak{s}}}\mathcal{A}^{\mathfrak{r},\mathfrak{s}}_{k,j}(\sigma_{k,j}\cdot\nabla)e^{2\pi i(l^{\mathfrak{r}}-k^{\mathfrak{s}})\cdot\mathfrak{z}}.$$

On the other hand, from (3.11), it follows

$$(\sigma_{k,j}\cdot\nabla)e^{2\pi i(l^{\mathfrak{r}}-k^{\mathfrak{s}})\cdot\mathfrak{z}} = \sqrt{2}\pi i\sum_{\mathfrak{t}=\pm}e^{2\pi i(k^{\mathfrak{t}}+l^{\mathfrak{r}}-k^{\mathfrak{s}})\cdot\mathfrak{z}}a^{\mathfrak{t}}_{k,j}\cdot(l^{\mathfrak{r}}-k^{\mathfrak{s}}).$$

Thus, we introduce the decomposition

$$S_{\theta^N,\mathrm{bulk}}\varphi = S_{\theta^N,\mathrm{bulk},1}\varphi + S_{\theta^N,\mathrm{bulk},2}\varphi, \tag{3.28}$$

where

$$S_{\theta^N,\mathrm{bulk},1}\varphi := \sqrt{2}\kappa\Pi\sum_{k\neq 0,j=1,2}\sum_{\mathfrak{r},\mathfrak{s}=\pm,\mathfrak{t}=\mathfrak{s}}(\theta^N_k)^2\mathcal{P}^\perp_{l^{\mathfrak{r}}-k^{\mathfrak{s}}}\mathcal{A}^{\mathfrak{r},\mathfrak{s}}_{k,j}e^{2\pi i l^{\mathfrak{r}}\cdot\mathfrak{z}}a^{\mathfrak{s}}_{k,j}\cdot(l^{\mathfrak{r}}-k^{\mathfrak{s}}) \tag{3.29}$$

is the non-oscillatory bulk part, corresponding to the matched interactions $\mathfrak{t}=\mathfrak{s}$, and

$$\begin{aligned}S_{\theta^N,\mathrm{bulk},2}\varphi := \sqrt{2}\kappa\Pi\sum_{k\neq 0,j=1,2}\sum_{\mathfrak{r},\mathfrak{s}=\pm,\mathfrak{t}\neq\mathfrak{s}}(\theta^N_k)^2\mathcal{P}^\perp_{l^{\mathfrak{r}}-k^{\mathfrak{s}}}\mathcal{A}^{\mathfrak{r},\mathfrak{s}}_{k,j}\\ \times e^{2\pi i(k^{\mathfrak{t}}+l^{\mathfrak{r}}-k^{\mathfrak{s}})\cdot\mathfrak{z}}a^{\mathfrak{t}}_{k,j}\cdot(l^{\mathfrak{r}}-k^{\mathfrak{s}})\end{aligned} \tag{3.30}$$

is the oscillatory bulk part, corresponding to the unmatched interactions $\mathfrak{t}\neq\mathfrak{s}$. The latter is oscillatory in the vertical direction.

Step IV. In this step, we justify the scaling limit of the non-oscillatory bulk part $S_{\theta^N,\mathrm{bulk},1}\varphi$. By the definition (3.27) of $\mathcal{A}^{\mathfrak{r},\mathfrak{s}}_{k,j}$ and (1.3), it follows that

$$S_{\theta^N,\mathrm{bulk},1}\varphi = -2\pi^2\kappa\Pi\sum_{k\neq 0,j=1,2}\sum_{\mathfrak{r},\mathfrak{s}=\pm}(\theta^N_k)^2|a^{\mathfrak{s}}_{k,j}\cdot l^{\mathfrak{r}}|^2\mathcal{P}^\perp_{l^{\mathfrak{r}}-k^{\mathfrak{s}}}(e^{2\pi i l^{\mathfrak{r}}\cdot\mathfrak{z}}\mathfrak{d}_{\mathfrak{r}}b^{\mathfrak{r}}).$$

Since l, b are given, it suffices to compute the coefficient matrix

$$\mathcal{S}^{\mathfrak{r}}_{\theta^N} := -2\pi^2\kappa \sum_{k\neq 0, j=1,2} \sum_{\mathfrak{s}=\pm} (\theta^N_k)^2 |a^{\mathfrak{s}}_{k,j} \cdot l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{l^{\mathfrak{r}}-k^{\mathfrak{s}}}.$$

By applying the estimate

$$|\mathcal{P}_{l^{\mathfrak{r}}-k^{\mathfrak{s}}} - \mathcal{P}_{k^{\mathfrak{s}}}| \lesssim \frac{|l|}{|k|}, \tag{3.31}$$

see Lemma 5.5 in [FL21], one has

$$\mathcal{S}^{\mathfrak{r}}_{\theta^N} = -2\pi^2\kappa \sum_{k\neq 0, j=1,2} \sum_{\mathfrak{s}=\pm} (\theta^N_k)^2 |a^{\mathfrak{s}}_{k,j} \cdot l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{k^{\mathfrak{s}}} + o(1).$$

Furthermore, by using the radial symmetry of $\theta^N_{\cdot}$, for any $\mathfrak{s}$, one has

$$\sum_{k\neq 0, j=1,2} (\theta^N_k)^2 |a^{\mathfrak{s}}_{k,j} \cdot l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{k^{\mathfrak{s}}} = \sum_{k\neq 0, j=1,2} (\theta^N_k)^2 |a^{\mathfrak{s}}_{k^{\mathfrak{s}},j} \cdot l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{k},$$

where, by using (1.4), it follows

$$\sum_{k\neq 0, j=1,2} (\theta^N_k)^2 |a^{+}_{k^{+},j} \cdot l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{k} = \sum_{k\neq 0} (\theta^N_k)^2 |\mathcal{P}^{\perp}_k l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{k},$$

and

$$\begin{aligned}
\sum_{k\neq 0, j=1,2} (\theta^N_k)^2 |a^{-}_{k^{-},j} \cdot l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{k} &= \sum_{k\neq 0, j=1,2} (\theta^N_k)^2 |a_{k^{-},j} \cdot l^{-\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{k} \\
&= \sum_{k\neq 0} (\theta^N_k)^2 (l^{-\mathfrak{r}})^T \left(I_3 - \frac{k^- \otimes k^-}{|k|^2} \right) l^{-\mathfrak{r}} \mathcal{P}^{\perp}_{k} \\
&= \sum_{k\neq 0} (\theta^N_k)^2 |\mathcal{P}^{\perp}_k l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{k}.
\end{aligned}$$

Therefore,

$$\mathcal{S}^{\mathfrak{r}}_{\theta^N} = -4\pi^2\kappa \sum_{k\neq 0} (\theta^N_k)^2 |\mathcal{P}^{\perp}_k l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{k} + o(1). \tag{3.32}$$

For any $\mathfrak{r}$, by proceeding as in the proof of Lemma 5.6 in [FL21] and using the polar coordinates, it follows

$$\lim_{N\to\infty} \sum_{k\neq 0} (\theta^N_k)^2 |\mathcal{P}^{\perp}_k l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{k} = \frac{1}{4\pi} \int_{|\zeta|=1} |\mathcal{P}^{\perp}_{\zeta} l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{\zeta} \mathrm{d}S. \tag{3.33}$$

Let Q be the orthogonal matrix such that $Q^T \frac{l^{\mathfrak{r}}}{|l^{\mathfrak{r}}|} = e_1$, where e_1 is the unit vector in $\mathbb{R}^3$ with the first component being 1. Using the coordinate transform $\zeta \mapsto Q\zeta$, one gets

$$\begin{aligned}
\frac{1}{4\pi} \int_{|\zeta|=1} |\mathcal{P}^{\perp}_{\zeta} l^{\mathfrak{r}}|^2 \mathcal{P}^{\perp}_{\zeta} \mathrm{d}S &= \frac{|l^{\mathfrak{r}}|^2}{4\pi} Q \left(\int_{|\zeta|=1} |\mathcal{P}^{\perp}_{\zeta} e_1|^2 \mathcal{P}^{\perp}_{\zeta} \mathrm{d}S \right) Q^T \\
&= \frac{|l^{\mathfrak{r}}|^2}{4\pi} Q \left(\int_{|\zeta|=1} (1-\zeta_1^2)(I_3 - \zeta\otimes\zeta) \mathrm{d}S \right) Q^T =: \frac{|l^{\mathfrak{r}}|^2}{4\pi} Q \mathcal{M} Q^T.
\end{aligned} \tag{3.34}$$

The matrix $\mathcal{M}$ is obtained in the following way. By symmetry, for $i \neq j$,

$$\mathcal{M}_{ij} = \int_{|\zeta|=1} (1-\zeta_1^2)(\delta_{ij} - \zeta_i\zeta_j) \mathrm{d}S = 0.$$

Through direct computation, one has

$$\mathcal{M}_{11} = \int_{|\zeta|=1} (1-\zeta_1^2)^2 \mathrm{d}S = \frac{32\pi}{15}, \qquad \mathcal{M}_{22} = \int_{|\zeta|=1} (1-\zeta_1^2)(1-\zeta_2^2) \mathrm{d}S = \frac{8\pi}{5}.$$

By symmetry again, $\mathcal{M}_{33} = \mathcal{M}_{22} = \frac{8\pi}{5}$. Hence,

$$\mathcal{M} = 4\pi \operatorname{diag}\left\{\frac{8}{15}, \frac{2}{5}, \frac{2}{5}\right\},$$

which, together with (3.34), yields

$$\frac{1}{4\pi}\int_{|\zeta|=1} |\mathcal{P}_\zeta^\perp l^{\mathfrak{r}}|^2 \mathcal{P}_\zeta^\perp \mathrm{d}S = \frac{2}{5}|l^{\mathfrak{r}}|^2 I_3 + \frac{2}{15} l^{\mathfrak{r}} \otimes l^{\mathfrak{r}}.$$

Combining this with (3.32) and (3.33) and using the fact $b^{\mathfrak{r}} \cdot l^{\mathfrak{r}} = 0$, it follows that

$$\begin{aligned}\lim_{N\to\infty} S_{\theta^N,\mathrm{bulk},1}\varphi &= -4\pi^2\kappa\Pi \textstyle\sum_{\mathfrak{r}=\pm} \mathfrak{d}_{\mathfrak{r}} \left(\frac{2}{5}|l^{\mathfrak{r}}|^2 e^{2\pi i l^{\mathfrak{r}}\cdot \mathfrak{z}} b^{\mathfrak{r}} + \frac{2}{15} e^{2\pi i l^{\mathfrak{r}}\cdot\mathfrak{z}} l^{\mathfrak{r}} \otimes l^{\mathfrak{r}} b^{\mathfrak{r}}\right) \\ &= \frac{4\kappa}{5}\Pi\Delta\varphi = \frac{4\kappa}{5}\Delta\varphi, \end{aligned} \tag{3.35}$$

where the convergence holds in L^2.

Step V. We turn to the oscillatory bulk part $S_{\theta^N,\mathrm{bulk},2}$. Let $J := \operatorname{diag}\{1,1,-1\}$. Notice that $Jl^+ = l^-$, $Jk^+ = k^-$, $Ja_{k,j}^+ = a_{k,j}^-$,

$$\mathcal{A}_{k,j}^{+,-} = -\sqrt{2}\pi^2 (a_{-k,j}^- \cdot l^+) b^+ = \sqrt{2}\pi^2 (a_{-k,j}^+ \cdot l^-)\mathfrak{d}_- J b^- = -J\mathcal{A}_{k,j}^{-,+},$$

$\mathcal{A}_{k,j}^{+,+} = -J\mathcal{A}_{k,j}^{-,-}$, and

$$\mathcal{P}_{Jk}^\perp = \left(I_3 - \frac{Jk\otimes Jk}{|k|^2}\right) = J\mathcal{P}_k^\perp J.$$

Applying the above relations to (3.30), it follows that

$$\begin{aligned} S_{\theta^N,\mathrm{bulk},2}\varphi = \sqrt{2}\kappa\Pi \textstyle\sum_{k\neq 0, j=1,2}\sum_{\mathfrak{r}=\pm} (\theta_k^N)^2 (a_{k,j}^+ \cdot l^{\mathfrak{r}} + a_{k,j,3} m) e^{2\pi i l_h\cdot x} \\ \times \left(\mathcal{P}_{l^{\mathfrak{r}}-k^-}^\perp \mathcal{A}_{k,j}^{\mathfrak{r},-} e^{2\pi i(\frac{\mathfrak{r}n}{2}+m)y} - J(\mathcal{P}_{l^{\mathfrak{r}}-k^-}^\perp \mathcal{A}_{k,j}^{\mathfrak{r},-}) e^{-2\pi i(\frac{\mathfrak{r}n}{2}+m)y}\right). \end{aligned}$$

Define

$$\mathcal{V}_{k,j}^{\mathfrak{r}} := (\theta_k^N)^2 (a_{k,j}^+ \cdot l^{\mathfrak{r}} + a_{k,j,3} m)\mathcal{P}_{l^{\mathfrak{r}}-k^-}^\perp \mathcal{A}_{k,j}^{\mathfrak{r},-}, \tag{3.36}$$

so one has

$$\begin{aligned} S_{\theta^N,\mathrm{bulk},2}\varphi &= \sqrt{2}\kappa \textstyle\sum_{k\neq 0,j=1,2}\sum_{\mathfrak{r}=\pm} \Pi\left(\mathcal{V}_{k,j}^{\mathfrak{r}} e^{2\pi i(\frac{\mathfrak{r}n}{2}+m)y} - J\mathcal{V}_{k,j}^{\mathfrak{r}} e^{-2\pi i(\frac{\mathfrak{r}n}{2}+m)y}\right) e^{2\pi i l_h\cdot x} \\ &= 2\sqrt{2}\kappa \textstyle\sum_{k\neq 0,j=1,2}\sum_{\mathfrak{r}=\pm} \Pi\left(\mathcal{V}_{k,j,h}^{\mathfrak{r}} i\sin(\lambda_{m,\mathfrak{r}} y), \mathcal{V}_{k,j,3}^{\mathfrak{r}} \cos(\lambda_{m,\mathfrak{r}} y)\right) e^{2\pi i l_h\cdot x} \\ &=: 2\sqrt{2}\kappa \textstyle\sum_{k\neq 0,j=1,2}\sum_{\mathfrak{r}=\pm} \Pi\mathcal{U}_{k,j}^{\mathfrak{r}}, \end{aligned} \tag{3.37}$$

where $\lambda_{m,\mathfrak{r}} := 2\pi\left(\frac{\mathfrak{r}n}{2} + m\right)$. Furthermore, by applying Proposition A.1 and using the fact

$$\begin{aligned}\int_D \mathcal{U}_{k,j,3}^{\mathfrak{r}} \mathrm{d}x\mathrm{d}y &= \mathbb{I}_{\{l_h=0,\lambda_{m,\mathfrak{r}}=0\}} \mathcal{V}_{k,j,3}^{\mathfrak{r}} \\ &= \mathbb{I}_{\{l_h=0,\lambda_{m,\mathfrak{r}}=0\}} (\theta_k^N)^2 (a_{k,j}^+ \cdot l^{\mathfrak{r}} + a_{k,j,3} m)(\mathcal{P}_{l^{\mathfrak{r}}-k^-}^\perp \mathcal{A}_{k,j}^{\mathfrak{r},-})_3 \\ &= \mathbb{I}_{\{l_h=0,\lambda_{m,\mathfrak{r}}=0\}} (\theta_k^N)^2 (a_{k,j,3}(\mathfrak{r}n/2) + a_{k,j,3} m)(\mathcal{P}_{l^{\mathfrak{r}}-k^-}^\perp \mathcal{A}_{k,j}^{\mathfrak{r},-})_3 = 0, \end{aligned}$$

it follows

$$\Pi\mathcal{U}_{k,j}^{\mathfrak{r}} = \mathcal{U}_{k,j}^{\mathfrak{r}} - \nabla\mathfrak{p}_{k,j}^{\mathfrak{r}}, \tag{3.38}$$

where $\mathfrak{p}^{\mathfrak{r}}_{k,j}$ satisfies

$$\begin{cases} \Delta \mathfrak{p}^{\mathfrak{r}}_{k,j} = \operatorname{div} \mathcal{U}^{\mathfrak{r}}_{k,j} = -e^{2\pi i l_h \cdot x} \sin(\lambda_{m,\mathfrak{r}} y)(2\pi l_h \cdot \mathcal{V}^{\mathfrak{r}}_{k,j,h} + \lambda_{m,\mathfrak{r}} \mathcal{V}^{\mathfrak{r}}_{k,j,3}), \\ \mathfrak{p}^{\mathfrak{r}}_{k,j}|_{y=0,1} = 0. \end{cases} \tag{3.39}$$

The above equation can be solved explicitly via the solution formula

$$\mathfrak{p}^{\mathfrak{r}}_{k,j} = \begin{cases} e^{2\pi i l_h \cdot x} \sin(\lambda_{m,\mathfrak{r}} y) \frac{2\pi l_h \cdot \mathcal{V}^{\mathfrak{r}}_{k,j,h} + \lambda_{m,\mathfrak{r}} \mathcal{V}^{\mathfrak{r}}_{k,j,3}}{4\pi^2 |l_h|^2 + \lambda^2_{m,\mathfrak{r}}}, & \text{if } \lambda_{m,\mathfrak{r}} \neq 0, \\ 0, & \text{if } \lambda_{m,\mathfrak{r}} = 0. \end{cases} \tag{3.40}$$

We claim that

$$\lim_{N\to\infty} 2\sqrt{2}\kappa \sum_{k\neq 0, j=1,2} \sum_{\mathfrak{r}=\pm} (\mathcal{U}^{\mathfrak{r}}_{k,j,3} - \partial_y \mathfrak{p}^{\mathfrak{r}}_{k,j}) = 0 \tag{3.41}$$

in L^2. Indeed, notice that

$$\partial_y \mathfrak{p}^{\mathfrak{r}}_{k,j} = e^{2\pi i l_h \cdot x} \cos(\lambda_{m,\mathfrak{r}} y) \frac{2\pi \lambda_{m,\mathfrak{r}} l_h \cdot \mathcal{V}^{\mathfrak{r}}_{k,j,h} + \lambda^2_{m,\mathfrak{r}} \mathcal{V}^{\mathfrak{r}}_{k,j,3}}{4\pi^2 |l_h|^2 + \lambda^2_{m,\mathfrak{r}}},$$

which gives

$$\mathcal{U}^{\mathfrak{r}}_{k,j,3} - \partial_y \mathfrak{p}^{\mathfrak{r}}_{k,j} = e^{2\pi i l_h \cdot x} \cos(\lambda_{m,\mathfrak{r}} y) \frac{4\pi^2 |l_h|^2 \mathcal{V}^{\mathfrak{r}}_{k,j,3} - 2\pi \lambda_{m,\mathfrak{r}} l_h \cdot \mathcal{V}^{\mathfrak{r}}_{k,j,h}}{4\pi^2 |l_h|^2 + \lambda^2_{m,\mathfrak{r}}}.$$

Since l_h is given as the tangential frequency of φ, by the Bessel inequality in the vertical direction combined with the estimate

$$|\mathcal{V}^{\mathfrak{r}}_{k,j}| \lesssim (\theta^N_k)^2 |m| \lesssim N^{-3} |m|, \tag{3.42}$$

it follows

$$\begin{aligned} &\left\| \sum_{k\neq 0, j=1,2} \sum_{\mathfrak{r}=\pm} (\mathcal{U}^{\mathfrak{r}}_{k,j,3} - \partial_y \mathfrak{p}^{\mathfrak{r}}_{k,j}) \right\| \\ \lesssim &\sum_{k_h \in \mathbb{Z}^2, j=1,2, \mathfrak{r}=\pm} \left\| \sum_{m\in\mathbb{Z}} \mathbb{I}_{\{N\leq |k| \leq 2N\}} \frac{4\pi^2 |l_h|^2 \mathcal{V}^{\mathfrak{r}}_{k,j,3} - 2\pi \lambda_{m,\mathfrak{r}} l_h \cdot \mathcal{V}^{\mathfrak{r}}_{k,j,h}}{4\pi^2 |l_h|^2 + \lambda^2_{m,\mathfrak{r}}} \cos(\lambda_{m,\mathfrak{r}} y) \right\|_{L^2_y} \\ \lesssim &N^{-3} \sum_{k_h \in \mathbb{Z}^2} \left(\sum_{m\in\mathbb{Z}} \mathbb{I}_{\{N\leq |k| \leq 2N\}} \right)^{\frac{1}{2}} \lesssim N^{-\frac{1}{2}} \to 0, \end{aligned} \tag{3.43}$$

as $N \to \infty$.

Next, we turn to the horizontal components. Notice that

$$\nabla_x \mathfrak{p}^{\mathfrak{r}}_{k,j} := 2\pi i e^{2\pi i l_h \cdot x} \sin(\lambda_{m,\mathfrak{r}} y) \frac{2\pi l_h \cdot \mathcal{V}^{\mathfrak{r}}_{k,j,h} + \lambda_{m,\mathfrak{r}} \mathcal{V}^{\mathfrak{r}}_{k,j,3}}{4\pi^2 |l_h|^2 + \lambda^2_{m,\mathfrak{r}}} l_h,$$

which gives

$$\begin{aligned} \mathcal{U}^{\mathfrak{r}}_{k,j,h} - \nabla_x \mathfrak{p}^{\mathfrak{r}}_{k,j} &= i e^{2\pi i l_h \cdot x} \sin(\lambda_{m,\mathfrak{r}} y) \mathcal{V}^{\mathfrak{r}}_{k,j,h} - i e^{2\pi i l_h \cdot x} \sin(\lambda_{m,\mathfrak{r}} y) \frac{4\pi^2 l_h \otimes l_h}{4\pi^2 |l_h|^2 + \lambda^2_{m,\mathfrak{r}}} \mathcal{V}^{\mathfrak{r}}_{k,j,h} \\ &\quad - i e^{2\pi i l_h \cdot x} \sin(\lambda_{m,\mathfrak{r}} y) \frac{2\pi \lambda_{m,\mathfrak{r}} \mathcal{V}^{\mathfrak{r}}_{k,j,3}}{4\pi^2 |l_h|^2 + \lambda^2_{m,\mathfrak{r}}} l_h \\ &=: i e^{2\pi i l_h \cdot x} \sin(\lambda_{m,\mathfrak{r}} y) \mathcal{V}^{\mathfrak{r}}_{k,j,h} + \mathcal{R}^{\mathfrak{r}}_{k,j}. \end{aligned}$$

Repeating the derivation of (3.43), one has

$$\lim_{N\to\infty} \left\| \sum_{k\neq 0, j=1,2} \sum_{\mathfrak{r}=\pm} \mathcal{R}^{\mathfrak{r}}_{k,j} \right\| = 0, \tag{3.44}$$

so it remains to control the main part

$$\mathscr{S}_{\theta^N} := 2\sqrt{2}\kappa i \sum_{k\neq 0, j=1,2} \sum_{\mathfrak{r}=\pm} e^{2\pi i l_h\cdot x} \sin(\lambda_{m,\mathfrak{r}} y) \mathcal{V}^{\mathfrak{r}}_{k,j,h}.$$

Define the boundary trace

$$\mathscr{S}_{\theta^N,\mathrm{bd}} := \begin{cases} 2\sqrt{2}\kappa i \left(\sum_{k\neq 0,\lambda_{m,\mathfrak{r}}\neq 0} \sum_{j=1,2,\mathfrak{r}=\pm} \lambda^{-1}_{m,\mathfrak{r}} \mathcal{V}^{\mathfrak{r}}_{k,j,h}\right) e^{2\pi i l_h\cdot x}, & \text{on } \{y=0\}, \\ 2\sqrt{2}\kappa i (-1)^n \left(\sum_{k\neq 0,\lambda_{m,\mathfrak{r}}\neq 0} \sum_{j=1,2,\mathfrak{r}=\pm} \lambda^{-1}_{m,\mathfrak{r}} \mathcal{V}^{\mathfrak{r}}_{k,j,h}\right) e^{2\pi i l_h\cdot x}, & \text{on } \{y=1\}. \end{cases}$$

Let $\mathfrak{T}_h$ be the operator defined in (3.7). For any smooth function ϑ, by integrating by parts, it follows

$$\begin{aligned} \langle \mathscr{S}_{\theta^N}, \vartheta\rangle &= 2\sqrt{2}\kappa i \sum_{k\neq 0, j=1,2} \sum_{\mathfrak{r}=\pm} \langle e^{2\pi i l_h\cdot x} \sin(\lambda_{m,\mathfrak{r}} y), \mathcal{V}^{\mathfrak{r}}_{k,j,h}\cdot \vartheta\rangle \\ &= \int_{\mathbb{T}^2} \mathscr{S}_{\theta^N,\mathrm{bd}}\cdot \vartheta|_{y=0} \mathrm{d}x - \int_{\mathbb{T}^2} \mathscr{S}_{\theta^N,\mathrm{bd}}\cdot\vartheta|_{y=1}\mathrm{d}x \\ &\quad + 2\sqrt{2}\kappa i \sum_{k\neq 0,\lambda_{m,\mathfrak{r}}\neq 0} \sum_{j=1,2,\mathfrak{r}=\pm} \lambda^{-1}_{m,\mathfrak{r}} \langle e^{2\pi i l_h\cdot x}\cos(\lambda_{m,\mathfrak{r}} y), \mathcal{V}^{\mathfrak{r}}_{k,j,h}\cdot \partial_y\vartheta\rangle \\ &= \langle \mathfrak{T}_h \mathscr{S}_{\theta^N,\mathrm{bd}}, \vartheta\rangle \\ &\quad + 2\sqrt{2}\kappa i \sum_{k\neq 0,\lambda_{m,\mathfrak{r}}\neq 0} \sum_{j=1,2,\mathfrak{r}=\pm} \lambda^{-1}_{m,\mathfrak{r}} \langle e^{2\pi i l_h\cdot x}\cos(\lambda_{m,\mathfrak{r}} y), \mathcal{V}^{\mathfrak{r}}_{k,j,h}\cdot \partial_y\vartheta\rangle. \end{aligned}$$

Let

$$\tilde{\mathcal{R}} = 2\sqrt{2}\kappa i \sum_{k\neq 0,\lambda_{m,\mathfrak{r}}\neq 0} \sum_{j=1,2,\mathfrak{r}=\pm} \lambda^{-1}_{m,\mathfrak{r}} e^{2\pi i l_h\cdot x}\cos(\lambda_{m,\mathfrak{r}} y)\mathcal{V}^{\mathfrak{r}}_{k,j,h}.$$

Then,

$$|\langle \mathscr{S}_{\theta^N} - \mathfrak{T}_h \mathscr{S}_{\theta^N,\mathrm{bd}}, \vartheta\rangle| \leq \|\tilde{\mathcal{R}}\|_{H^{1-s}} \|\partial_y\vartheta\|_{H^{s-1}}.$$

Since $s\in(\frac{1}{2},1]$, the spaces H^{1-s}_0, H^{1-s} coincide, which implies

$$\|\partial_y\vartheta\|_{H^{s-1}} \lesssim \|\vartheta\|_{H^s},$$

and thus

$$\|\mathscr{S}_{\theta^N} - \mathfrak{T}_h\mathscr{S}_{\theta^N,\mathrm{bd}}\|_{H^{-s}} \leq \|\tilde{\mathcal{R}}\|_{H^{1-s}}.$$

The remainder term $\tilde{\mathcal{R}}$ is estimated in the same way as in the vertical case, which gives

$$\|\tilde{\mathcal{R}}\|_{H^{1-s}} \lesssim N^{\frac{1}{2}-s}.$$

Consequently, one gets

$$\|\mathscr{S}_{\theta^N} - \mathfrak{T}_h\mathscr{S}_{\theta^N,\mathrm{bd}}\|_{H^{-s}} \lesssim N^{\frac{1}{2}-s} \to 0, \tag{3.45}$$

as $N\to\infty$.

Now it remains to compute the scaling limit of the coefficient in $\mathscr{S}_{\theta^N,\mathrm{bd}}$

$$\begin{aligned} \tilde{\mathscr{S}}_{\theta^N} &:= 2\sqrt{2}\kappa i \sum_{k\neq 0,\lambda_{m,\mathfrak{r}}\neq 0} \sum_{j=1,2,\mathfrak{r}=\pm} \lambda^{-1}_{m,\mathfrak{r}} \mathcal{V}^{\mathfrak{r}}_{k,j,h} \\ &= 2\sqrt{2}\kappa i \sum_{k\neq 0,\lambda_{m,\mathfrak{r}}\neq 0} \sum_{j=1,2,\mathfrak{r}=\pm} (\theta^N_k)^2 \frac{2\pi a_{k,j,h}\cdot l_h + a_{k,j,3}\lambda_{m,\mathfrak{r}}}{2\pi\lambda_{m,\mathfrak{r}}} (\mathcal{P}^{\perp}_{l^{\mathfrak{r}}-k^-}\mathcal{A}^{\mathfrak{r},-}_{k,j})_h. \end{aligned}$$

Notice that by applying (3.23) and (3.27), it follows

$$\begin{aligned} &\sum_{k\neq 0,\lambda_{m,\mathfrak{r}}\neq 0} \sum_{j=1,2,\mathfrak{r}=\pm} (\theta^N_k)^2 \frac{a_{k,j,h}\cdot l_h}{\lambda_{m,\mathfrak{r}}} (\mathcal{P}^{\perp}_{l^{\mathfrak{r}}-k^-}\mathcal{A}^{\mathfrak{r},-}_{k,j})_h \lesssim \sum_{k\neq 0,\lambda_{m,\mathfrak{r}}\neq 0,\mathfrak{r}=\pm} (\theta^N_k)^2 |\lambda_{m,\mathfrak{r}}|^{-1} \\ &\lesssim \sum_{k\neq 0,\lambda_{m,\mathfrak{r}}\neq 0,|\lambda_{m,\mathfrak{r}}|\leq \epsilon N,\mathfrak{r}=\pm} (\theta^N_k)^2 |\lambda_{m,\mathfrak{r}}|^{-1} + \sum_{k\neq 0,|\lambda_{m,\mathfrak{r}}|>\epsilon N,\mathfrak{r}=\pm} (\theta^N_k)^2 |\lambda_{m,\mathfrak{r}}|^{-1} \\ &\lesssim \epsilon + \epsilon^{-1}N^{-1}. \end{aligned}$$

Consequently, by using (1.3), (3.27), (3.31), one has

$$\begin{aligned}\tilde{\mathscr{S}}_{\theta^N} &= \sqrt{2}\pi^{-1}\kappa i \sum_{k\neq 0,\lambda_{m,\mathfrak{r}}\neq 0}\sum_{j=1,2,\mathfrak{r}=\pm}(\theta_k^N)^2 a_{k,j,3}(\mathcal{P}^{\perp}_{l^{\mathfrak{r}}-k^-}\mathcal{A}^{\mathfrak{r},-}_{k,j})_h + O(\epsilon) + O_\epsilon(N^{-1})\\ &= -2\pi i\kappa \sum_{k\neq 0}\sum_{j=1,2,\mathfrak{r}=\pm}(\theta_k^N)^2 a_{k,j,3}(a^-_{k,j}\cdot l^{\mathfrak{r}})(\mathfrak{d}_{\mathfrak{r}}\mathcal{P}^{\perp}_{k^-}b^{\mathfrak{r}})_h + O(\epsilon) + O_\epsilon(N^{-1}).\end{aligned}$$

Here, the restriction $\lambda_{m,\mathfrak{r}}\neq 0$ is removed, since the condition $\{\lambda_{m,\mathfrak{r}}=0\}=\{m=-\mathfrak{r}n/2\}$ gives at most one horizontal layer, which contains $O(N^2)$ lattice points in the shell and contributes at most $O(N^{-1})$.

We turn to the first term on the right-hand side of the above identity. As the error term $O(\epsilon)+O_\epsilon(N^{-1})$ vanishes by taking first $N\to\infty$ and then $\epsilon\to 0$, we shall simply denote it by $o(1)$ in what follows. Applying (1.4), one has

$$\sum_{j=1,2} a_{k,j,3}(a^-_{k,j}\cdot l^{\mathfrak{r}}) = \sum_{j=1,2} a_{k,j,3}(a_{k,j}\cdot Jl^{\mathfrak{r}}) = \mathrm{Row}_3\,\mathcal{P}^{\perp}_k\cdot Jl^{\mathfrak{r}}.$$

Therefore, by setting $\hat{k}:=\frac{k}{|k|}$, it follows

$$\begin{aligned}\tilde{\mathscr{S}}_{\theta^N} &= -2\pi i\kappa\sum_{k\neq 0}\sum_{\mathfrak{r}=\pm}(\mathrm{Row}_3\,\mathcal{P}^{\perp}_k\cdot Jl^{\mathfrak{r}})(\theta_k^N)^2(\mathfrak{d}_{\mathfrak{r}}\mathcal{P}^{\perp}_{k^-}b^{\mathfrak{r}})_h + o(1)\\ &= 2\pi i\kappa\sum_{k\neq 0}\sum_{\mathfrak{r}=\pm}(\theta_k^N)^2\left(\hat{k}_3\hat{k}_h\cdot l_h + \frac{\mathfrak{r}n}{2}(1-\hat{k}_3^2)\right)\\ &\qquad\qquad\times\left(\mathfrak{r}\big(b_h-(\hat{k}_h\cdot b_h)\hat{k}_h\big)+\hat{k}_3 b_3\hat{k}_h\right)+o(1)\\ &= 4\pi i\kappa\sum_{k\neq 0}(\theta_k^N)^2\left(\hat{k}_3^2 b_3(\hat{k}_h\otimes\hat{k}_h)l_h + \frac{n}{2}(1-\hat{k}_3^2)(I_2-\hat{k}_h\otimes\hat{k}_h)b_h\right)+o(1).\end{aligned}$$

Proceeding as in Step Ⅳ, one gets

$$\begin{aligned}\lim_{N\to\infty}\tilde{\mathscr{S}}_{\theta^N} &= i\kappa\left(b_3 l_h\int_{|\zeta|=1}\zeta_3^2\zeta_h\otimes\zeta_h\mathrm{d}S + \frac{nb_h}{2}\int_{|\zeta|=1}(1-\zeta_3^2)(I_2-\zeta_h\otimes\zeta_h)\mathrm{d}S\right)\\ &= \frac{4\pi i\kappa}{15}(3nb_h+b_3l_h),\end{aligned}$$

which, together with the definition (3.10) of φ, implies

$$\lim_{N\to\infty}\mathscr{S}_{\theta^N,\mathrm{bd}} = \frac{4\kappa}{5}\partial_y\varphi_h|_{\partial D} + \frac{2\kappa}{15}\nabla_x\varphi_3|_{\partial D}$$

in $L^2(\partial D)$, and thus

$$\lim_{N\to\infty}\mathfrak{T}_h\mathscr{S}_{\theta^N,\mathrm{bd}} = \mathfrak{T}_h\left(\frac{4\kappa}{5}\partial_y\varphi_h|_{\partial D} + \frac{2\kappa}{15}\nabla_x\varphi_3|_{\partial D}\right) \tag{3.46}$$

in H^{-s}. To summarize, by (3.37), (3.38), (3.41), (3.44)–(3.46), one has

$$\lim_{N\to\infty}S_{\theta^N,\mathrm{bulk},2}\varphi = \mathfrak{T}\left(\frac{4\kappa}{5}\partial_y\varphi_h|_{\partial D} + \frac{2\kappa}{15}\nabla_x\varphi_3|_{\partial D}\right) \tag{3.47}$$

in H^{-s}.

Step Ⅵ. Combining (3.18), (3.26), (3.28), (3.35), and (3.47), one has the convergence

$$\lim_{N\to\infty}\left\|S_{\theta^N}\varphi - \frac{4\kappa}{5}\Delta\varphi - \mathfrak{T}\left(\frac{4\kappa}{5}\partial_y\varphi_h|_{\partial D} + \frac{2\kappa}{15}\nabla_x\varphi_3|_{\partial D}\right)\right\|_{H^{-s}} = 0 \tag{3.48}$$

for a single mode φ. The extension to general smooth functions follows from a standard argument as in [FL21]. Therefore, we give a brief sketch and omit the details. By choosing

suitable polarizations of b, the single modes introduced in (3.10) form an orthonormal basis of H_{curl}. Together with $\varphi_h \in C_c^\infty(D)$ and $\partial_y \varphi_3 = -\operatorname{div}_h \varphi_h \in C_c^\infty(D)$, this implies that

$$\varphi = \sum_{l_h \in \mathbb{Z}^2, n \geq 0} c_l \varphi_l,$$

where φ_l is a single mode in the form given in (3.10) and the coefficient c_l decays rapidly. For any integer $M \geq 1$, define

$$\varphi^{<M} := \sum_{l_h \in \mathbb{Z}^2, n \geq 0} \mathbb{I}_{\{|l|<M\}} c_l \varphi_l, \qquad \varphi^{\geq M} := \sum_{l_h \in \mathbb{Z}^2, n \geq 0} \mathbb{I}_{\{|l| \geq M\}} c_l \varphi_l.$$

It is clear that (3.48) holds for $\varphi^{<M}$. As for $\varphi^{\geq M}$, notice that from the single-mode estimates established in Steps I–V, one has

$$\sup_{N \geq 1} \|S_{\theta^N} \varphi_l\|_{H^{-s}} \leq C_l,$$

where C_l has a polynomial growth in l. Hence, by choosing M sufficiently large and using the rapid decay of c_l, it follows

$$\sup_{N \geq 1} \left\|S_{\theta^N} \varphi^{\geq M}\right\|_{H^{-s}} \leq \sum_{l_h \in \mathbb{Z}^2, n \geq 0} \mathbb{I}_{\{|l| \geq M\}} |c_l| C_l \leq \epsilon,$$

and

$$\left\| \frac{4\kappa}{5} \Delta \varphi^{\geq M} + \mathfrak{T} \left(\frac{4\kappa}{5} \partial_y \varphi_h^{\geq M}|_{\partial D} + \frac{2\kappa}{15} \nabla_x \varphi_3^{\geq M}|_{\partial D} \right) \right\|_{H^{-s}} \leq \epsilon.$$

This, together with the convergence of $\varphi^{<M}$, implies (3.48) in the general case. Since $\varphi_h \in C_c^\infty(D)$, we obtain (3.9). □

3.2.2 *The degenerate case*

Next, we consider the degenerate case. The main result is given below.

Proposition 3.3. *Let $\{\theta_\cdot^N\}_{N \geq 1}$ be defined in (3.2). For any $\varphi \in H_{\mathrm{curl}} \cap C^\infty$,*

$$\lim_{N \to \infty} \|S_{\theta^N} \varphi - \kappa \mathfrak{A}_{\mathrm{eff}} \varphi\| = 0, \tag{3.49}$$

where S_θ is defined in $(1.16)_2$ and

$$\mathfrak{A}_{\mathrm{eff}} \varphi := \Pi \left(\frac{1}{4} \Delta_x \varphi_h + \frac{1}{2} \nabla_x \operatorname{div}_h \varphi_h, \Delta_x \varphi_3 \right). \tag{3.50}$$

Proof. Notice that

$$S_\theta \varphi = 2\kappa \sum_{k \neq 0} \theta_k^2 \Pi(\sigma_k \cdot \nabla_x)(\sigma_{-k} \cdot \nabla_x)\varphi - 2\kappa \sum_{k \neq 0} \theta_k^2 \Pi(\sigma_k \cdot \nabla_x)\Pi^\perp(\sigma_{-k} \cdot \nabla_x)\varphi,$$

where, by using the definition (1.9) of σ_k and (1.8), it follows

$$\begin{aligned}
\sum_{k \neq 0} \theta_k^2 \Pi(\sigma_k \cdot \nabla_x)(\sigma_{-k} \cdot \nabla_x)\varphi &= \Pi \sum_{k \neq 0} \theta_k^2 \frac{(k^\perp \cdot \nabla_x)^2 \varphi}{|k|^2} \\
&= \Pi \sum_{k \neq 0} \theta_k^2 \frac{1}{|k|^2} (k_2^2 \partial_{x_1}^2 \varphi + k_1^2 \partial_{x_2}^2 \varphi - 2k_1 k_2 \partial_{x_1} \partial_{x_2} \varphi) \\
&= \frac{1}{2} \Pi \Delta_x \varphi = \frac{1}{2} \Delta_x \varphi.
\end{aligned}$$

This implies

$$S_\theta \varphi = \kappa \Delta_x \varphi - S_\theta^\perp \varphi, \tag{3.51}$$

where

$$S_\theta^\perp \varphi := 2\kappa \sum_{k\neq 0} \theta_k^2 \Pi(\sigma_k \cdot \nabla_x)\Pi^\perp(\sigma_{-k} \cdot \nabla_x)\varphi.$$

Therefore, it remains to compute the scaling limit of $S_\theta^\perp \varphi$. As in the proof of Proposition 3.2, the general case can be reduced to a single tangential Fourier mode of the form

$$\varphi(x, y) = e^{2\pi i l\cdot x} b(y) \in H_{\mathrm{curl}}, \tag{3.52}$$

where $b = (b_h, b_3) = (b_1, b_2, b_3)$ is a smooth vector field satisfying

$$2\pi i l \cdot b_h(y) + b_3'(y) = 0. \tag{3.53}$$

Moreover, if $l = 0$, by the horizontal zero-mean condition for the vertical component in (1.2), one has $b_3 \equiv 0$, which implies that φ is a shear flow and $S_{\theta^N}\varphi = \mathfrak{A}_{\mathrm{eff}}\varphi = 0$. Thus, we assume $l \neq 0$ in what follows. The rest of the proof is divided into the following steps.

Step I. Define

$$\mathfrak{d}_k = \begin{cases} 1, & k \in \mathbb{Z}_{0,+}, \\ -1, & k \in \mathbb{Z}_{0,-}. \end{cases}$$

Then, from (1.9), one has

$$(\sigma_{-k} \cdot \nabla_x)\varphi = -2\pi i \mathfrak{d}_{-k} \frac{k^\perp \cdot l}{|k|} e^{2\pi i(l-k)\cdot x} b(y).$$

Together with the divergence free condition (3.53), this gives

$$\begin{aligned} \mathrm{div}((\sigma_{-k} \cdot \nabla_x)\varphi) &= -2\pi i \mathfrak{d}_{-k} \frac{k^\perp \cdot l}{|k|} e^{2\pi i(l-k)\cdot x}(2\pi i(l-k)\cdot b_h + b_3') \\ &= 4\pi^2 \mathfrak{d}_k \frac{k^\perp \cdot l}{|k|} e^{2\pi i(l-k)\cdot x}(k \cdot b_h). \end{aligned}$$

Combining this with Proposition A.1, one gets

$$\Pi^\perp(\sigma_{-k} \cdot \nabla_x)\varphi = \nabla(\mathfrak{d}_k \phi_k(y) e^{2\pi i(l-k)\cdot x}),$$

where ϕ_k solves the Dirichlet–Poisson problem

$$\begin{cases} \phi_k'' - \lambda_k^2 \phi_k = 4\pi^2 \frac{k^\perp \cdot l}{|k|}(k \cdot b_h(y)) =: f_k(y), \\ \phi_k(0) = \phi_k(1) = 0 \end{cases} \tag{3.54}$$

with $\lambda_k := 2\pi|l - k|$. Therefore,

$$S_\theta^\perp \varphi = 4\pi\kappa i \Pi \sum_{k\neq 0} \theta_k^2 \frac{k^\perp \cdot l}{|k|} e^{2\pi i l\cdot x}(2\pi i(l_1 - k_1)\phi_k, 2\pi i(l_2 - k_2)\phi_k, \phi_k').$$

To study the asymptotic behavior of ϕ_k as $|k| \to \infty$, we introduce the decomposition

$$\phi_k = \phi_{\mathrm{bulk},k} + \phi_{\mathrm{bd},k} + \phi_{\mathrm{rem},k}, \tag{3.55}$$

where

$$\phi_{\mathrm{bulk},k} := -\lambda_k^{-2} f_k, \tag{3.56}$$

and $\phi_{\mathrm{bd},k}, \phi_{\mathrm{rem},k}$ satisfy the following problems

$$\begin{cases} \phi_{\mathrm{bd},k}'' - \lambda_k^2 \phi_{\mathrm{bd},k} = 0, \\ \phi_{\mathrm{bd},k}(0) = \lambda_k^{-2} f_k(0), \\ \phi_{\mathrm{bd},k}(1) = \lambda_k^{-2} f_k(1), \end{cases} \tag{3.57}$$

and

$$\begin{cases} \phi''_{\mathrm{rem},k} - \lambda_k^2 \phi_{\mathrm{rem},k} = \lambda_k^{-2} f''_k, \\ \phi_{\mathrm{rem},k}(0) = \phi_{\mathrm{rem},k}(1) = 0. \end{cases} \tag{3.58}$$

Compared with the decomposition (3.15) in the non-degenerate case, the decomposition (3.55) contains an additional remainder term. The reason is that the source term in (3.54) is not a full trigonometric mode in all variables. Consequently, the one-dimensional elliptic problem does not admit a simple explicit particular solution. We therefore use an approximate bulk term and introduce a remainder, which accounts for the error made by replacing the full one-dimensional resolvent by its leading-order high-frequency approximation.

Step II. Define

$$S^{\perp}_{\theta,\mathrm{bulk}}\varphi := 4\pi\kappa i \Pi \textstyle\sum_{k\neq 0} \theta_k^2 \dfrac{k^{\perp}\cdot l}{|k|} e^{2\pi i l\cdot x}(2\pi i(l_1-k_1)\phi_{\mathrm{bulk},k}, 2\pi i(l_2-k_2)\phi_{\mathrm{bulk},k}, \phi'_{\mathrm{bulk},k}),$$

and

$$S^{\perp}_{\theta,\mathrm{bd}}\varphi := 4\pi\kappa i \Pi \textstyle\sum_{k\neq 0} \theta_k^2 \dfrac{k^{\perp}\cdot l}{|k|} e^{2\pi i l\cdot x}(2\pi i(l_1-k_1)\phi_{\mathrm{bd},k}, 2\pi i(l_2-k_2)\phi_{\mathrm{bd},k}, \phi'_{\mathrm{bd},k}),$$

and

$$S^{\perp}_{\theta,\mathrm{rem}}\varphi := 4\pi\kappa i \Pi \textstyle\sum_{k\neq 0} \theta_k^2 \dfrac{k^{\perp}\cdot l}{|k|} e^{2\pi i l\cdot x}(2\pi i(l_1-k_1)\phi_{\mathrm{rem},k}, 2\pi i(l_2-k_2)\phi_{\mathrm{rem},k}, \phi'_{\mathrm{rem},k}).$$

In this step, we prove that

$$\lim_{N\to\infty} \|S^{\perp}_{\theta^N,\mathrm{bd}}\varphi + S^{\perp}_{\theta^N,\mathrm{rem}}\varphi\| = 0. \tag{3.59}$$

Indeed, notice that $\phi_{\mathrm{bd},k}$ has an explicit solution formula

$$\phi_{\mathrm{bd},k}(y) = \lambda_k^{-2} f_k(0) \frac{\sinh(\lambda_k(1-y))}{\sinh\lambda_k} + \lambda_k^{-2} f_k(1) \frac{\sinh(\lambda_k y)}{\sinh\lambda_k},$$

which, together with (3.20), yields

$$\|\phi_{\mathrm{bd},k}\| \lesssim \frac{|k||l|\|b_h\|_{H^1_y}}{\lambda_k^{5/2}}, \qquad \|\phi'_{\mathrm{bd},k}\| \lesssim \frac{|k||l|\|b_h\|_{H^1_y}}{\lambda_k^{3/2}}. \tag{3.60}$$

On the other hand, the remainder term $\phi_{\mathrm{rem},k}$ can be bounded through a direct energy estimate

$$\|\phi'_{\mathrm{rem},k}\|^2 + \lambda_k^2 \|\phi_{\mathrm{rem},k}\|^2 \le \lambda_k^{-2} \|f''_{k,l}\| \|\phi_{\mathrm{rem},k}\|,$$

which implies

$$\|\phi_{\mathrm{rem},k}\| \lesssim \frac{|k||l|\|b_h\|_{H^2_y}}{\lambda_k^4}, \qquad \|\phi'_{\mathrm{rem},k}\| \lesssim \frac{|k||l|\|b_h\|_{H^2_y}}{\lambda_k^3}. \tag{3.61}$$

Together with (3.60), this implies

$$\begin{aligned} \|S^{\perp}_{\theta^N,\mathrm{bd}}\varphi\| &\lesssim \kappa \textstyle\sum_{k\neq 0} (\theta_k^N)^2 |l| \left(|l-k| \|\phi_{\mathrm{bd},k}\| + \|\phi'_{\mathrm{bd},k}\| \right) \\ &\lesssim \kappa |l|^2 \|b_h\|_{H^1_y} \textstyle\sum_{k\neq 0} (\theta_k^N)^2 |k| \lambda_k^{-3/2} \lesssim \left(\kappa |l|^2 \|b_h\|_{H^1_y} \right) N^{-1/2} \to 0, \end{aligned}$$

and

$$\begin{aligned} \|S^{\perp}_{\theta^N,\mathrm{rem}}\varphi\| &\lesssim \kappa \textstyle\sum_{k\neq 0} (\theta_k^N)^2 |l| \left(|l-k| \|\phi_{\mathrm{rem},k}\| + \|\phi'_{\mathrm{rem},k}\| \right) \\ &\lesssim \kappa |l|^2 \|b_h\|_{H^2_y} \textstyle\sum_{k\neq 0} (\theta_k^N)^2 |k| \lambda_k^{-3} \lesssim \left(\kappa |l|^2 \|b_h\|_{H^2_y} \right) N^{-2} \to 0, \end{aligned}$$

as $N \to \infty$.

Step Ⅲ. Now we turn to $S^{\perp}_{\theta^N,\mathrm{bulk}}\varphi$. Since Π is a continuous linear operator, it suffices to study

$$\mathfrak{S}_{\theta^N} := 4\pi\kappa i \sum_{k\neq 0} \theta_k^2 \frac{k^\perp \cdot l}{|k|} e^{2\pi i l\cdot x}(2\pi i(l_1-k_1)\phi_{\mathrm{bulk},k}, 2\pi i(l_2-k_2)\phi_{\mathrm{bulk},k}, \phi'_{\mathrm{bulk},k}).$$

For the vertical component, it follows

$$\begin{aligned}\|\mathfrak{S}_{\theta^N,3}\|_{L^2} &\lesssim \kappa \sum_{k\neq 0}(\theta_k^N)^2|l|\|\phi'_{\mathrm{bulk},k}\|_{L^2_y} \lesssim \kappa \sum_{k\neq 0}(\theta_k^N)^2|l|\lambda_k^{-2}\|f'_k\| \\ &\lesssim \kappa \sum_{k\neq 0}(\theta_k^N)^2|l|^2\lambda_k^{-2}|k|\|b_h\|_{H^1_y} \lesssim \left(\kappa|l|^2\|b_h\|_{H^1_y}\right) N^{-1} \to 0. \end{aligned} \tag{3.62}$$

As for the horizontal components, one has

$$\begin{aligned}\mathfrak{S}_{\theta^N,h} &= -8\pi^2\kappa e^{2\pi i l\cdot x}\sum_{k\neq 0}(\theta_k^N)^2\left(\frac{k^\perp\cdot l}{|k|}\right)(l-k)\phi_{\mathrm{bulk},k}(y) \\ &= 8\pi^2\kappa e^{2\pi i l\cdot x}\sum_{k\neq 0}(\theta_k^N)^2\left(\frac{k^\perp\cdot l}{|k|}\right)(l-k)\lambda_k^{-2}\left(4\pi^2\frac{k^\perp\cdot l}{|k|}k\cdot b_h(y)\right) \\ &= 8\pi^2\kappa e^{2\pi i l\cdot x}\left(\sum_{k\neq 0}(\theta_k^N)^2\left(\frac{k^\perp\cdot l}{|k|}\right)^2\frac{(l-k)\otimes k}{|l-k|^2}\right) b_h(y).\end{aligned}$$

Thus, it remains to justify the convergence of the coefficient matrix

$$\mathcal{M}_{\theta^N} := \sum_{k\neq 0}(\theta_k^N)^2\left(\frac{k^\perp\cdot l}{|k|}\right)^2\frac{(l-k)\otimes k}{|l-k|^2}.$$

Since this can be done in the same way as in Step Ⅳ of the proof of Proposition 3.2, we only give a brief sketch. By using a two-dimensional version of the estimate (3.31)

$$\left|\frac{(l-k)\otimes(l-k)}{|l-k|^2} - \frac{k\otimes k}{|k|^2}\right| \lesssim \frac{|l|}{|k|},$$

and repeating the derivation of (3.33), one gets

$$\mathcal{M}_{\theta^N} = -\sum_{k\neq 0}(\theta_k^N)^2\left(\frac{k^\perp\cdot l}{|k|}\right)^2\frac{k\otimes k}{|k|^2} + o(1) \to -\frac{1}{2\pi}\int_{|\xi|=1}\left(\frac{\xi^\perp\cdot l}{|\xi|}\right)^2\frac{\xi\otimes\xi}{|\xi|^2}\mathrm{d}S.$$

Let Q be the two-dimensional orthogonal matrix such that $Q^T\frac{l}{|l|} = e_1$. By applying the coordinate transform $\xi \to Q\xi$, it follows

$$\begin{aligned}\int_{|\xi|=1}\left(\frac{\xi^\perp\cdot l}{|\xi|}\right)^2\frac{\xi\otimes\xi}{|\xi|^2}\mathrm{d}S &= |l|^2 Q\left(\int_{|\xi|=1}\left(\frac{\xi^\perp\cdot e_1}{|\xi|}\right)^2\frac{\xi\otimes\xi}{|\xi|^2}\mathrm{d}S\right)Q^T \\ &= \pi|l|^2 Q\,\mathrm{diag}\left\{\frac{1}{4},\frac{3}{4}\right\}Q^T,\end{aligned}$$

which implies that

$$\begin{aligned}\lim_{N\to\infty}\mathfrak{S}_{\theta^N,h} &= -4\pi^2|l|^2\kappa\left(Q\,\mathrm{diag}\left\{\frac{1}{4},\frac{3}{4}\right\}Q^T\right)e^{2\pi i l\cdot x}b_h(y) \\ &= -4\pi^2|l|^2\kappa\left(\frac{3}{4}I_2 - \frac{1}{2}\frac{l\otimes l}{|l|^2}\right)e^{2\pi i l\cdot x}b_h(y) = \frac{3\kappa}{4}\Delta_x\varphi_h - \frac{\kappa}{2}\nabla_x\,\mathrm{div}_h\,\varphi_h,\end{aligned}$$

and

$$\lim_{N\to\infty} S^{\perp}_{\theta^N,\mathrm{bulk}}\varphi = \kappa\Pi\left(\frac{3}{4}\Delta_x\varphi_h - \frac{1}{2}\nabla_x\,\mathrm{div}_h\,\varphi_h, 0\right)$$

in L^2. Combining this with (3.51), (3.59), one obtains (3.49). □

3.3 *Scaling limit of the cut-off solutions*

Let w_{cut}^N be the solution of the cut-off equation (3.1), corresponding to the noise coefficients $\theta_\cdot^N$ given in (3.2). The goal of this subsection is to justify the limit of w_{cut}^N, as $N \to \infty$. We shall mainly focus on the non-degenerate case. Since the degenerate case is obtained by the same tightness argument, combined with its own corrector limit, we only state the result of that case at the end.

Proposition 3.4. *Let $\alpha, \nu, \kappa, R > 0$, and*

$$w_{\mathrm{in}} \in L^\infty(\Omega; H_{\mathrm{curl}} \cap H^1)$$

be $\mathscr{F}_0$-measurable. Suppose that W is the non-degenerate random field defined in (1.7). Then, there exists δ_0 depending on α, ν, κ such that for any $T > 0$, $\epsilon \in (0,1)$, and $\delta < \delta_0$,

$$\lim_{N\to\infty} \mathbb{P}\left\{ \sup_{t\le T} \|w_{\mathrm{cut}}^N(t) - w_{\mathrm{cut}}^{\det}(t)\|_{H^\delta} > \epsilon \right\} = 0, \tag{3.63}$$

where $w_{\mathrm{cut}}^{\det}$ satisfies the following NS-type equation with cut-off and boundary feedback

$$\begin{cases} \partial_t w + \eta_R(\|w\|)[(u\cdot\nabla)w - (w\cdot\nabla)u] = \left(\nu + \frac{4\kappa}{5}\right)\Delta w + \frac{14\alpha\kappa}{15}\nabla\mathcal{H}[n_3 w_3], \\ u = \mathrm{curl}^{-1}\, w, \\ (\partial_y w_3 - \alpha w_3)|_{y=0} = (w_h - \alpha u_h^\perp)|_{y=0} = 0, \\ (\partial_y w_3 + \alpha w_3)|_{y=1} = (w_h + \alpha u_h^\perp)|_{y=1} = 0, \\ w|_{t=0} = w_{\mathrm{in}}. \end{cases} \tag{3.64}$$

Here, $n = (n_1, n_2, n_3)$ denotes the unit outward normal vector of the boundary, and $\mathcal{H}$ is the harmonic extension operator, that is, for any boundary data h, $\mathcal{H}[h]$ satisfies

$$\begin{cases} \Delta\mathcal{H}[h] = 0, \\ \mathcal{H}[h]|_{y=0,1} = h|_{y=0,1}. \end{cases} \tag{3.65}$$

Proof. **Step I.** First, we assume that w_{in} is deterministic. Let φ be a test function satisfying (1.12). Notice that $\varphi_h, \partial_y\varphi_3 \in C_c^\infty(D)$ and $u_3|_{y=0,1} = 0$, one has

$$\langle \nabla w, \nabla\varphi\rangle = -\langle w, \Delta\varphi\rangle,$$

and

$$\begin{aligned} \langle (u\cdot\nabla)w - (w\cdot\nabla)u, \varphi\rangle &= -\langle w, (u\cdot\nabla)\varphi\rangle + \langle u, (w\cdot\nabla)\varphi\rangle \\ &= \langle u\otimes w - w\otimes u, \nabla\varphi\rangle. \end{aligned}$$

Therefore, from (1.13), the weak formulation of (3.1) is given by

$$\begin{aligned} &\langle w_{\mathrm{cut}}^N(t), \varphi\rangle - \langle w_{\mathrm{in}}, \varphi\rangle + \int_0^t \left(-\nu\langle w_{\mathrm{cut}}^N, \Delta\varphi\rangle + \nu\alpha\langle w_{\mathrm{cut},3}^N, \varphi_3\rangle_{L^2(\partial D)}\right)\mathrm{d}s \\ &\quad + \int_0^t \eta_R(\|w_{\mathrm{cut}}^N\|)\langle u_{\mathrm{cut}}^N \otimes w_{\mathrm{cut}}^N - w_{\mathrm{cut}}^N \otimes u_{\mathrm{cut}}^N, \nabla\varphi\rangle \mathrm{d}s \\ &= \int_0^t \langle w_{\mathrm{cut}}^N, S_{\theta^N}\varphi\rangle \mathrm{d}s + \sqrt{2\kappa}\sum_{k\neq 0}\sum_{j=1,2}\int_0^t \theta_k^N \langle \Pi(\sigma_{k,j}\cdot\nabla)w_{\mathrm{cut}}^N, \varphi\rangle \mathrm{d}B_t^{k,j} \end{aligned} \tag{3.66}$$

$\mathbb{P}$-almost surely for any $t \in [0,T]$. Applying Proposition 3.1 with sufficiently small $\delta > 0$ combined with the Prokhorov theorem and the Skorokhod theorem, passing to a subsequence and changing the underlying probability space and noise if necessary, it follows

$$w_{\mathrm{cut}}^N \to w_{\mathrm{cut}}^{\det} \qquad \mathbb{P}\text{-almost surely in } \chi_{\delta,T} \tag{3.67}$$

for some process $w_{\mathrm{cut}}^{\mathrm{det}}$. This implies

$$\langle w_{\mathrm{cut}}^N(t),\varphi\rangle+\int_0^t\left(-\nu\langle w_{\mathrm{cut}}^N,\Delta\varphi\rangle+\nu\alpha\langle w_{\mathrm{cut},3}^N,\varphi_3\rangle_{L^2(\partial D)}\right)\mathrm{d}s$$
$$\to\langle w_{\mathrm{cut}}^{\mathrm{det}}(t),\varphi\rangle+\int_0^t\left(-\nu\langle w_{\mathrm{cut}}^{\mathrm{det}},\Delta\varphi\rangle+\nu\alpha\langle w_{\mathrm{cut},3}^{\mathrm{det}},\varphi_3\rangle_{L^2(\partial D)}\right)\mathrm{d}s \tag{3.68}$$

and

$$\int_0^t\eta_R(\|w_{\mathrm{cut}}^N\|)\langle u_{\mathrm{cut}}^N\otimes w_{\mathrm{cut}}^N-w_{\mathrm{cut}}^N\otimes u_{\mathrm{cut}}^N,\nabla\varphi\rangle\mathrm{d}s$$
$$\to\int_0^t\eta_R(\|w_{\mathrm{cut}}^{\mathrm{det}}\|)\langle u_{\mathrm{cut}}^{\mathrm{det}}\otimes w_{\mathrm{cut}}^{\mathrm{det}}-w_{\mathrm{cut}}^{\mathrm{det}}\otimes u_{\mathrm{cut}}^{\mathrm{det}},\nabla\varphi\rangle\mathrm{d}s \tag{3.69}$$

$\mathbb{P}$-almost surely in $L^\infty(0,T)$.

Step II. It remains to deal with the Itô–Stratonovich corrector and the martingale term. Indeed, by using (3.67) with sufficiently small δ and applying Proposition 3.2, one has

$$\int_0^t\langle w_{\mathrm{cut}}^N,S_{\theta^N}\varphi\rangle\mathrm{d}s\to\frac{4\kappa}{5}\int_0^t\langle w_{\mathrm{cut}}^{\mathrm{det}},\Delta\varphi\rangle\mathrm{d}s-\frac{2\kappa}{15}\int_0^t\int_{\mathbb{T}^2}\nabla_x\varphi_3\cdot w_{\mathrm{cut},h}^{\mathrm{det}}|_0^1\mathrm{d}x\mathrm{d}s \tag{3.70}$$

$\mathbb{P}$-almost surely in $L^\infty(0,T)$, where we used the notation $g|_0^1:=g|_{y=1}-g|_{y=0}$. Since the tangential Navier boundary conditions in (3.1) are preserved in the above convergence, by integrating by parts, it follows

$$-\frac{2\kappa}{15}\int_{\mathbb{T}^2}\nabla_x\varphi_3\cdot w_{\mathrm{cut},h}^{\mathrm{det}}|_0^1\mathrm{d}x=\frac{2\alpha\kappa}{15}\langle w_{\mathrm{cut},3}^{\mathrm{det}},\varphi_3\rangle_{L^2(\partial D)}. \tag{3.71}$$

We turn to the martingale term. By the Burkholder–Davis–Gundy inequality, integrating by parts, the Bessel inequality[1], (3.23), and (3.6) with $s=0,p=2$, it follows that

$$\begin{aligned}\sqrt{2\kappa}\mathbb{E}\sup_{t\le T}\left|\textstyle\sum_{k\neq0}\sum_{j=1,2}\int_0^t\theta_k^N\langle\Pi(\sigma_{k,j}\cdot\nabla)w_{\mathrm{cut}}^N,\varphi\rangle\mathrm{d}B_t^{k,j}\right|&\\
\lesssim_\kappa\mathbb{E}\left(\textstyle\sum_{k\neq0}\sum_{j=1,2}\int_0^T(\theta_k^N)^2|\langle\nabla\varphi w_{\mathrm{cut}}^N,\sigma_{k,j}\rangle|^2\mathrm{d}t\right)^{\frac12}&\\
\lesssim_\kappa N^{-\frac32}\mathbb{E}\left(\int_0^T\|\nabla\varphi w_{\mathrm{cut}}^N\|^2\mathrm{d}t\right)^{\frac12}&\\
\lesssim_{\alpha,\nu,\kappa,T,R}N^{-\frac32}\|\nabla\varphi\|_\infty(1+\|w_{\mathrm{in}}\|_{H^1})\to0,&\end{aligned} \tag{3.72}$$

as $N\to\infty$.

Now, by combining (3.68)–(3.72), it follows that any convergent subsequence of $\mathcal{D}(w_{\mathrm{cut}}^N)$ converges weakly to $\delta_{w_{\mathrm{cut}}^{\mathrm{det}}}$, where $w_{\mathrm{cut}}^{\mathrm{det}}$ satisfies the weak formulation

$$\langle w_{\mathrm{cut}}^{\mathrm{det}}(t),\varphi\rangle-\langle w_{\mathrm{in}},\varphi\rangle+\int_0^t\left(\left(\nu+\frac{4\kappa}{5}\right)\langle\nabla w_{\mathrm{cut}}^{\mathrm{det}},\nabla\varphi\rangle+\nu\alpha\langle w_{\mathrm{cut},3}^{\mathrm{det}},\varphi_3\rangle_{L^2(\partial D)}\right)\mathrm{d}s$$
$$-\int_0^t\frac{2\alpha\kappa}{15}\langle w_{\mathrm{cut},3}^{\mathrm{det}},\varphi_3\rangle_{L^2(\partial D)}\mathrm{d}s+\int_0^t\eta_R(\|w_{\mathrm{cut}}^{\mathrm{det}}\|)\langle u_{\mathrm{cut}}^{\mathrm{det}}\otimes w_{\mathrm{cut}}^{\mathrm{det}}-w_{\mathrm{cut}}^{\mathrm{det}}\otimes u_{\mathrm{cut}}^{\mathrm{det}},\nabla\varphi\rangle\mathrm{d}s=0$$

[1]Splitting the summation according to the sign of the vertical frequency m, the Bessel inequality can be applied to the corresponding subfamilies.

for any $t \in [0,T]$ and test function φ satisfying (1.12). Notice that

$$\begin{aligned}
&\left(\nu + \frac{4\kappa}{5}\right)\langle \nabla w_{\mathrm{cut}}^{\mathrm{det}}, \nabla\varphi\rangle + \left(\nu\alpha - \frac{2\alpha\kappa}{15}\right)\langle w_{\mathrm{cut},3}^{\mathrm{det}}, \varphi_3\rangle_{L^2(\partial D)} \\
&\qquad = \left(\nu + \frac{4\kappa}{5}\right)\Big(\langle \nabla w_{\mathrm{cut}}^{\mathrm{det}}, \nabla\varphi\rangle + \alpha\langle w_{\mathrm{cut},3}^{\mathrm{det}}, \varphi_3\rangle_{L^2(\partial D)}\Big) - \frac{14\alpha\kappa}{15}\langle w_{\mathrm{cut},3}^{\mathrm{det}}, \varphi_3\rangle_{L^2(\partial D)} \\
&\qquad = \left(\nu + \frac{4\kappa}{5}\right)\Big(\langle \nabla w_{\mathrm{cut}}^{\mathrm{det}}, \nabla\varphi\rangle + \alpha\langle w_{\mathrm{cut},3}^{\mathrm{det}}, \varphi_3\rangle_{L^2(\partial D)}\Big) - \frac{14\alpha\kappa}{15}\langle \nabla\mathcal{H}[n_3 w_{\mathrm{cut},3}^{\mathrm{det}}], \varphi\rangle
\end{aligned}$$

where $\mathcal{H}$ is defined in (3.65) and $n = (n_1, n_2, n_3)$ denotes the unit outward normal vector of the boundary. Hence, we conclude that $w_{\mathrm{cut}}^{\mathrm{det}}$ is a weak solution of (3.64).

Step III. We turn to the uniqueness of the limiting equation (3.64). Since the argument is standard, we shall give a brief sketch. Indeed, by using the decomposition (A.1), the limiting weak formulation can be extended, by density and orthogonality, from the test functions in Definition 1.1 to the mixed space $H_0^1(D)^2 \times H^1(D)$. Thus, the boundary-corrected energy method used in Proposition 2.1 can be applied to the difference of two solutions. This, together with the harmonic lifting estimate

$$\|\nabla\mathcal{H}[n_3 w_3]\| \lesssim \|w_3\|_{H^1}, \tag{3.73}$$

and the cut-off in the nonlinear term, implies the uniqueness of weak solutions of (3.64).

Consequently, all subsequential limits of $\mathcal{D}(w_{\mathrm{cut}}^N)$ coincide with the Dirac measure $\delta_{w_{\mathrm{cut}}^{\mathrm{det}}}$, which ensures the convergence of the whole family and thereby the convergence in probability of the cut-off solutions w_{cut}^N. The case of general initial data follows from the standard Gyöngy–Krylov argument combined with the uniqueness of the limiting equation (3.64), we thus omit the details. This completes the proof. □

The degenerate case is obtained by the same tightness argument, combined with the scaling limit of its own corrector. The corresponding result is presented below.

Proposition 3.5. *Let $\alpha, \nu, \kappa, R > 0$, and*

$$w_{\mathrm{in}} \in L^\infty(\Omega; H_{\mathrm{curl}} \cap H^1)$$

be $\mathscr{F}_0$-measurable. Suppose that W is the degenerate random field defined in (1.10). Then, there exists δ_0 depending on α, ν, κ such that for any $T > 0$, $\epsilon \in (0,1)$, and $\delta < \delta_0$,

$$\lim_{N\to\infty} \mathbb{P}\left\{\sup_{t\le T} \|w_{\mathrm{cut}}^N(t) - w_{\mathrm{cut}}^{\mathrm{det}}(t)\|_{H^\delta} > \epsilon\right\} = 0, \tag{3.74}$$

where $w_{\mathrm{cut}}^{\mathrm{det}}$ satisfies the following NS-type equation with cut-off and anisotropic nonlocal dissipation

$$\begin{cases}
\partial_t w + \eta_R(\|w\|)[(u\cdot\nabla)w - (w\cdot\nabla)u] = \nu\Delta w + \kappa\mathfrak{A}_{\mathrm{eff}} w, \\
u = \mathrm{curl}^{-1} w, \\
(\partial_y w_3 - \alpha w_3)|_{y=0} = (w_h - \alpha u_h^\perp)|_{y=0} = 0, \\
(\partial_y w_3 + \alpha w_3)|_{y=1} = (w_h + \alpha u_h^\perp)|_{y=1} = 0, \\
w|_{t=0} = w_{\mathrm{in}}.
\end{cases} \tag{3.75}$$

Here, $\mathfrak{A}_{\mathrm{eff}}$ is defined in (3.50).

4 Global well-posedness of the limiting equations without cut-off

In this section, we establish the global well-posedness of the limiting equations without cut-off for small initial data.

4.1 The non-degenerate case

First, we consider the non-degenerate case. The corresponding limiting equation without cut-off is given by

$$\begin{cases} \partial_t w + (u\cdot\nabla)w - (w\cdot\nabla)u = \left(\nu + \frac{4\kappa}{5}\right)\Delta w + \frac{14\alpha\kappa}{15}\nabla\mathcal{H}[n_3 w_3], \\ u = \operatorname{curl}^{-1} w, \\ (\partial_y w_3 - \alpha w_3)|_{y=0} = (w_h - \alpha u_h^{\perp})|_{y=0} = 0, \\ (\partial_y w_3 + \alpha w_3)|_{y=1} = (w_h + \alpha u_h^{\perp})|_{y=1} = 0, \\ w|_{t=0} = w_{\mathrm{in}}, \end{cases} \tag{4.1}$$

where $\mathcal{H}$ is the harmonic extension operator defined in (3.65) and $n = (n_1, n_2, n_3)$ denotes the unit outward normal vector of the boundary. Throughout this section, the parameters $\alpha, \nu > 0$ are fixed. Denote

$$\nu_\kappa := \nu + \frac{4\kappa}{5}, \qquad \beta_\kappa := \frac{14\alpha\kappa}{15}.$$

Although the limiting equation exhibits an enhanced viscosity, the boundary feedback term also modifies the boundary vorticity generation mechanism. This modification creates an additional difficulty, since the enhanced bulk dissipation does not immediately imply stability of the full linearized operator. To overcome this issue, we establish a resolvent estimate for the linear operator and use it to derive an exponential decay estimate for the corresponding semigroup. This linear decay estimate is then combined with a standard bootstrap argument to obtain global well-posedness for small initial data.

4.1.1 Linear decay estimate

Define the effective dissipation operator

$$\mathfrak{L}_{\mathrm{eff}} w := \nu_\kappa \Delta w + \beta_\kappa \nabla\mathcal{H}[n_3 w_3]$$

on the domain

$$\mathscr{D}(\mathfrak{L}_{\mathrm{eff}}) := \{w \in H^2 \cap H_{\mathrm{curl}} |\ w \text{ satisfies the boundary conditions } (4.1)_3 \text{ and } (4.1)_4\}.$$

The main result is stated below.

Proposition 4.1. *Let $\alpha, \nu, \kappa > 0$. Then, there are constants $C_\alpha, c_\alpha > 0$ depending only on α such that*

$$\sup_{\operatorname{Re}\lambda \geq -c_\alpha \nu_\kappa} \|(\lambda - \mathfrak{L}_{\mathrm{eff}})^{-1}\|_{\mathcal{L}(H_{\mathrm{curl}})} \leq \frac{C_\alpha}{\nu_\kappa} \tag{4.2}$$

Proof. By rescaling, it suffices to prove the following uniform-in-κ resolvent estimate

$$\sup_{\operatorname{Re}\lambda \geq -c_\alpha} \|(\lambda - \nu_\kappa^{-1}\mathfrak{L}_{\mathrm{eff}})^{-1}\|_{\mathcal{L}(H_{\mathrm{curl}})} \leq C_\alpha. \tag{4.3}$$

For this purpose, we consider the normalized effective dissipation operator

$$\nu_\kappa^{-1}\mathfrak{L}_{\mathrm{eff}} w = \Delta w + \mathfrak{c}_\kappa \nabla\mathcal{H}[n_3 w_3], \qquad \mathfrak{c}_\kappa = \frac{\beta_\kappa}{\nu_\kappa} \in \left(0, \frac{7\alpha}{6}\right),$$

and the resolvent equation

$$\lambda w - \Delta w - \mathfrak{c}\nabla\mathcal{H}[n_3 w_3] = f. \tag{4.4}$$

Here and in what follows, we omit the dependence of $\mathfrak{c}_\kappa$ on κ, and establish estimates for the above resolvent equation uniformly in $\mathfrak{c} \in [0, \frac{7\alpha}{6}]$. The proof is divided into the following steps.

Step I. We start with the vertical component w_3, which satisfies a closed equation

$$\begin{cases} \lambda w_3 - \Delta w_3 - \mathfrak{c}\partial_y \mathcal{H}[n_3 w_3] = f_3, \\ (\partial_y w_3 - \alpha w_3)|_{y=0} = (\partial_y w_3 + \alpha w_3)|_{y=1} = 0. \end{cases} \tag{4.5}$$

Consider the tangential Fourier expansion

$$w_3(t,x,y) = \textstyle\sum_{k\neq 0} \hat{w}_{3,k}(t,y) e^{2\pi i k\cdot x},$$

where we used the horizontal zero-mean condition in (1.2). From (4.5), it follows that

$$\begin{cases} (\lambda + 4\pi^2|k|^2)\hat{w}_{3,k} - \hat{w}_{3,k}'' - \mathfrak{c}\partial_y \mathcal{H}_k = \hat{f}_{3,k}, \\ (\hat{w}_{3,k}' - \alpha \hat{w}_{3,k})|_{y=0} = (\hat{w}_{3,k}' + \alpha \hat{w}_{3,k})|_{y=1} = 0, \end{cases} \tag{4.6}$$

where $\mathcal{H}_k := \mathcal{H}_k[n_3 \hat{w}_{3,k}]$ solves

$$\begin{cases} \mathcal{H}_k'' - 4\pi^2|k|^2 \mathcal{H}_k = 0, \\ \mathcal{H}_k|_{y=0,1} = n_3 \hat{w}_{3,k}|_{y=0,1}. \end{cases} \tag{4.7}$$

After translating the vertical variable $y \mapsto y - \frac{1}{2}$, we regard the above equations as being posed on the interval $(-\frac{1}{2}, \frac{1}{2})$. Notice that if $\hat{f}_{3,k}$ is odd in y, then the corresponding solution $\hat{w}_{3,k}$ remains odd; the same is true for the even part. Therefore, by linearity and by decomposing $\hat{f}_{3,k}$ into its odd and even parts, it suffices to treat the odd and even cases separately.

Step II. We first give the details for the odd case. The even case can be handled by a similar argument, with the necessary modifications indicated afterwards. In the odd case, the equation (4.6) reduces to

$$\begin{cases} (\lambda + 4\pi^2|k|^2)\hat{w}_{3,k} - \hat{w}_{3,k}'' - \mathfrak{c}\mathcal{H}_k' = \hat{f}_{3,k}, \\ \hat{w}_{3,k}|_{y=0} = (\hat{w}_{3,k}' + \alpha \hat{w}_{3,k})|_{y=\frac{1}{2}} = 0. \end{cases} \tag{4.8}$$

Let $\mathfrak{q}_k, \mathfrak{u}_k$ be the solutions of the following auxiliary Dirichlet problems, respectively:

$$(\lambda + 4\pi^2|k|^2)\mathfrak{q}_k - \mathfrak{q}_k'' = \hat{f}_{3,k}, \qquad \mathfrak{q}_k(0) = \mathfrak{q}_k(1/2) = 0, \tag{4.9}$$

and

$$(\lambda + 4\pi^2|k|^2)\mathfrak{u}_k - \mathfrak{u}_k'' - \mathfrak{c}\mathcal{H}_k'[n_3 \mathfrak{u}_k] = 0, \qquad \mathfrak{u}_k(0) = 0, \qquad \mathfrak{u}_k(1/2) = 1. \tag{4.10}$$

Here, the notation $\mathcal{H}_k[n_3\mathfrak{u}_k]$ is understood by first taking odd extension to $\mathfrak{u}_k$ and then applying the full harmonic extension operator $\mathcal{H}_k$ to the boundary data $n_3\mathfrak{u}_k$. We shall choose a constant $\mathfrak{a}_k$ such that the solution $\hat{w}_{3,k}$ of the resolvent equation (4.8) satisfies the decomposition

$$\hat{w}_{3,k} = \mathfrak{q}_k + \mathfrak{a}_k \mathfrak{u}_k. \tag{4.11}$$

Indeed, notice that

$$\mathcal{H}_k[n_3\mathfrak{q}_k] = \mathcal{H}_k[0] = 0,$$

the equation $(4.8)_1$ is satisfied by construction. Therefore, it remains to match the boundary condition $(4.8)_2$ at $y = \frac{1}{2}$. To this end, one needs to solve the equations (4.9) and (4.10) explicitly. As (4.9) can be solved by using the Green representation formula, see (4.27), we turn to (4.10). By the odd symmetry of $\mathfrak{u}_k$, one has

$$\mathcal{H}_k[n_3\mathfrak{u}_k] = \mathcal{H}_k[1] = \frac{\cosh(2\pi|k|y)}{\cosh(\pi|k|)}, \qquad \mathcal{H}_k'[n_3\mathfrak{u}_k] = 2\pi|k|\frac{\sinh(2\pi|k|y)}{\cosh(\pi|k|)},$$

which reduces the nonlocal equation (4.10) to the following local Dirichlet problem

$$(\lambda + 4\pi^2|k|^2)\mathfrak{u}_k - \mathfrak{u}_k'' = 2\pi|k|\mathfrak{c}\frac{\sinh(2\pi|k|y)}{\cosh(\pi|k|)}, \qquad \mathfrak{u}_k(0) = 0, \qquad \mathfrak{u}_k(1/2) = 1. \tag{4.12}$$

For $\lambda \neq 0$, notice that

$$(\lambda + 4\pi^2|k|^2)\frac{\mathfrak{c}}{\lambda}\mathcal{H}_k'[n_3\mathfrak{u}_k] - \frac{\mathfrak{c}}{\lambda}\mathcal{H}_k'''[n_3\mathfrak{u}_k] = 2\pi|k|\mathfrak{c}\frac{\sinh(2\pi|k|y)}{\cosh(\pi|k|)},$$

and the general solutions of the corresponding homogeneous equation are linear combinations of $\sinh \rho_{k,\lambda} y$ and $\cosh \rho_{k,\lambda} y$, where

$$\rho_{k,\lambda} := (\lambda + 4\pi^2|k|^2)^{\frac{1}{2}}$$

is chosen with $\operatorname{Re} \rho_{k,\lambda} > 0$. Therefore, by matching the boundary conditions, the solution of (4.10) is given by

$$\mathfrak{u}_k = C_{\neq}\sinh(\rho_{k,\lambda}y) + \frac{2\pi|k|\mathfrak{c}}{\lambda}\frac{\sinh(2\pi|k|y)}{\cosh(\pi|k|)}, \qquad \lambda \neq 0 \tag{4.13}$$

with $C_{\neq}$ satisfying

$$C_{\neq}\sinh(\rho_{k,\lambda}/2) = 1 - \frac{2\pi|k|\mathfrak{c}}{\lambda}\tanh(\pi|k|).$$

If $\lambda = 0$, then equation (4.12) reduces to

$$4\pi^2|k|^2\mathfrak{u}_k - \mathfrak{u}_k'' = 2\pi|k|\mathfrak{c}\frac{\sinh(2\pi|k|y)}{\cosh(\pi|k|)}.$$

Since

$$4\pi^2|k|^2(y\cosh(2\pi|k|y)) - (y\cosh(2\pi|k|y))'' = -4\pi|k|\sinh(2\pi|k|y),$$

the corresponding solution of (4.10) is given by

$$\mathfrak{u}_k = C_0\sinh(2\pi|k|y) - \frac{\mathfrak{c}}{2}\frac{y\cosh(2\pi|k|y)}{\cosh(\pi|k|)}, \tag{4.14}$$

where C_0 satisfies

$$C_0\sinh(\pi|k|) = 1 + \frac{\mathfrak{c}}{4}.$$

Plugging (4.13) and (4.14) into the boundary condition $(4.8)_2$ at $y = \frac{1}{2}$ yields

$$\begin{aligned} -\mathfrak{q}_k'(1/2) &= \mathfrak{a}_k\left(\rho_{k,\lambda}\coth(\rho_{k,\lambda}/2)\left(1 - \frac{2\pi|k|\mathfrak{c}}{\lambda}\tanh(\pi|k|)\right) + \frac{4\pi^2|k|^2\mathfrak{c}}{\lambda} + \alpha\right) \\ &= \mathfrak{a}_k\left(\mathfrak{m}_k(\lambda) - \mathfrak{n}_k\mathfrak{c}\frac{\mathfrak{m}_k(\lambda) - \mathfrak{m}_k(0)}{\lambda} + \alpha\right) =: \mathfrak{D}(\lambda)\mathfrak{a}_k \end{aligned} \tag{4.15}$$

for $\lambda \neq 0$, where

$$\mathfrak{m}_k(\lambda) = \rho_{k,\lambda}\coth(\rho_{k,\lambda}/2), \qquad \mathfrak{n}_k = 2\pi|k|\tanh(\pi|k|). \tag{4.16}$$

Similarly, for $\lambda = 0$, one has

$$-\mathfrak{q}_k'(1/2) = \mathfrak{a}_k\left(\mathfrak{m}_k(0) - \mathfrak{n}_k\mathfrak{c}\mathfrak{m}_k'(0) + \alpha\right) =: \mathfrak{D}(0)\mathfrak{a}_k. \tag{4.17}$$

Hence, the validity of the decomposition (4.11) reduces to appropriate lower bounds for the function $\mathfrak{D}(\lambda)$.

Step III. To bound $\mathfrak{D}(\lambda)$, we establish some auxiliary estimates for $\mathfrak{m}_k$. Introduce

$$\mathfrak{h}_{k,\lambda}(y) := \frac{\sinh(\rho_{k,\lambda} y)}{\sinh(\rho_{k,\lambda}/2)},$$

which solves the following Dirichlet problem

$$(\lambda + 4\pi^2|k|^2)\mathfrak{h}_{k,\lambda} - \mathfrak{h}''_{k,\lambda} = 0, \qquad \mathfrak{h}_{k,\lambda}(0) = 0, \qquad \mathfrak{h}_{k,\lambda}(1/2) = 1, \tag{4.18}$$

and satisfies the relation

$$\mathfrak{m}_k(\lambda) = \mathfrak{h}'_{k,\lambda}(1/2).$$

Applying the standard energy estimate to (4.18) and using the above relation, one has

$$\operatorname{Re}\mathfrak{m}_k(\lambda) = \int_0^{1/2} |\mathfrak{h}'_{k,\lambda}|^2 \mathrm{d}y + \left(\operatorname{Re}\lambda + 4\pi^2|k|^2\right) \int_0^{1/2} |\mathfrak{h}_{k,\lambda}|^2 \mathrm{d}y \geq 0, \tag{4.19}$$

provided that

$$\operatorname{Re}\lambda \geq -2\pi^2. \tag{4.20}$$

Next, we turn to $\mathfrak{m}'_k$. Using (4.18), one has

$$(\mathfrak{h}'_{k,\lambda}\mathfrak{h}_{k,0} - \mathfrak{h}_{k,\lambda}\mathfrak{h}'_{k,0})' = \lambda \mathfrak{h}_{k,\lambda}\mathfrak{h}_{k,0},$$

which gives

$$\frac{\mathfrak{m}_k(\lambda) - \mathfrak{m}_k(0)}{\lambda} = \int_0^{1/2} \mathfrak{h}_{k,\lambda}\mathfrak{h}_{k,0} \mathrm{d}y. \tag{4.21}$$

On the other hand, define

$$\Delta_k := \partial_y^2 - 4\pi^2|k|^2, \qquad \mathcal{D}(\Delta_k) := H^2(0,1/2) \cap H_0^1(0,1/2). \tag{4.22}$$

From (4.18), it follows

$$(\lambda - \Delta_k)(\mathfrak{h}_{k,\lambda} - \mathfrak{h}_{k,0}) = -\lambda \mathfrak{h}_{k,0},$$

which yields

$$\mathfrak{h}_{k,\lambda} = \mathfrak{h}_{k,0} - (\lambda - \Delta_k)^{-1} \lambda \mathfrak{h}_{k,0} \tag{4.23}$$

for any λ satisfying (4.20). Let $\{e_j\}_{j\geq 1}$ be the normalized eigenfunctions of Δ_k, namely,

$$\Delta_k e_j = -4\pi^2(|k|^2 + j^2)e_j, \qquad j \geq 1.$$

Combining this with (4.23), one obtains

$$\mathfrak{h}_{k,\lambda} = \textstyle\sum_{j\geq 1} \dfrac{4\pi^2(|k|^2+j^2)}{\lambda + 4\pi^2(|k|^2+j^2)} \langle \mathfrak{h}_{k,0}, e_j\rangle e_j.$$

Since $\mathfrak{h}_{k,0}$ is real-valued, by plugging the above identity into (4.21) and using the Parseval identity, it follows

$$\frac{\mathfrak{m}_k(\lambda) - \mathfrak{m}_k(0)}{\lambda} = \textstyle\sum_{j\geq 1} \dfrac{4\pi^2(|k|^2+j^2)}{\lambda + 4\pi^2(|k|^2+j^2)} |\langle \mathfrak{h}_{k,0}, e_j\rangle|^2,$$

which, together with (4.16) and (4.20), implies

$$\begin{aligned} \mathfrak{n}_k \left| \frac{\mathfrak{m}_k(\lambda) - \mathfrak{m}_k(0)}{\lambda} \right| &\leq \mathfrak{n}_k \textstyle\sum_{j\geq 1} \dfrac{4\pi^2(|k|^2+j^2)}{\operatorname{Re}\lambda + 4\pi^2(|k|^2+j^2)} |\langle \mathfrak{h}_{k,0}, e_j\rangle|^2 \\ &\leq \frac{4\mathfrak{n}_k}{3} \|\mathfrak{h}_{k,0}\|_{L^2_y}^2 = \frac{2}{3}\left(1 - \frac{2\pi|k|}{\sinh(2\pi|k|)}\right) \leq \frac{2}{3}. \end{aligned} \tag{4.24}$$

Combining this with the definition of $\mathfrak{D}(\lambda)$ given in (4.15), the fact $\mathfrak{c} \in [0, \frac{7\alpha}{6}]$, and (4.19), one gets

$$|\mathfrak{D}(\lambda)| \geq |\mathfrak{m}_k(\lambda)+\alpha| \left(1 - \frac{7\alpha}{9|\mathfrak{m}_k(\lambda)+\alpha|}\right) \geq \frac{2|\mathfrak{m}_k(\lambda)+\alpha|}{9} \geq \frac{\sqrt{2}(\alpha+|\mathfrak{m}_k(\lambda)|)}{9}.$$

Moreover, from (4.20), one has

$$\operatorname{Re}\rho_{k,\lambda} = \sqrt{\frac{|\rho_{k,\lambda}^2| + \operatorname{Re}\rho_{k,\lambda}^2}{2}} \geq \sqrt{2}\pi > 0, \tag{4.25}$$

which ensures

$$|\mathfrak{m}_k(\lambda)| = |\rho_{k,\lambda}| \left|\frac{1+e^{-\rho_{k,\lambda}}}{1-e^{-\rho_{k,\lambda}}}\right| \geq |\rho_{k,\lambda}| \frac{1-e^{-\operatorname{Re}\rho_{k,\lambda}}}{1+e^{-\operatorname{Re}\rho_{k,\lambda}}} \geq \frac{1-e^{-\sqrt{2}\pi}}{2}|\rho_{k,\lambda}|.$$

Consequently, for any $\lambda \neq 0$ satisfying (4.20), it follows

$$|\mathfrak{D}(\lambda)| \geq C_\alpha(1+|\rho_{k,\lambda}|). \tag{4.26}$$

By taking $\lambda \to 0$ in the above estimate, the restriction $\lambda \neq 0$ can be removed, and thus (4.26) holds for any λ satisfying (4.20).

Step IV. We are in a position to close the estimate for equation (4.8). Indeed, by applying the decomposition (4.11) and matching the boundary condition $(4.8)_2$ at $y = \frac{1}{2}$, one has

$$\mathfrak{a}_k = -\mathfrak{D}^{-1}(\lambda)\mathfrak{q}_k'(1/2),$$

where, by using the Green kernel of the problem (4.9), it follows

$$\mathfrak{q}_k(y) = \int_0^{1/2} \frac{\sinh(\rho_{k,\lambda}\min\{y,y'\})\sinh(\rho_{k,\lambda}(1/2-\max\{y,y'\}))}{\rho_{k,\lambda}\sinh(\rho_{k,\lambda}/2)} \hat{f}_{3,k}(y')\mathrm{d}y'. \tag{4.27}$$

This, together with (4.25), (4.26), and the Cauchy–Schwarz inequality, implies

$$\begin{aligned}
|\mathfrak{a}_k| &\lesssim_\alpha (1+|\rho_{k,\lambda}|)^{-1}|\mathfrak{q}_k'(1/2)| \\
&\lesssim_\alpha (1+|\rho_{k,\lambda}|)^{-1} \int_0^{1/2} \left|\frac{\sinh(\rho_{k,\lambda}y')}{\sinh(\rho_{k,\lambda}/2)}\hat{f}_{3,k}(y')\right| \mathrm{d}y' \\
&\lesssim_\alpha (1+|\rho_{k,\lambda}|)^{-1} \left\|\frac{\sinh(\rho_{k,\lambda}\cdot)}{\sinh(\rho_{k,\lambda}/2)}\right\|_{L_y^2} \|\hat{f}_{3,k}\|_{L_y^2} \lesssim_\alpha (1+|\rho_{k,\lambda}|)^{-\frac{3}{2}} \|\hat{f}_{3,k}\|_{L_y^2}.
\end{aligned} \tag{4.28}$$

We turn to the estimate of $\mathfrak{u}_k$. Suppose that $\lambda \neq 0$ satisfies (4.20). Then,

$$\begin{aligned}
\mathfrak{u}_k &= \frac{\sinh\rho_{k,\lambda}y}{\sinh(\rho_{k,\lambda}/2)} + \frac{\mathfrak{cn}_k}{\lambda}\left(\frac{\sinh 2\pi|k|y}{\sinh\pi|k|} - \frac{\sinh\rho_{k,\lambda}y}{\sinh(\rho_{k,\lambda}/2)}\right) \\
&= \mathfrak{f}_{\rho_{k,\lambda}}(y) - \frac{\mathfrak{cn}_k}{\rho_{k,\lambda}+2\pi|k|}\frac{\mathfrak{f}_{\rho_{k,\lambda}}-\mathfrak{f}_{2\pi|k|}}{\rho_{k,\lambda}-2\pi|k|},
\end{aligned} \tag{4.29}$$

where we used the notation

$$\mathfrak{f}_\mathfrak{l}(y) := \frac{\sinh\mathfrak{l}y}{\sinh(\mathfrak{l}/2)}.$$

As $\mathfrak{n}_k = 2\pi|k|\tanh(\pi|k|) \sim |k|$, it remains to bound the function $\mathfrak{f}_\mathfrak{l}(y)$. Due to the fact

$$\operatorname{Re}\rho_{k,\lambda}^2 = \operatorname{Re}\lambda + 4\pi^2|k|^2 \geq 2\pi^2 > 0 \qquad \Rightarrow \qquad |\arg\rho_{k,\lambda}| < \frac{\pi}{4}, \tag{4.30}$$

and (4.25), one may assume $\mathfrak{l} \in \{|\arg \mathfrak{l}| < \frac{\pi}{4}, \operatorname{Re} \mathfrak{l} \geq \sqrt{2}\pi\}$. In this case, one has

$$\operatorname{Re} \mathfrak{l} \geq \frac{\sqrt{2}}{2}|\mathfrak{l}|, \tag{4.31}$$

which implies

$$|\mathfrak{f}_{\mathfrak{l}}(y)| = \left| \frac{e^{-(1/2-y)\mathfrak{l}} - e^{-(1/2+y)\mathfrak{l}}}{1-e^{-\mathfrak{l}}} \right| \leq \frac{e^{-\frac{\sqrt{2}}{2}(1/2-y)|\mathfrak{l}|} + e^{-\frac{\sqrt{2}}{2}(1/2+y)|\mathfrak{l}|}}{1-e^{-\sqrt{2}\pi}} \lesssim e^{-\frac{\sqrt{2}}{2}(1/2-y)|\mathfrak{l}|}. \tag{4.32}$$

Similarly, for $j = 1, 2$,

$$|\partial_y^j \mathfrak{f}_{\mathfrak{l}}(y)| \lesssim |\mathfrak{l}|^j e^{-\frac{\sqrt{2}}{2}(1/2-y)|\mathfrak{l}|}. \tag{4.33}$$

To estimate the difference quotient in (4.29), one needs to address

$$\partial_{\mathfrak{l}} \mathfrak{f}_{\mathfrak{l}} = \frac{-(1/2-y)e^{-(1/2-y)\mathfrak{l}} + (1/2+y)e^{-(1/2+y)\mathfrak{l}}}{1-e^{-\mathfrak{l}}} - e^{-\mathfrak{l}} \frac{e^{-(1/2-y)\mathfrak{l}} - e^{-(1/2+y)\mathfrak{l}}}{(1-e^{-\mathfrak{l}})^2},$$

which is bounded in the following way:

$$\begin{aligned} |\partial_{\mathfrak{l}} \mathfrak{f}_{\mathfrak{l}}| &\lesssim (1/2-y)e^{-\frac{\sqrt{2}}{2}(1/2-y)|\mathfrak{l}|} + (1/2+y)e^{-\frac{\sqrt{2}}{2}(1/2+y)|\mathfrak{l}|} + e^{-\frac{\sqrt{2}}{2}|\mathfrak{l}|} e^{-\frac{\sqrt{2}}{2}(1/2-y)|\mathfrak{l}|} \\ &\lesssim |\mathfrak{l}|^{-1} e^{-\frac{\sqrt{2}}{4}(1/2-y)|\mathfrak{l}|} + |\mathfrak{l}|^{-1} e^{-\frac{\sqrt{2}}{4}(1/2+y)|\mathfrak{l}|} + |\mathfrak{l}|^{-1} e^{-\frac{\sqrt{2}}{2}(1/2-y)|\mathfrak{l}|} \lesssim |\mathfrak{l}|^{-1} e^{-\frac{\sqrt{2}}{4}(1/2-y)|\mathfrak{l}|}. \end{aligned}$$

Thus, for $j = 0, 1, 2$,

$$|\partial_y^j \partial_{\mathfrak{l}} \mathfrak{f}_{\mathfrak{l}}| \lesssim |\mathfrak{l}|^{j-1} e^{-\frac{\sqrt{2}}{4}(1/2-y)|\mathfrak{l}|}. \tag{4.34}$$

Combining (4.29) with (4.32)–(4.34) and the mean value theorem, one has

$$\begin{aligned} \|\partial_y^j \mathfrak{u}_k\| &\lesssim \|\partial_y^j \mathfrak{f}_{\rho_{k,\lambda}}\| + \mathfrak{c} \left\| \frac{\partial_y^j \mathfrak{f}_{\rho_{k,\lambda}} - \partial_y^j \mathfrak{f}_{2\pi|k|}}{\rho_{k,\lambda} - 2\pi|k|} \right\| \\ &\lesssim \|\partial_y^j \mathfrak{f}_{\rho_{k,\lambda}}\| + \alpha \int_0^1 \|\partial_y^j \partial_{\mathfrak{l}} \mathfrak{f}_{\mathfrak{l}}|_{\mathfrak{l}=2\pi|k|+(\rho_{k,\lambda}-2\pi|k|)s}\| \mathrm{d}s \\ &\lesssim_\alpha |\rho_{k,\lambda}|^{j-\frac{1}{2}} + \int_0^1 |2\pi|k| + (\rho_{k,\lambda} - 2\pi|k|)s|^{j-\frac{3}{2}} \mathrm{d}s, \end{aligned}$$

where, by applying (4.30), it follows

$$|2\pi|k| + (\rho_{k,\lambda} - 2\pi|k|)s| = |2\pi|k|(1-s) + \rho_{k,\lambda} s| \geq \frac{1}{\sqrt{2}} \big(|\rho_{k,\lambda}| s + 2\pi|k|(1-s) \big).$$

Together with (4.25), this implies

$$\|\partial_y^j \mathfrak{u}_k\| \lesssim_\alpha |\rho_{k,\lambda}|^{j-\frac{1}{2}},$$

and thus

$$(|\lambda| + 4\pi^2|k|^2)\|\mathfrak{u}_k\| + 2\pi|k|\|\mathfrak{u}_k'\| + \|\mathfrak{u}_k''\| \lesssim_\alpha |\rho_{k,\lambda}|^{\frac{3}{2}}. \tag{4.35}$$

The estimate for $\mathfrak{u}_k$ with $\lambda = 0$ can be obtained by taking $\lambda \to 0$ in (4.29) and repeating the above argument, so we omit the details. Moreover, the standard resolvent estimate for the operator Δ_k defined in (4.22) gives the following estimate for $\mathfrak{q}_k$

$$(|\lambda| + 4\pi^2|k|^2)\|\mathfrak{q}_k\| + 2\pi|k|\|\mathfrak{q}_k'\| + \|\mathfrak{q}_k''\| \lesssim \|\hat{f}_{3,k}\|_{L^2_y},$$

provided that (4.20) holds. Combining this with (4.11), (4.28), and (4.35), one gets

$$(|\lambda| + 4\pi^2|k|^2)\|\hat{w}_{3,k}\| + 2\pi|k|\|\hat{w}'_{3,k}\| + \|\hat{w}''_{3,k}\| \lesssim_\alpha \|\hat{f}_{3,k}\|_{L^2_y}, \tag{4.36}$$

when $\hat{f}_{3,k}$ is odd.

The even case can be treated in the same way. The auxiliary problems are now defined by

$$(\lambda + 4\pi^2|k|^2)\mathfrak{q}_k - \mathfrak{q}''_k = \hat{f}_{3,k}, \qquad \mathfrak{q}'_k(0) = \mathfrak{q}_k(1/2) = 0,$$

and

$$(\lambda + 4\pi^2|k|^2)\mathfrak{u}_k - \mathfrak{u}''_k - \mathfrak{c}\mathcal{H}'_k[n_3\mathfrak{u}_k] = 0, \qquad \mathfrak{u}'_k(0) = 0, \qquad \mathfrak{u}_k(1/2) = 1.$$

The corresponding solutions are given by

$$\mathfrak{q}_k(y) = \int_0^{1/2} \frac{\cosh(\rho_{k,\lambda}\min\{y,y'\})\sinh(\rho_{k,\lambda}(1/2-\max\{y,y'\}))}{\rho_{k,\lambda}\cosh(\rho_{k,\lambda}/2)}\hat{f}_{3,k}(y')\mathrm{d}y',$$

and

$$\mathfrak{u}_k = \begin{cases} C_{\neq}\cosh(\rho_{k,\lambda}y) + \frac{2\pi|k|\mathfrak{c}}{\lambda}\frac{\cosh(2\pi|k|y)}{\sinh(\pi|k|)}, & \lambda \neq 0, \\ C_0\cosh(2\pi|k|y) - \frac{\mathfrak{c}}{2}\frac{y\sinh(2\pi|k|y)}{\sinh(\pi|k|)}, & \lambda = 0, \end{cases}$$

where the constants $C_{\neq}, C_0$ satisfy

$$C_{\neq}\cosh(\rho_{k,\lambda}/2) = 1 - \frac{2\pi|k|\mathfrak{c}}{\lambda}\coth(\pi|k|), \qquad C_0\cosh(\pi|k|) = 1 + \frac{\mathfrak{c}}{4}.$$

Therefore, the decomposition (4.11) holds with $\mathfrak{a}_k$ given by

$$\mathfrak{a}_k := -\mathfrak{D}^{-1}(\lambda)\mathfrak{q}'_k(1/2),$$

where

$$\mathfrak{D}(\lambda) := \mathfrak{m}_k(\lambda) - \mathfrak{n}_k\mathfrak{c}\frac{\mathfrak{m}_k(\lambda)-\mathfrak{m}_k(0)}{\lambda} + \alpha$$

with

$$\mathfrak{m}_k(\lambda) := \rho_{k,\lambda}\tanh(\rho_{k,\lambda}/2), \qquad \mathfrak{n}_k := 2\pi|k|\coth(\pi|k|).$$

Repeating the derivation of (4.24), one gets

$$\mathfrak{n}_k\left|\frac{\mathfrak{m}_k(\lambda)-\mathfrak{m}_k(0)}{\lambda}\right| \le \frac{5}{3}\mathfrak{m}'_k(0)\mathfrak{n}_k = \frac{5}{6}\left(1 + \frac{2\pi|k|}{\sinh(2\pi|k|)}\right) \le \frac{5}{6}\left(1 + \frac{2\pi}{\sinh(2\pi)}\right) =: c_{\mathrm{even}}.$$

Since $c_{\mathrm{even}} < \frac{6}{7}$, one has the estimate

$$|\mathfrak{D}(\lambda)| \ge |\mathfrak{m}_k(\lambda)+\alpha|\left(1 - \frac{c_{\mathrm{even}}\mathfrak{c}}{|\mathfrak{m}_k(\lambda)+\alpha|}\right) \ge \left(1 - \frac{7c_{\mathrm{even}}}{6}\right)|\mathfrak{m}_k(\lambda)+\alpha|,$$

which implies

$$|\mathfrak{D}(\lambda)| \ge C_\alpha(1 + |\rho_{k,\lambda}|).$$

This, together with the arguments used in the derivation of (4.28) and (4.35), leads to (4.36) in the even case.

Hence, the following estimate

$$|\lambda|\|w_3\| + \|w_3\|_{H^2} \lesssim_\alpha \|f_3\| \tag{4.37}$$

holds for equation (4.5) with λ satisfying (4.20).

Step V. Finally, we turn to the horizontal components w_h, which satisfy

$$\begin{cases} \lambda w_h - \Delta w_h - \mathfrak{c}\nabla_x \mathcal{H}[n_3 w_3] = f_h, & u = \operatorname{curl}^{-1} w, \\ (w_h - \alpha u_h^{\perp})|_{y=0} = (w_h + \alpha u_h^{\perp})|_{y=1} = 0. \end{cases} \tag{4.38}$$

We claim that there exist constants $c_\alpha, C_\alpha > 0$ such that for any λ satisfying $\operatorname{Re}\lambda \geq -c_\alpha$ and $\mathfrak{c} \in [0, \frac{7\alpha}{6}]$, it follows

$$\|w_h\| \leq C_\alpha \|f\|. \tag{4.39}$$

By contradiction, suppose that there are $\lambda_n, \mathfrak{c}_n$ such that

$$\operatorname{Re}\lambda_n \geq -c_\alpha, \qquad \mathfrak{c}_n \in \left[0, \frac{7\alpha}{6}\right],$$

and functions w_h^n, f^n satisfying (4.38) and

$$\|w_h^n\| = 1, \qquad \lim_{n\to\infty} \|f^n\| = 0. \tag{4.40}$$

Applying $\mathfrak{J}$ to both sides of (4.38), where $\mathfrak{J}$ is the boundary correction operator defined in (2.3), one gets

$$\lambda \tilde{w}_h - \Delta \tilde{w}_h + ([\mathcal{B}, \partial_y^2] w)_h = \tilde{f}_h + \mathfrak{c}\big(\mathfrak{J}\nabla\mathcal{H}[n_3 w_3]\big)_h, \qquad \tilde{w}_h|_{y=0,1} = 0, \tag{4.41}$$

where $\tilde{w} := \mathfrak{J} w$ and $\tilde{f} := \mathfrak{J} f$. By using (4.37) and (3.73), it follows

$$\|w_3^n\|_{H^2} \to 0, \qquad \big\|\mathfrak{c}_n\big(\mathfrak{J}\nabla\mathcal{H}[n_3 w_3^n]\big)_h\big\| \to 0, \tag{4.42}$$

as $n \to \infty$. We split the rest of the argument into two cases.

Case I. Suppose that the sequence $\{\lambda_n\}_{n\geq 1}$ is bounded. Applying the standard elliptic estimate to (4.41) and using (4.40), (4.42), and Proposition B.2, one has

$$\begin{aligned} \|\tilde{w}_h^n\|_{H^2} &\leq |\lambda_n| \|\tilde{w}_h^n\| + \|[\mathcal{B}, \partial_y^2] w^n\| + \|\tilde{f}_h^n + \mathfrak{c}_n\big(\mathfrak{J}\nabla\mathcal{H}[n_3 w_3^n]\big)_h\| \\ &\lesssim_\alpha \sup_{n\geq 1} \Big(\|w^n\| + \|\tilde{f}_h^n + \mathfrak{c}_n\big(\mathfrak{J}\nabla\mathcal{H}[n_3 w_3^n]\big)_h\| \Big) < \infty. \end{aligned}$$

Combining this with the compact Sobolev embedding theorem and passing to a subsequence if necessary, it follows

$$\mathfrak{c}_n \to \mathfrak{c}_\infty, \qquad \lambda_n \to \lambda_\infty, \qquad w^n \to w^\infty,$$

as $n \to \infty$, where $\mathfrak{c}_\infty, \lambda_\infty$ satisfy

$$\mathfrak{c}_\infty \in [0, \frac{7\alpha}{6}], \qquad \operatorname{Re}\lambda_\infty \geq -c_\alpha,$$

and the convergence of w^n holds in the strong topology of H^1 and the weak topology of H^2. Consequently, w^∞ solves the limiting equation

$$\lambda_\infty \tilde{w}_h^\infty - \Delta \tilde{w}_h^\infty + ([\mathcal{B}, \partial_y^2] w^\infty)_h = 0, \qquad w_3^\infty = 0, \qquad \tilde{w}_h^\infty|_{y=0,1} = 0.$$

On the other hand, by the definition (2.4) of $\mathcal{B}$, one has $([\mathcal{B}, \partial_y^2] w^\infty)_3 \equiv 0$. Therefore, the above equation is equivalent with

$$\begin{cases} \lambda_\infty w_h^\infty - \Delta w_h^\infty = 0, & w_3^\infty = 0, & u^\infty = \operatorname{curl}^{-1} w^\infty, \\ (w_h^\infty - \alpha (u_h^\infty)^\perp)|_{y=0} = (w_h^\infty + \alpha (u_h^\infty)^\perp)|_{y=1} = 0. \end{cases}$$

Notice that $\operatorname{curl}(\lambda_\infty u^\infty - \Delta u^\infty) = 0$. Then, by applying Lemma A.4, one has

$$\begin{cases} \lambda_\infty u^\infty - \Delta u^\infty + \nabla p + (C_1, C_2, 0) = 0, \\ (\partial_y u_h^\infty - \alpha u_h^\infty)|_{y=0} = (\partial_y u_h^\infty + \alpha u_h^\infty)|_{y=1} = u_3^\infty|_{y=0,1} = 0 \end{cases}$$

for some constants C_1, C_2 and function p. By the standard energy method combined with the fact $\int_D u_h^\infty \mathrm{d}x = 0$, it follows

$$\operatorname{Re}\lambda_\infty \|u^\infty\|^2 + \|\nabla u^\infty\|^2 + \alpha \|u_h^\infty\|^2_{L^2(\partial D)} = 0.$$

Together with choosing

$$c_\alpha \le \frac{C_p}{2}, \tag{4.43}$$

where C_p is the constant in the Poincaré inequality, this leads to $u^\infty \equiv 0$ and $w_h^\infty \equiv 0$. Since $\|w_h^n\| \equiv 1$ and w^n converges to w^∞ in the strong topology of H^1, this is a contradiction.

Case II. Now we assume that $|\lambda_n| \to \infty$, as $n \to \infty$. From (4.41), it follows

$$\tilde{w}_h^n = (\lambda_n - \Delta_D)^{-1} \left(\tilde{f}_h^n + \mathfrak{c}_n \big(\mathfrak{J} \nabla \mathcal{H}[n_3 w_3^n] \big)_h - ([\mathcal{B}, \partial_y^2] w^n)_h \right),$$

where Δ_D denotes the Dirichlet–Laplacian operator. By the spectral theorem for the positive self-adjoint operator $-\Delta_D$, one has

$$\|(\lambda - \Delta_D)^{-1}\|_{\mathcal{L}(L^2)} = \sup_{j \ge 1} |\lambda + \mathfrak{b}_j|^{-1},$$

where $0 < \mathfrak{b}_1 < \mathfrak{b}_2 < \dots$ are the eigenvalues of $-\Delta_D$. Combining the above results with Proposition B.2 and (4.42), one obtains

$$\begin{aligned} \|\tilde{w}_h^n\| &\lesssim_\alpha |\lambda_n + \mathfrak{b}_1|^{-1} \left(\|\tilde{f}_h^n + \mathfrak{c}_n \big(\mathfrak{J} \nabla \mathcal{H}[n_3 w_3^n] \big)_h\| + \|w_3^n\| + \|\tilde{w}_h^n\| \right) \\ &\lesssim_\alpha |\lambda_n + \mathfrak{b}_1|^{-1} \|\tilde{w}_h^n\| + |\lambda_n + \mathfrak{b}_1|^{-1} \sup_{n \ge 1} \left(\|\tilde{f}_h^n + \mathfrak{c}_n \big(\mathfrak{J} \nabla \mathcal{H}[n_3 w_3^n] \big)_h\| + \|w_3^n\| \right). \end{aligned}$$

By taking n sufficiently large, it follows

$$\|\tilde{w}_h^n\| \lesssim_\alpha |\lambda_n + \mathfrak{b}_1|^{-1} \sup_{n \ge 1} \left(\|\tilde{f}_h^n + \mathfrak{c}_n \big(\mathfrak{J} \nabla \mathcal{H}[n_3 w_3^n] \big)_h\| + \|w_3^n\| \right) \to 0,$$

as $n \to \infty$. Together with (4.42) and Proposition B.1, this contradicts $\|w_h^n\| = 1$. Hence, the estimate (4.39) holds as claimed.

Now set $c_\alpha := \min\{2\pi^2, C_p/2\}$. By combining (4.20), (4.37), (4.39), and (4.43), one has

$$\|(\lambda - \nu_\kappa^{-1} \mathfrak{L}_{\text{eff}})^{-1} f\| = \|w\| \lesssim_\alpha \|f\|.$$

This implies (4.3), and by rescaling, completes the proof. □

We now turn the resolvent estimate into an exponential decay estimate for the linear semigroup. To this end, we use the following exponential stability criterion, cf. Theorem 1.4 in [HSV24] and references therein.

Theorem 4.2. *Let $S(t)$ be a strongly continuous semigroup on a Hilbert space H with the generator A. Suppose that there is $\lambda_0 \in \mathbb{R}$ such that*

$$r_0^{-1} := \sup_{\operatorname{Re}\lambda > \lambda_0} \|(\lambda - A)^{-1}\|_{\mathcal{L}(H)} < \infty.$$

Assume that

$$\|S(t)\|_{\mathcal{L}(H)} \le m(t), \qquad \forall t \ge 0$$

for some continuous positive function $m(t)$. Then, for any $t, a, b > 0$ such that $t \geq a + b$,

$$\|S(t)\|_{\mathcal{L}(H)} \leq \frac{e^{\lambda_0 t - r_0(t-a-b)}}{r_0 \|\frac{1}{m}\|_{e^{-\lambda_0}L^2(0,a)} \|\frac{1}{m}\|_{e^{-\lambda_0}L^2(0,b)}},$$

where

$$\|f\|^2_{e^{-\lambda_0}L^2(0,a)} := \int_0^a |f(t)|^2 e^{2\lambda_0 t} \mathrm{d}t.$$

Now we are in a position to present the following linear decay estimate.

Proposition 4.3. *Let $\alpha, \nu, \kappa > 0$. Then, there are constants $C_\alpha, c_\alpha > 0$ depending only on α such that*

$$\|e^{t\mathfrak{L}_{\mathrm{eff}}}\|_{\mathcal{L}(H_{\mathrm{curl}})} \leq C_\alpha e^{-c_\alpha \nu_\kappa t}. \tag{4.44}$$

Proof. **Step I.** In this step, we derive a rough semigroup bound. Consider the linear evolutionary problem

$$\begin{cases} \partial_t w = \mathfrak{L}_{\mathrm{eff}} w, \\ w|_{t=0} = w_{\mathrm{in}}. \end{cases}$$

Let $\tilde{w} := \mathfrak{J} w$, where $\mathfrak{J}$ is the boundary correction operator defined in (2.3). Applying $\mathfrak{J}$ on both sides of the above equation, it follows

$$\begin{cases} \partial_t \tilde{w} = \nu_\kappa \Delta \tilde{w} - \nu_\kappa [\mathcal{B}, \partial_y^2] w + \beta_\kappa \mathfrak{J} \nabla \mathcal{H}[n_3 w_3], \\ (\partial_y \tilde{w}_3 - \alpha \tilde{w}_3)|_{y=0} = (\partial_y \tilde{w}_3 + \alpha \tilde{w}_3)|_{y=1} = \tilde{w}_h|_{y=0,1} = 0, \\ \tilde{w}|_{t=0} = \tilde{w}_{\mathrm{in}}. \end{cases} \tag{4.45}$$

By the standard energy estimate combined with Proposition B.2 and (3.73), it follows that

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\tilde{w}\|^2 + \nu_\kappa \left(\|\nabla \tilde{w}\|^2 + \alpha \|\tilde{w}_3\|^2_{L^2(\partial D)} \right) \lesssim_\alpha \nu_\kappa \|\tilde{w}\|^2, \tag{4.46}$$

which implies

$$\|\tilde{w}(t)\| \leq e^{C_\alpha \nu_\kappa t} \|\tilde{w}_{\mathrm{in}}\|,$$

and thus

$$\|e^{t\mathfrak{L}_{\mathrm{eff}}}\|_{\mathcal{L}(H_{\mathrm{curl}})} \leq C_\alpha e^{C_\alpha \nu_\kappa t}. \tag{4.47}$$

Step II. Now we apply Theorem 4.2 with

$$\lambda_0 := -c_\alpha \nu_\kappa, \qquad m(t) := C_\alpha e^{C_\alpha \nu_\kappa t}, \qquad a = b = r_0^{-1},$$

where c_α is given in Proposition 4.1. Notice that

$$\begin{aligned} r_0 \|m^{-1}\|^2_{e^{-\lambda_0}L^2(0,r_0^{-1})} &= C_\alpha^{-2} r_0 \int_0^{r_0^{-1}} \exp\left(-(2c_\alpha \nu_\kappa + 2C_\alpha \nu_\kappa)t\right) \mathrm{d}t \\ &= \frac{r_0}{2C_\alpha^2 \nu_\kappa (c_\alpha + C_\alpha)} \left(1 - \exp\left(-\frac{2\nu_\kappa (c_\alpha + C_\alpha)}{r_0}\right)\right), \end{aligned}$$

where, by using (4.2), one has

$$\frac{2\nu_\kappa (c_\alpha + C_\alpha)}{r_0} \leq 2(c_\alpha + C_\alpha) C_\alpha =: C'_\alpha.$$

Since the function $t \mapsto t^{-1}(1-e^{-t})$ is decreasing on $\mathbb{R}_+$, this gives

$$r_0 \|m^{-1}\|^2_{e^{-\lambda_0} L^2(0, r_0^{-1})} \geq \frac{1-e^{-C'_\alpha}}{C_\alpha^2 C'_\alpha},$$

and thus

$$\|e^{t\mathfrak{L}_{\mathrm{eff}}}\|_{\mathcal{L}(H_{\mathrm{curl}})} \leq \frac{C_\alpha^2 C'_\alpha}{1-e^{-C'_\alpha}} e^{-c_\alpha \nu_\kappa t}, \qquad t \geq \frac{2}{r_0}. \tag{4.48}$$

For $t \in [0, 2/r_0)$, notice that from (4.2), one has

$$\frac{\nu_\kappa}{r_0} \leq C_\alpha.$$

This, together with the rough bound (4.47), gives

$$\|e^{t\mathfrak{L}_{\mathrm{eff}}}\|_{\mathcal{L}(H_{\mathrm{curl}})} \leq C_\alpha e^{(C_\alpha + c_\alpha)\nu_\kappa t} e^{-c_\alpha \nu_\kappa t} \leq C_\alpha \exp(2(C_\alpha + c_\alpha)C_\alpha) e^{-c_\alpha \nu_\kappa t}.$$

Combining the above estimate with (4.48), we obtain (4.44). □

Let $\mathcal{X}^m$ denote the anisotropic Sobolev space defined in (1.14). The following linear estimate in $\mathcal{X}^m$ follows directly from the above semigroup estimate; for completeness, its proof is postponed to Appendix C.

Corollary 4.4. *Let m be a positive integer and $f \in L^2_{\mathrm{loc}}(0,\infty; H_{\mathrm{curl}})$. Suppose that w solves*

$$\begin{cases} \partial_t w = \nu_\kappa \Delta w + \beta_\kappa \nabla \mathcal{H}[n_3 w_3] + f, \qquad u = \mathrm{curl}^{-1} w, \\ (\partial_y w_3 - \alpha w_3)|_{y=0} = (w_h - \alpha u_h^\perp)|_{y=0} = 0, \\ (\partial_y w_3 + \alpha w_3)|_{y=1} = (w_h + \alpha u_h^\perp)|_{y=1} = 0, \\ w|_{t=0} = w_{\mathrm{in}}. \end{cases} \tag{4.49}$$

Then, there is a constant $C_\alpha > 0$ depending only on α such that for any $T > 0$,

$$\sup_{t \leq T} \|w(t)\|^2_{m,0} + \nu_\kappa \int_0^T \left(\|\nabla w\|^2_{m,0} + \|w\|^2_{m,0} \right) \mathrm{d}t \leq C_\alpha \|w_{\mathrm{in}}\|^2_{m,0} + \frac{C_\alpha}{\nu_\kappa} \int_0^T \|f\|^2_{m-1,0} \mathrm{d}t.$$

4.1.2 *Nonlinear estimate*

In this subsection, we prove the global well-posedness of (4.1) with small initial data. The main result is given below.

Proposition 4.5. *Let $\alpha, \nu, \kappa > 0$ and $m > 2$ be an integer. Then, there is a constant $C_{\alpha,m} > 0$ such that if*

$$w_{\mathrm{in}} \in H_{\mathrm{curl}} \cap \mathcal{X}^m, \qquad \|w_{\mathrm{in}}\|_{m,0} \leq \frac{\nu_\kappa}{C_{\alpha,m}}, \tag{4.50}$$

the equation (4.1) admits a unique global solution satisfying

$$\sup_{t \geq 0} \|w(t)\|^2_{m,0} + \nu_\kappa \int_0^\infty \|\nabla w(t)\|^2_{m,0} \mathrm{d}t \lesssim_{\alpha,m} \|w_{\mathrm{in}}\|^2_{m,0}. \tag{4.51}$$

Proof. **Step I.** In this step, we estimate the nonlinear terms. By applying Corollary A.5, the Sobolev embedding $H^1_y \hookrightarrow L^\infty_y$, and the divergence-free condition of w, it follows

$$\begin{aligned} \|(w \cdot \nabla) u\|_{m-1,0} &\lesssim_m \| \|w_h\|_{H^{m-1}_x} \|\nabla_x u\|_{H^{m-1}_x} \|_{L^2_y} + \| \|w_3\|_{H^{m-1}_x} \|\partial_y u\|_{H^{m-1}_x} \|_{L^2_y} \\ &\lesssim_m \|w_h\|_{m-1,0} \|\nabla_x u\|_{H^1_y H^{m-1}_x} + \|w_3\|_{H^1_y H^{m-1}_x} \|\partial_y u\|_{m-1,0} \\ &\lesssim_m \|w\|^2_{m,0} + \|w_3\|_{m-1,0} \|w\|_{m-1,0} + \|\partial_y w_3\|_{m-1,0} \|w\|_{m-1,0} \\ &\lesssim_m \|w\|^2_{m,0} + \|\operatorname{div} w_h\|_{m-1,0} \|w\|_{m-1,0} \\ &\lesssim_m \|w\|^2_{m,0}. \end{aligned} \tag{4.52}$$

Similarly,

$$\|(u\cdot\nabla)w\|_{m-1,0}\lesssim_m \|u\|_{H^1_yH^{m-1}_x}\|\nabla w\|_{m-1,0}\lesssim_m \|w\|_{m,0}\|\nabla w\|_{m,0}. \tag{4.53}$$

Step II. Define

$$\mathcal{E}_m(T):=\sup_{t\le T}\|w(t)\|^2_{m,0}+\nu_\kappa\int_0^T(\|\nabla w\|^2_{m,0}+\|w\|^2_{m,0})\mathrm{d}t.$$

By applying Corollary 4.4 combined with (4.52) and (4.53), it follows

$$\begin{aligned}\mathcal{E}_m(T)&\le C_\alpha\|w_{\mathrm{in}}\|^2_{m,0}+\frac{C_{\alpha,m}}{\nu_\kappa}\int_0^T\|w\|^2_{m,0}(\|\nabla w\|^2_{m,0}+\|w\|^2_{m,0})\mathrm{d}t\\&\le C_\alpha\|w_{\mathrm{in}}\|^2_{m,0}+\frac{C_{\alpha,m}}{\nu_\kappa^2}\mathcal{E}_m^2(T).\end{aligned}$$

Let

$$T^*:=\sup\{T>0|\mathcal{E}_m(T)\le 2C_\alpha\|w_{\mathrm{in}}\|^2_{m,0}\}>0.$$

For any $T<T^*$, one has

$$\mathcal{E}_m(T)\le C_\alpha\|w_{\mathrm{in}}\|^2_{m,0}+\frac{4C_{\alpha,m}C_\alpha^2}{\nu_\kappa^2}\|w_{\mathrm{in}}\|^4_{m,0}.$$

Hence, if

$$\|w_{\mathrm{in}}\|^2_{m,0}\le\frac{\nu_\kappa^2}{8C_{\alpha,m}C_\alpha},$$

one obtains

$$\mathcal{E}_m(T)\le\frac{3C_\alpha}{2}\|w_{\mathrm{in}}\|^2_{m,0}<2C_\alpha\|w_{\mathrm{in}}\|^2_{m,0},\qquad\forall T<T^*,$$

which implies $T^*=\infty$ and (4.51). □

4.2 The degenerate case

Next, we turn to the degenerate case. After correcting the boundary conditions and removing the cut-off function in (3.75), the corresponding limiting equation without cut-off is given by

$$\begin{cases}\partial_t\tilde w+(u\cdot\nabla)\tilde w-(\tilde w\cdot\nabla)u=\nu\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde w+\kappa\mathfrak{J}\mathfrak{A}_{\mathrm{eff}}\mathfrak{J}^{-1}\tilde w+\mathcal{R}[w],\\(\partial_y\tilde w_3-\alpha\tilde w_3)|_{y=0}=(\partial_y\tilde w_3+\alpha\tilde w_3)|_{y=1}=\tilde w_h|_{y=0,1}=0,\\\tilde w|_{t=0}=\tilde w_{\mathrm{in}},\end{cases}\tag{4.54}$$

where $\mathfrak{J}$ is the boundary correction operator introduced in (2.3), $\tilde w:=\mathfrak{J}w$ is the corrected vorticity, $\mathfrak{A}_{\mathrm{eff}}$ is defined in (3.50), and

$$\mathcal{R}[w]:=\mathcal{B}(u\cdot\nabla)w-\mathcal{B}(w\cdot\nabla)u-(u\cdot\nabla)\mathcal{B}w+(\mathcal{B}w\cdot\nabla)u.\tag{4.55}$$

Define the effective dissipation operator

$$\mathfrak{L}_{\mathrm{eff}}\tilde w:=\nu\mathfrak{J}\Delta\mathfrak{J}^{-1}\tilde w+\kappa\mathfrak{J}\mathfrak{A}_{\mathrm{eff}}\mathfrak{J}^{-1}\tilde w$$

on the domain

$$\mathscr{D}(\mathfrak{L}_{\mathrm{eff}}):=\{\tilde w\in H^2\cap\widetilde{H}_{\mathrm{curl}}|\ \tilde w\text{ satisfies the corrected boundary condition (4.54)}_2\},$$

where $\widetilde{H}_{\mathrm{curl}}$ is the corrected vorticity class introduced in (B.1). Consider the tangential Fourier expansion

$$\tilde w=\textstyle\sum_k e^{2\pi ik\cdot x}\hat{\tilde w}(k,y),$$

and decompose $\tilde{w}$ into its tangential zeroth mode and non-zero modes

$$\tilde{w}_{=} := \hat{\tilde{w}}(0,y), \qquad \tilde{w}_{\neq} := \textstyle\sum_{k\neq 0} e^{2\pi i k\cdot x}\hat{\tilde{w}}(k,y).$$

Since

$$\mathfrak{J}\mathfrak{A}_{\text{eff}}\mathfrak{J}^{-1}\tilde{w}_{=} = 0,$$

the effective dissipation operator $\mathcal{L}_{\text{eff}}$ does not produce enhanced dissipation on the zeroth mode $\tilde{w}_{=}$. Therefore, we shall treat the zeroth mode and the non-zero modes separately. Notice that

$$\partial_y u_{=,3} = -\int_{\mathbb{T}^2} \operatorname{div} u_h \mathrm{d}x \equiv 0.$$

Together with the boundary condition $u_3|_{y=0,1} = 0$, this gives $u_{=,3} \equiv 0$. Hence, the zeroth modes of u, w, and $\tilde{w}$ are shear flows. In particular, the zero–zero interactions in

$$\mathcal{N} := (u\cdot\nabla)\tilde{w} - (\tilde{w}\cdot\nabla)u$$

vanish identically. The same observation applies to the remainder $\mathcal{R}[w]$ in (4.54). As a result, any possible growth of $\tilde{w}_{=}$ can only be generated by interactions involving the non-zero modes $\tilde{w}_{\neq}$. This structure allows us to control $\tilde{w}_{=}$ through the exponential stability estimate for the non-zero modes $\tilde{w}_{\neq}$.

4.2.1 Linear decay estimate for the non-zero modes

We establish a semigroup estimate for $e^{t\mathfrak{L}_{\text{eff}}}\mathrm{P}_{\neq}$, where $\mathrm{P}_{\neq}$ denotes the projection onto the non-zero modes

$$\mathrm{P}_{\neq} f := f - \mathrm{P}_{=} f, \qquad \mathrm{P}_{=} f := \int_{\mathbb{T}^2} f(x,y)\mathrm{d}x.$$

To this end, for each $k \neq 0$, we define

$$H_{\text{curl},k} := \{f = f(y) \in L^2 | 2\pi i k\cdot f_h + f_3' = 0\},$$

and

$$\widetilde{H}_{\text{curl},k} := \{f = f(y) \in L^2 | 2\pi i k\cdot f_h + f_3' = \alpha(1-2y)f_3\}.$$

Let $\mathfrak{J}_k, \Pi_k, \mathcal{B}_k, \Delta_k$ be the operators induced by $\mathfrak{J}, \Pi, \mathcal{B}, \Delta$ on the k-th tangential Fourier mode, respectively. More precisely, they are defined by

$$\mathfrak{J}(fe^{2\pi i k\cdot x}) = e^{2\pi i k\cdot x}\mathfrak{J}_k f, \qquad \Pi(fe^{2\pi i k\cdot x}) = e^{2\pi i k\cdot x}\Pi_k f, \qquad \mathcal{B}(fe^{2\pi i k\cdot x}) = e^{2\pi i k\cdot x}\mathcal{B}_k f, \tag{4.56}$$

and

$$\Delta_k f := -4\pi^2|k|^2 f + f''.$$

These operators are well-defined, since $\mathfrak{J}$, Π, $\mathcal{B}$, Δ commute with the tangential Fourier transform. In particular, they inherit the bounds of the corresponding full operators, e.g.,

$$\sup_{k\neq 0}\left(\|\mathfrak{J}_k\|_{\mathcal{L}(H_{\text{curl},k};\widetilde{H}_{\text{curl},k})} + \|\mathfrak{J}_k^{-1}\|_{\mathcal{L}(\widetilde{H}_{\text{curl},k};H_{\text{curl},k})}\right) + \sup_{k\neq 0}\left(\|\Pi_k\|_{\mathcal{L}(L^2_y;H_{\text{curl},k})} + \|\mathcal{B}_k\|_{\mathcal{L}(H_{\text{curl},k};H^1_y)}\right) < \infty. \tag{4.57}$$

Let

$$\mathcal{M}_k := \operatorname{diag}\left\{\frac{1}{4}I_2 + \frac{1}{2}\frac{k\otimes k}{|k|^2}, 1\right\},$$

and

$$\mathfrak{L}_{\mathrm{eff},k} := \nu \mathfrak{J}_k \Delta_k \mathfrak{J}_k^{-1} - 4\pi^2 |k|^2 \kappa \mathfrak{J}_k \Pi_k \mathcal{M}_k \mathfrak{J}_k^{-1}.$$

By construction, $\mathfrak{L}_{\mathrm{eff},k}$ is the restriction of $\mathfrak{L}_{\mathrm{eff}}$ to the k-th mode. The following resolvent estimate holds for $\mathfrak{L}_{\mathrm{eff},k}$.

Proposition 4.6. *There are $C_\alpha, c_\alpha > 0$ depending only on α such that for any $\kappa \geq C_\alpha \nu$ and $k \neq 0$,*

$$\sup_{\mathrm{Re}\,\lambda \geq -c_\alpha \kappa |k|^2} \|(\lambda - \mathfrak{L}_{\mathrm{eff},k})^{-1}\|_{\mathcal{L}(\widetilde{H}_{\mathrm{curl},k})} \leq \frac{C_\alpha}{\kappa}. \tag{4.58}$$

Proof. Let

$$\mathfrak{G}_{\epsilon,k} := \epsilon \mathfrak{J}_k \Delta_k \mathfrak{J}_k^{-1} - 4\pi^2 |k|^2 \mathfrak{J}_k \Pi_k \mathcal{M}_k \mathfrak{J}_k^{-1}$$

be the normalized effective dissipation operator. By rescaling, it suffices to prove

$$\sup_{\mathrm{Re}\,\lambda \geq -c_\alpha |k|^2} \|(\lambda - \mathfrak{G}_{\epsilon,k})^{-1}\|_{\mathcal{L}(\widetilde{H}_{\mathrm{curl},k})} \leq C_\alpha \tag{4.59}$$

for any $\epsilon \in (0, C_\alpha^{-1}]$ and $k \neq 0$. The proof is divided into the following steps.

Step I. First, we establish an auxiliary estimate. Let $\{g_n\}_{n\geq 1} \subset \widetilde{H}_{\mathrm{curl},k}$ satisfy

$$\|g_n\| = 1 \qquad \text{and} \qquad g_n \rightharpoonup 0, \qquad \text{as } n \to \infty. \tag{4.60}$$

Then,

$$\liminf_{n\to\infty} \mathrm{Re}\langle \mathfrak{J}_k \Pi_k \mathcal{M}_k \mathfrak{J}_k^{-1} g_n, g_n \rangle \geq \frac{1}{4}. \tag{4.61}$$

To see this, notice that $\mathcal{B}_k$ is a compact linear operator from $H_{\mathrm{curl},k}$ to L^2. Consequently, for $\bar{g}_n := \mathfrak{J}_k^{-1} g_n$, one has

$$\bar{g}_n - g_n = (I - \mathfrak{J}_k)\mathfrak{J}_k^{-1} g_n = \mathcal{B}_k \mathfrak{J}_k^{-1} g_n \to 0$$

strongly in L^2, as $n \to \infty$. This implies

$$\lim_{n\to\infty} \|\bar{g}_n\| = 1.$$

Combining this with the compactness of $\mathcal{B}_k$ and the fact that $g_n \rightharpoonup 0$ as $n \to \infty$, it follows

$$\begin{aligned}
\langle \mathfrak{J}_k \Pi_k \mathcal{M}_k \mathfrak{J}_k^{-1} g_n, g_n \rangle &= \langle \mathfrak{J}_k \Pi_k \mathcal{M}_k \bar{g}_n, \mathfrak{J}_k \bar{g}_n \rangle \\
&= \langle \Pi_k \mathcal{M}_k \bar{g}_n, \bar{g}_n \rangle - \langle \mathcal{B}_k \Pi_k \mathcal{M}_k \bar{g}_n, \bar{g}_n \rangle - \langle \mathfrak{J}_k \Pi_k \mathcal{M}_k \bar{g}_n, \mathcal{B}_k \bar{g}_n \rangle \\
&= \langle \mathcal{M}_k \bar{g}_n, \bar{g}_n \rangle + o(1).
\end{aligned} \tag{4.62}$$

As $\mathcal{M}_k$ is orthogonally similar to $\mathrm{diag}\{\frac{3}{4}, \frac{1}{4}, 1\}$, this yields (4.61).

Step II. In this step, we establish the resolvent estimate (4.59) with large $|\lambda|$. Consider the resolvent equation

$$\begin{aligned}
f = (\lambda - \mathfrak{G}_{\epsilon,k})g &= \lambda g - \epsilon \mathfrak{J}_k \Delta_k \mathfrak{J}_k^{-1} g + 4\pi^2 |k|^2 \mathfrak{J}_k \Pi_k \mathcal{M}_k \mathfrak{J}_k^{-1} g \\
&= \lambda g - \epsilon \Delta_k g - \epsilon [\mathfrak{J}_k, \partial_y^2] \mathfrak{J}_k^{-1} g + 4\pi^2 |k|^2 \mathfrak{J}_k \Pi_k \mathcal{M}_k \mathfrak{J}_k^{-1} g.
\end{aligned} \tag{4.63}$$

Here, we view Δ_k as a linear operator defined on the domain

$$\mathscr{D}(\Delta_k) := \{g \in H_y^2 | g \text{ satisfies the corrected boundary condition } (4.54)_2\}.$$

Since

$$\langle -\Delta_k g, g\rangle = 4\pi^2|k|^2\|g\|^2 + \|g'\|^2 + \alpha|g_3(0)|^2 + \alpha|g_3(1)|^2 \geq (4\pi^2|k|^2 + C_\alpha)\|g\|^2,$$

one has

$$\sigma(-\Delta_k) \subset (4\pi^2, \infty). \tag{4.64}$$

On the other hand, by the spectral theorem for the positive self-adjoint operator $-\Delta_k$, one has

$$\|(\lambda - \epsilon\Delta_k)^{-1}\|_{\mathcal{L}(L^2_y)} = \sup_{j\geq 1} |\lambda + \epsilon\mathfrak{b}_j|^{-1},$$

where $\mathfrak{b}_1 < \mathfrak{b}_2 < \dots$ are the eigenvalues of $-\Delta_k$. This, together with (4.64), gives

$$\|(\lambda - \epsilon\Delta_k)^{-1}\|_{\mathcal{L}(L^2_y)} \leq \sup_{\mathfrak{b}>4\pi^2} |\lambda + \epsilon\mathfrak{b}|^{-1}. \tag{4.65}$$

Now assume that $\operatorname{Re}\lambda \geq -\frac{\pi^2}{2}|k|^2$ and $|\lambda| \geq 2\pi^2|k|^2$. Notice that if $|\operatorname{Im}\lambda| > \frac{1}{2}|\lambda|$, then

$$|\lambda + \epsilon\mathfrak{b}|^{-1} \leq |\operatorname{Im}\lambda|^{-1} \leq 2|\lambda|^{-1};$$

if $|\operatorname{Re}\lambda| > \frac{1}{2}|\lambda|$, then $|\operatorname{Re}\lambda| > \pi^2|k|^2$, which implies $\operatorname{Re}\lambda > 0$ and thus

$$|\lambda + \epsilon\mathfrak{b}|^{-1} \leq (\operatorname{Re}\lambda + \epsilon\mathfrak{b})^{-1} \leq 2|\lambda|^{-1}.$$

As $|\operatorname{Re}\lambda| + |\operatorname{Im}\lambda| \geq |\lambda|$, one of the above two cases must hold. Consequently, from (4.65), it follows

$$\|(\lambda - \epsilon\Delta_k)^{-1}\|_{\mathcal{L}(L^2_y)} \leq 2|\lambda|^{-1},$$

which, together with Propositions B.1 and B.2, leads to

$$\begin{aligned}
\|g\| &= \|(\lambda - \epsilon\Delta_k)^{-1}(f + \epsilon[\mathfrak{J}_k, \partial_y^2]\mathfrak{J}_k^{-1}g - 4\pi^2|k|^2\mathfrak{J}_k\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g)\| \\
&\leq 2|\lambda|^{-1}\|f\| + 2|\lambda|^{-1}\|\epsilon[\mathfrak{J}_k, \partial_y^2]\mathfrak{J}_k^{-1}g - 4\pi^2|k|^2\mathfrak{J}_k\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g\| \\
&\leq 2|\lambda|^{-1}\|f\| + C_\alpha|\lambda|^{-1}(|k|^2+1)\|g\|.
\end{aligned}$$

By taking

$$\operatorname{Re}\lambda \geq -\frac{\pi^2}{2}|k|^2, \qquad |\lambda| \geq (2\pi^2 + 4C_\alpha)|k|^2, \tag{4.66}$$

in the above estimate, one gets the a priori estimate

$$\|g\| \leq 4|\lambda|^{-1}\|f\|,$$

which ensures

$$\|(\lambda - \mathfrak{G}_{\epsilon,k})^{-1}\|_{\mathcal{L}(\widetilde{H}_{\mathrm{curl},k})} \leq 4|\lambda|^{-1} \tag{4.67}$$

for any $\epsilon \in (0, C_\alpha^{-1}]$ and $k \neq 0$.

Step III. By contradiction, suppose that there are no c_α, C_α such that (4.59) holds for any $\epsilon \in (0, C_\alpha^{-1}]$ and $k \neq 0$. Then, for any $c_\alpha \in (0, \frac{\pi^2}{2})$ and $n \geq 1$, there are $\epsilon_n, k_n, \lambda_n$ and functions f_n, g_n satisfying

$$\epsilon_n \in \left(0, \frac{1}{2n}\right], \qquad k_n \neq 0, \qquad \operatorname{Re}\lambda_n \geq -c_\alpha|k_n|^2, \tag{4.68}$$

and

$$\|f_n\| \leq \frac{1}{n}, \qquad \|g_n\| = 1, \qquad (\lambda_n - \mathfrak{G}_{\epsilon_n,k_n})g_n = f_n. \tag{4.69}$$

We claim that $\sup_{n\geq 1}\frac{|\lambda_n|}{|k_n|^2}<\infty$. If not, passing to a subsequence if necessary, one has

$$\lim_{n\to\infty}|\lambda_n|\geq\lim_{n\to\infty}\frac{|\lambda_n|}{|k_n|^2}=\infty.$$

Together with (4.67), this implies

$$\|g_n\|\leq\frac{4}{n}|\lambda_n|^{-1}\to 0,$$

which contradicts (4.69). Therefore, without loss of generality, one may assume $\frac{\lambda_n}{|k_n|^2}\to\mu$ for some μ satisfying

$$\operatorname{Re}\mu\geq -c_\alpha. \tag{4.70}$$

Next, we show that $\sup_{n\geq 1}|k_n|<\infty$. To see this, we need the following auxiliary estimate

$$\|\mathcal{B}_k\|_{\mathcal{L}(H_{\mathrm{curl},k};L^2_y)}\lesssim_\alpha |k|^{-1}. \tag{4.71}$$

Indeed, by applying Corollary A.5 and the definition (4.56) of $\mathcal{B}_k$, one has

$$|k|\|\mathcal{B}_k\tilde{g}\|\lesssim\|\mathcal{B}(\nabla_x(\tilde{g}e^{2\pi ik\cdot x}))\|\lesssim_\alpha\|\tilde{g}\|,$$

which ensures (4.71). By contradiction, suppose that $|k_n|\to\infty$. Then, from the resolvent equation

$$\lambda_n g_n-\epsilon_n\Delta_k g_n-\epsilon_n[\mathfrak{J}_k,\partial_y^2]\mathfrak{J}_k^{-1}g_n+4\pi^2|k_n|^2\mathfrak{J}_k\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g_n=f_n, \tag{4.72}$$

it follows

$$\frac{\lambda_n}{|k_n|^2}g_n-\frac{\epsilon_n}{|k_n|^2}\Delta_k g_n-\frac{\epsilon_n}{|k_n|^2}[\mathfrak{J}_k,\partial_y^2]\mathfrak{J}_k^{-1}g_n+4\pi^2\mathfrak{J}_k\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g_n=\frac{f_n}{|k_n|^2}. \tag{4.73}$$

Here, to avoid double subscripts, we omit the dependence of k in $\mathfrak{J}_k,\Delta_k,\Pi_k,\mathcal{M}_k$ on n. Repeating the derivation of (4.62) and using (4.57), (4.69), and (4.71), it follows

$$\operatorname{Re}\langle\mathfrak{J}_k\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g_n,g_n\rangle=\operatorname{Re}\langle\mathcal{M}_k\mathfrak{J}_k^{-1}g_n,\mathfrak{J}_k^{-1}g_n\rangle+o(1),$$

which implies

$$\liminf_{n\to\infty}4\pi^2\operatorname{Re}\langle\mathfrak{J}_k\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g_n,g_n\rangle\geq\pi^2\|\mathfrak{J}_k^{-1}g_n\|^2\geq c_\alpha^* \tag{4.74}$$

for some constant $c_\alpha^*>0$, where the last inequality follows from the uniform boundedness of $\mathfrak{J}_k$ and $\mathfrak{J}_k^{-1}$ in (4.57). Moreover, by applying Proposition B.2, one has

$$\frac{\epsilon_n}{|k_n|^2}|\langle[\mathfrak{J}_k,\partial_y^2]\mathfrak{J}_k^{-1}g_n,g_n\rangle|\leq\frac{C_\alpha\epsilon_n}{|k_n|^2}\|g_n\|^2=\frac{C_\alpha\epsilon_n}{|k_n|^2}\to 0. \tag{4.75}$$

Therefore, by taking inner product with g_n on both sides of (4.73), sending $n\to\infty$, and using (4.68), (4.70), (4.74), and (4.75), it follows

$$c_\alpha^*-c_\alpha\leq\operatorname{Re}\mu+c_\alpha^*\leq 0.$$

By taking $c_\alpha<c_\alpha^*$, one gets a contradiction.

Step IV. Since $\{k_n\}_{n\geq 1}$ is bounded, we may pass to a subsequence and assume that $k_n\equiv k$ for some fixed $k\neq 0$. Therefore, one has $\lambda_n\to\mu|k|^2$ with μ satisfying (4.70). Since g_n is uniformly bounded in L^2, there is a function g_∞ such that

$$g_n\rightharpoonup g_\infty.$$

We claim that $g_\infty \neq 0$. Indeed, by contradiction, suppose that $g_\infty = 0$. Applying the standard energy estimate to the resolvent equation (4.72), it follows

$$\operatorname{Re}\lambda_n\|g_n\|^2 + \epsilon_n\|g_n'\|^2 + 4\pi^2|k|^2\epsilon_n\|g_n\|^2 + 4\pi^2|k|^2\operatorname{Re}\langle\mathfrak{J}_k\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g_n, g_n\rangle \\ \leq (\|f_n\| + \epsilon_n\|[\mathfrak{J}_k, \partial_y^2]\mathfrak{J}_k^{-1}g_n\|)\|g_n\|, \quad (4.76)$$

which, together with sending $n \to \infty$ and using (4.61), (4.68), (4.69), and Proposition B.2, yields

$$(\pi^2 - c_\alpha)|k|^2 = \operatorname{Re}\mu|k|^2 + \pi^2|k|^2 \leq 0.$$

As $c_\alpha < \frac{\pi^2}{2} \wedge c_\alpha^*$, one gets a contradiction.

Now we are in a position to conclude the proof. Indeed, from (4.76), it follows

$$\epsilon_n\|g_n'\|^2 \leq 4\pi^2|k|^2|\langle\mathfrak{J}_k\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g_n, g_n\rangle| + (\|f_n\| + \epsilon_n\|[\mathfrak{J}_k, \partial_y^2]\mathfrak{J}_k^{-1}g_n\|)\|g_n\| \leq C_{\alpha,k}.$$

Thus, for any test function $\varphi \in C_0^\infty(0,1)$

$$\begin{aligned}\epsilon_n|\langle -\Delta_k g_n, \varphi\rangle| &\leq \epsilon_n(4\pi^2|k|^2\|g_n\|\|\varphi\| + \|g_n'\|\|\varphi'\|) \\ &\leq 4\pi^2|k|^2\epsilon_n\|\varphi\| + \sqrt{\epsilon_n}C_{\alpha,k}\|\varphi'\| \to 0.\end{aligned}$$

Notice that

$$f_n + \epsilon_n[\mathfrak{J}_k, \partial_y^2]\mathfrak{J}_k^{-1}g_n \to 0 \qquad \text{in } L^2,$$

and

$$\lambda_n g_n + 4\pi^2|k|^2\mathfrak{J}_k\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g_n \rightharpoonup \mu|k|^2 g_\infty + 4\pi^2|k|^2\mathfrak{J}_k\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g_\infty \qquad \text{in } L^2.$$

By taking $n \to \infty$ in the resolvent equation (4.72), it follows

$$\mu|k|^2 g_\infty + 4\pi^2|k|^2\mathfrak{J}_k\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g_\infty = 0,$$

where the equality holds in the sense of distributions. Moreover, by density argument, the above equation holds in $\widetilde{H}_{\mathrm{curl},k}$, as the left-hand side belongs to $\widetilde{H}_{\mathrm{curl},k}$. This yields

$$\mu|k|^2\mathfrak{J}_k^{-1}g_\infty + 4\pi^2|k|^2\Pi_k\mathcal{M}_k\mathfrak{J}_k^{-1}g_\infty = 0,$$

and thus

$$\mu|k|^2 \in \sigma(-4\pi^2|k|^2\Pi_k\mathcal{M}_k).$$

However, since $-4\pi^2|k|^2\Pi_k\mathcal{M}_k$ is symmetric on $H_{\mathrm{curl},k}$ and

$$\operatorname{Re}\langle -4\pi^2|k|^2\Pi_k\mathcal{M}_k\phi, \phi\rangle \leq -\pi^2|k|^2\|\phi\|^2, \qquad \forall\phi \in H_{\mathrm{curl},k},$$

it follows

$$\mu|k|^2 \in \sigma(-4\pi^2|k|^2\Pi_k\mathcal{M}_k) \subset (-\infty, -\pi^2|k|^2].$$

This contradicts the fact $\operatorname{Re}\mu \geq -c_\alpha > -\frac{\pi^2}{2}$ and thus completes the proof. □

Together with the Parseval theorem, the above resolvent estimate ensures

$$\sup_{\operatorname{Re}\lambda \geq -c_\alpha\kappa} \|(\lambda - \mathfrak{L}_{\mathrm{eff}})^{-1}\mathrm{P}_{\neq}\|_{\mathcal{L}(\widetilde{H}_{\mathrm{curl}})} \leq \frac{C_\alpha}{\kappa}.$$

Combining this with the exponential stability criterion in Theorem 4.2, the following decay estimate for the semigroup $e^{t\mathfrak{L}_{\mathrm{eff}}}\mathrm{P}_{\neq}$ holds. The proof is identical to that of Proposition 4.3, and is therefore omitted.

Proposition 4.7. *Let $\alpha, \nu > 0$. There exist constants $C_\alpha, c_\alpha > 0$ depending only on α such that for any $\kappa \geq C_\alpha \nu$,*

$$\|e^{t\mathfrak{L}_{\mathrm{eff}}} \mathrm{P}_{\neq}\|_{\mathcal{L}(\widetilde{H}_{\mathrm{curl}})} \leq C_\alpha e^{-c_\alpha \kappa t}.$$

As a direct consequence, we obtain the following linear estimate for the non-zero modes in the anisotropic space $\mathcal{X}^m$. The proof is identical to that of Corollary 4.4; we therefore omit the details and refer the reader to Appendix C.

Corollary 4.8. *Let $\alpha, \nu > 0$, m be a positive integer, and $f \in L^2_{\mathrm{loc}}(0, \infty; \widetilde{H}_{\mathrm{curl}})$. Suppose that $\tilde{w}$ solves*

$$\begin{cases} \partial_t \tilde{w} = \mathfrak{L}_{\mathrm{eff}} \tilde{w} + \mathrm{P}_{\neq} f, \\ (\partial_y \tilde{w}_3 - \alpha \tilde{w}_3)|_{y=0} = (\partial_y \tilde{w}_3 + \alpha \tilde{w}_3)|_{y=1} = \tilde{w}_h|_{y=0,1} = 0, \\ \tilde{w}|_{t=0} = \mathrm{P}_{\neq} \tilde{w}_{\mathrm{in}}. \end{cases}$$

Then, there is a constant $C_\alpha > 0$ depending only on α such that for any $\kappa \geq C_\alpha \nu$ and $T > 0$,

$$\sup_{t \leq T} \|\tilde{w}(t)\|^2_{m,0} + \kappa \int_0^T \|\nabla_x \tilde{w}\|^2_{m,0} \mathrm{d}t + \nu \int_0^T \|\partial_y \tilde{w}\|^2_{m,0} \mathrm{d}t \\ \leq C_\alpha \|\mathrm{P}_{\neq} \tilde{w}_{\mathrm{in}}\|^2_{m,0} + \frac{C_\alpha}{\kappa} \int_0^T \|\mathrm{P}_{\neq} f\|^2_{m-1,0} \mathrm{d}t.$$

4.2.2 Nonlinear estimate

In this subsection, we prove the global well-posedness of (4.54) with small initial data. The main result is given below.

Proposition 4.9. *Let $\alpha, \nu > 0$ and $m > 2$ be an integer. Then, there is a constant $C_{\alpha,\nu,m} > 0$ such that if $\kappa \geq C_{\alpha,\nu,m}$ and the initial data $\tilde{w}_{\mathrm{in}}$ is given by*

$$\tilde{w}_{\mathrm{in}} := \mathfrak{J} w_{\mathrm{in}} \qquad \textit{with} \qquad w_{\mathrm{in}} \in H_{\mathrm{curl}} \cap \mathcal{X}^m, \qquad \|w_{\mathrm{in}}\|_{m,0} \leq \frac{\sqrt{\kappa}}{C_{\alpha,\nu,m}},$$

the equation (4.54) admits a unique global solution satisfying

$$\sup_{t \geq 0} \|\tilde{w}(t)\|^2_{m,0} + \kappa \int_0^\infty \|\nabla_x \tilde{w}\|^2_{m,0} \mathrm{d}t + \nu \int_0^\infty \|\partial_y \tilde{w}\|^2_{m,0} \mathrm{d}t \lesssim_{\alpha,\nu,m} \|w_{\mathrm{in}}\|^2_{m,0}.$$

Proof. The proof follows the same strategy as that of Proposition 4.5, so we only outline the main steps.

Step I. In this step, we estimate the nonlinear terms. First, we decompose the convection terms into

$$\mathrm{P}_{\neq} \mathcal{N} = \mathrm{P}_{\neq}[(u \cdot \nabla)\tilde{w} - (\tilde{w} \cdot \nabla) u] = \mathrm{P}_{\neq} \mathcal{N}_1 + \mathrm{P}_{\neq} \mathcal{N}_2 + \mathrm{P}_{\neq} \mathcal{N}_3, \tag{4.77}$$

where

$$\mathcal{N}_1 := (u_{\neq} \cdot \nabla)\tilde{w}_{\neq} - (\tilde{w}_{\neq} \cdot \nabla) u_{\neq},$$
$$\mathcal{N}_2 := (u_{=} \cdot \nabla)\tilde{w}_{\neq} - (\tilde{w}_{=} \cdot \nabla) u_{\neq},$$
$$\mathcal{N}_3 := (u_{\neq} \cdot \nabla)\tilde{w}_{=} - (\tilde{w}_{\neq} \cdot \nabla) u_{=}.$$

Repeating the derivation of (4.52) and (4.53), for $\mathcal{N}_1$, one has

$$\|\mathcal{N}_1\|_{m-1,0} \lesssim_{\alpha,m} \|\tilde{w}_{\neq}\|_{m,0} \left(\|\tilde{w}_{\neq}\|_{m,0} + \|\partial_y \tilde{w}_{\neq}\|_{m,0} \right). \tag{4.78}$$

Notice that $\tilde{w}_=$ and $u_=$ are shear flows, then it follows

$$\|\mathcal{N}_2\|_{m-1,0} \lesssim_{\alpha,m} \|\tilde{w}_=\|_{L^2_y}\|\tilde{w}_{\neq}\|_{m,0}, \tag{4.79}$$

and

$$\|\mathcal{N}_3\|_{m-1,0} \lesssim_{\alpha,m} \|\tilde{w}_=\|_{H^1_y}\|\tilde{w}_{\neq}\|_{m,0}. \tag{4.80}$$

Combining (4.77)–(4.80), one gets

$$\|\mathrm{P}_{\neq}\mathcal{N}\|_{m-1,0} \lesssim_{\alpha,m} \|\tilde{w}_{\neq}\|_{m,0}\left(\|\tilde{w}_{\neq}\|_{m,0} + \|\partial_y\tilde{w}_{\neq}\|_{m,0}\right) + \|\tilde{w}_=\|_{H^1_y}\|\tilde{w}_{\neq}\|_{m,0}. \tag{4.81}$$

Next, we turn to the remainder term

$$\begin{aligned}\mathcal{R}[w] &= \mathcal{B}(u\cdot\nabla)w - \mathcal{B}(w\cdot\nabla)u - (u\cdot\nabla)\mathcal{B}w + (\mathcal{B}w\cdot\nabla)u\\ &= \mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4.\end{aligned} \tag{4.82}$$

For $\mathcal{R}_1$, by using the divergence-free condition for $u_{\neq}, u_=$ and the fact that $w_=, u_=$ are shear, it follows

$$\begin{aligned}\|\mathcal{R}_1\|_{m-1,0} &\le \|\mathcal{B}(u_{\neq}\cdot\nabla)w_{\neq}\|_{m-1,0} + \|\mathcal{B}(u_=\cdot\nabla)w_{\neq}\|_{m-1,0} + \|\mathcal{B}(u_{\neq}\cdot\nabla)w_=\|_{m-1,0}\\ &\le \|\mathcal{B}\,\mathrm{div}(u_{\neq}\otimes w_{\neq})\|_{m-1,0} + \|\mathcal{B}\,\mathrm{div}(u_=\otimes w_{\neq})\|_{m-1,0} + \|\mathcal{B}(u_{\neq,3}\partial_y w_=)\|_{m-1,0}\\ &\lesssim_{\alpha,m} \|\tilde{w}_{\neq}\|^2_{m,0} + \|\tilde{w}_=\|_{L^2_y}\|\tilde{w}_{\neq}\|_{m,0} + \|u_{\neq,3}w_=\|_{m-1,0} + \|\mathcal{B}\,\mathrm{div}_h(u_{\neq,h}\otimes w_=)\|_{m-1,0}\\ &\lesssim_{\alpha,m} \|\tilde{w}_{\neq}\|^2_{m,0} + \|\tilde{w}_=\|_{L^2_y}\|\tilde{w}_{\neq}\|_{m,0}.\end{aligned} \tag{4.83}$$

Similarly,

$$\|\mathcal{R}_2\|_{m-1,0} \lesssim_{\alpha,m} \|\tilde{w}_{\neq}\|^2_{m,0} + \|\tilde{w}_=\|_{L^2_y}\|\tilde{w}_{\neq}\|_{m,0}. \tag{4.84}$$

Since $\mathcal{B}w_=$ is also a shear flow, for $\mathcal{R}_3$, one has

$$\begin{aligned}\|\mathcal{R}_3\|_{m-1,0} &\le \|(u_{\neq}\cdot\nabla)\mathcal{B}w_{\neq}\|_{m-1,0} + \|(u_=\cdot\nabla)\mathcal{B}w_{\neq}\|_{m-1,0} + \|(u_{\neq}\cdot\nabla)\mathcal{B}w_=\|_{m-1,0}\\ &\lesssim_{\alpha,m} \|\tilde{w}_{\neq}\|^2_{m,0} + \|\tilde{w}_=\|_{L^2_y}\|\tilde{w}_{\neq}\|_{m,0} + \|u_{\neq,3}\partial_y\mathcal{B}w_=\|_{m-1,0}\\ &\lesssim_{\alpha,m} \|\tilde{w}_{\neq}\|^2_{m,0} + \|\tilde{w}_=\|_{L^2_y}\|\tilde{w}_{\neq}\|_{m,0}.\end{aligned} \tag{4.85}$$

Proceeding similarly, for $\mathcal{R}_4$, one gets

$$\|\mathcal{R}_4\|_{m-1,0} \lesssim_{\alpha,m} \|\tilde{w}_{\neq}\|^2_{m,0} + \|\tilde{w}_=\|_{L^2_y}\|\tilde{w}_{\neq}\|_{m,0}. \tag{4.86}$$

By summarizing the estimates (4.82)–(4.86), it follows

$$\|\mathrm{P}_{\neq}\mathcal{R}[w]\|_{m-1,0} \lesssim_{\alpha,m} \|\tilde{w}_{\neq}\|^2_{m,0} + \|\tilde{w}_=\|_{L^2_y}\|\tilde{w}_{\neq}\|_{m,0}. \tag{4.87}$$

Step Ⅱ. Now we estimate the zeroth mode. Since

$$\tilde{w}_= := \mathfrak{J}_0 w_= = \mathfrak{J}_0(\partial_y u^{\perp}_{=,h}, 0), \tag{4.88}$$

we establish the estimate for $\tilde{w}_=$ through the velocity formulation of (4.54):

$$\begin{cases}\partial_t u + (u\cdot\nabla)u + \nabla p + (c_1(t), c_2(t), 0) = \nu\Delta u + \kappa\,\mathrm{curl}^{-1}\,\mathfrak{A}_{\mathrm{eff}}\,\mathrm{curl}\,u,\\ (\partial_y u_h - \alpha u_h)|_{y=0} = (\partial_y u_h + \alpha u_h)|_{y=1} = u_3|_{y=0,1} = 0,\\ u|_{t=0} = u_{\mathrm{in}}.\end{cases}$$

In particular, since $\mathrm{curl}^{-1}\,\mathfrak{A}_{\mathrm{eff}}$ commutes with $\mathrm{P}_=$ and $u_{=,3}\equiv 0$, $\int_{\mathbb{T}^2} u_{\neq,3}\mathrm{d}x \equiv 0$, by applying the projection $\mathrm{P}_=$, the equation for $u_{=,h}$ is given by

$$\begin{cases}\partial_t u_{=,h} + \mathrm{P}_=\partial_y(u_{\neq,3}u_{\neq,h}) + (c_1(t), c_2(t)) = \nu\partial_y^2 u_{=,h},\\ (\partial_y u_{=,h} - \alpha u_{=,h})|_{y=0} = (\partial_y u_{=,h} + \alpha u_{=,h})|_{y=1} = 0,\\ u_{=,h}|_{t=0} = \mathrm{P}_= u_{\mathrm{in},h}.\end{cases} \tag{4.89}$$

Notice that $\int_0^1 u_{=,h}\mathrm{d}y = \int_D u_h \mathrm{d}x\mathrm{d}y = 0$, by taking L^2_y inner product with $u_{=,h}$ on both sides, it follows

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|u_{=,h}\|^2 + \nu\|\partial_y u_{=,h}\|^2 + \alpha\nu|u_{=,h}|_{y=0}|^2 + \alpha\nu|u_{=,h}|_{y=1}|^2 = -\langle \partial_y(u_{\neq,3}u_{\neq,h}), u_{=,h}\rangle,$$

which gives

$$\nu\int_0^t \|\partial_y u_{=,h}\|^2\mathrm{d}s \le \|\mathrm{P}_= u_{\mathrm{in},h}\|^2 + C_\nu\int_0^t \|\mathrm{P}_=(u_{\neq,3}u_{\neq,h})\|^2\mathrm{d}s. \tag{4.90}$$

On the other hand, by taking integration with respect to $y\in(0,1)$ on both sides of (4.89) and using the identity

$$\int_0^1 \mathrm{P}_=\partial_y(u_{\neq,3}u_{\neq,h})\mathrm{d}y = \int_{\mathbb{T}^2}\int_0^1 \partial_y(u_{\neq,3}u_{\neq,h})\mathrm{d}y\mathrm{d}x = 0,$$

one has

$$(c_1(t), c_2(t)) = \nu\int_0^1 \partial_y^2 u_{=,h}\mathrm{d}y = -\alpha\nu(u_{=,h}|_{y=1} + u_{=,h}|_{y=0}).$$

Therefore, by testing (4.89) against $-\partial_y^2 u_{=,h}$ and using the Poincaré inequality, it follows

$$\begin{aligned}
&\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left(\|\partial_y u_{=,h}\|^2 + \alpha|u_{=,h}|_{y=0}|^2 + \alpha|u_{=,h}|_{y=1}|^2\right) + \nu\|\partial_y^2 u_{=,h}\|^2\\
&\qquad = \langle \partial_y(u_{\neq,3}u_{\neq,h}), \partial_y^2 u_{=,h}\rangle + \alpha^2\nu|u_{=,h}|_{y=1} + u_{=,h}|_{y=0}|^2\\
&\qquad \le |\langle \partial_y(u_{\neq,3}u_{\neq,h}), \partial_y^2 u_{=,h}\rangle| + C_{\alpha,\nu}\|\partial_y u_{=,h}\|^2,
\end{aligned}$$

which, together with (4.90), implies

$$\|\partial_y u_{=,h}(t)\|^2 + \nu\int_0^t \|\partial_y^2 u_{=,h}\|^2\mathrm{d}s \lesssim_{\alpha,\nu} \|\mathrm{P}_= u_{\mathrm{in},h}\|^2_{H^1_y} + \int_0^t \|\mathrm{P}_=(u_{\neq,3}u_{\neq,h})\|^2_{H^1_y}\mathrm{d}s.$$

Combining this, (4.88), and (4.90), one gets

$$\begin{aligned}
\|\tilde{w}_=(t)\|^2 + \nu\int_0^t \|\partial_y\tilde{w}_=\|^2\mathrm{d}s &\lesssim_{\alpha,\nu} \|\tilde{w}_{\mathrm{in}}\|^2 + \int_0^t \|\mathrm{P}_=(u_{\neq,3}u_{\neq,h})\|^2_{H^1_y}\mathrm{d}s\\
&\lesssim_{\alpha,\nu} \|\tilde{w}_{\mathrm{in}}\|^2 + \int_0^t \|\tilde{w}_{\neq}\|^4_{m,0}\mathrm{d}s.
\end{aligned} \tag{4.91}$$

Step Ⅲ. Define

$$\begin{aligned}
\mathcal{E}_m(T) := \sup_{t\le T}\|\tilde{w}_{\neq}(t)\|^2_{m,0} + \sup_{t\le T}\|\tilde{w}_=(t)\|^2_{L^2_y} + \kappa\int_0^T \|\nabla_x\tilde{w}_{\neq}\|^2_{m,0}\mathrm{d}t\\
+ \nu\int_0^T \|\partial_y\tilde{w}_{\neq}\|^2_{m,0}\mathrm{d}t + \nu\int_0^T \|\partial_y\tilde{w}_=\|^2_{L^2_y}\mathrm{d}t.
\end{aligned}$$

By applying Corollary 4.8 combined with (4.81), (4.87), and the Poincaré inequalities

$$\|\tilde{w}_{\neq}\|_{m,0} \le \|\nabla_x\tilde{w}_{\neq}\|_{m,0}, \qquad \|\tilde{w}_{=,h}\|_{L^2_y} \lesssim \|\partial_y\tilde{w}_{=,h}\|_{L^2_y},$$

it follows

$$\sup_{t\le T}\|\tilde{w}_{\neq}(t)\|^2_{m,0} + \kappa\int_0^T \|\nabla_x\tilde{w}_{\neq}\|^2_{m,0}\mathrm{d}t + \nu\int_0^T \|\partial_y\tilde{w}_{\neq}\|^2_{m,0}\mathrm{d}t \lesssim_{\alpha,\nu,m} \|\tilde{w}_{\mathrm{in}}\|^2_{m,0} + \frac{1}{\kappa}\mathcal{E}_m^2(T).$$

Similarly, from (4.91), one has

$$\sup_{t\le T}\|\tilde{w}_{=}(t)\|_{L^2_y}^2+\nu\int_0^T\|\partial_y\tilde{w}_{=}\|_{L^2_y}^2\mathrm{d}t\lesssim_{\alpha,\nu,m}\|\tilde{w}_{\mathrm{in}}\|_{m,0}^2+\frac{1}{\kappa}\mathcal{E}_m^2(T)$$

Therefore,

$$\mathcal{E}_m(T)\lesssim_{\alpha,\nu,m}\|\tilde{w}_{\mathrm{in}}\|_{m,0}^2+\frac{1}{\kappa}\mathcal{E}_m^2(T).$$

This, together with a standard bootstrap argument, completes the proof; see Step II of the proof of Proposition 4.5. □

5 Proof of Theorem 1.2

In this section, we combine the results obtained in Sections 3 and 4 to prove Theorem 1.2. We prove the result only in the non-degenerate case. The degenerate case is treated in the same way, using the corresponding degenerate versions of the scaling limit result and the deterministic global well-posedness result.

Let

$$M:=\|w_{\mathrm{in}}\|_{L^\infty(\Omega;H^1\cap\mathcal{X}^m)}.$$

Choose $\kappa>0$ sufficiently large so that

$$M\le\frac{\nu_\kappa}{2C_{\alpha,m}},\qquad \nu_\kappa:=\nu+\frac{4\kappa}{5},$$

where $C_{\alpha,m}$ is the constant given in Proposition 4.5. Then, there is a unique global solution w^{det} of the limiting equation (4.1) satisfying

$$\sup_{t\ge0}\|w^{\mathrm{det}}(t)\|_{m,0}^2+\nu_\kappa\int_0^\infty\|\nabla w^{\mathrm{det}}(t)\|_{m,0}^2\mathrm{d}t\le C'^2_{\alpha,m}M^2.$$

In particular, since $m>2$, it follows

$$\sup_{t\ge0}\|w^{\mathrm{det}}(t)\|\le C'_{\alpha,m}M.$$

Let the cut-off parameter $R>0$ be sufficiently large so that

$$R\ge2C'_{\alpha,m}M.$$

Therefore, the cut-off function η_R is inactive along w^{det}, and the deterministic cut-off solution coincides with the deterministic solution without cut-off, that is,

$$w^{\mathrm{det}}_{\mathrm{cut}}(t)=w^{\mathrm{det}}(t),\qquad\forall t\in[0,T].$$

Let w^N_{cut} be the solution of the cut-off equation (3.1), corresponding to the noise coefficients $\theta^N_\cdot$ given in (3.2). By Proposition 3.4, after choosing N sufficiently large, it follows

$$\mathbb{P}\left\{\sup_{t\le T}\|w^N_{\mathrm{cut}}(t)-w^{\mathrm{det}}_{\mathrm{cut}}(t)\|_{H^\delta}<R-C'_{\alpha,m}M\right\}\ge1-\epsilon.$$

Notice that on this event, one has

$$\sup_{t\le T}\|w^N_{\mathrm{cut}}(t)\|\le\sup_{t\le T}\|w^N_{\mathrm{cut}}(t)-w^{\mathrm{det}}_{\mathrm{cut}}(t)\|_{H^\delta}+\sup_{t\le T}\|w^{\mathrm{det}}_{\mathrm{cut}}(t)\|<R.$$

Define

$$\tau^N:=\inf\{t|\|w^N_{\mathrm{cut}}(t)\|\ge R\}.$$

The preceding estimate implies

$$\mathbb{P}\{\tau^N \geq T\} \geq 1 - \epsilon. \tag{5.1}$$

Moreover, on the time interval $[0, \tau^N)$, the cut-off is inactive, so w^N_{cut} coincides with a local solution of (1.1). This, together with (5.1), ensures the existence of a solution

$$(w, \tau) := (w^N_{\mathrm{cut}}, \tau^N)$$

of (1.1), which exists up to T with high probability.

A The curl-admissible class and Biot–Savart operator in the periodic channel

In this section, we collect some basic properties of the curl-admissible class H_{curl} defined in (1.2) and the Biot–Savart operator curl^{-1}. Define

$$H^{\perp}_{\mathrm{curl}} := \{\nabla \mathfrak{p} | \mathfrak{p} \in H^1,\ \mathfrak{p}|_{y=0,1} \text{ are constants}\}.$$

Proposition A.1. *The following orthogonal decomposition holds*

$$L^2 = H_{\mathrm{curl}} \oplus H^{\perp}_{\mathrm{curl}}.$$

Moreover, for any $f \in L^2$,

$$\Pi f = f - \nabla \mathfrak{p}, \tag{A.1}$$

where $\Pi : L^2 \to H_{\mathrm{curl}}$ is the projection operator and $\mathfrak{p}$ satisfies the non-homogeneous Dirichlet–Poisson problem

$$\begin{cases} \Delta \mathfrak{p} = \operatorname{div} f, \\ \mathfrak{p}|_{y=0} = 0, \qquad \mathfrak{p}|_{y=1} = \int_D f_3 \mathrm{d}x\mathrm{d}y. \end{cases} \tag{A.2}$$

Proof. **Step I.** Since

$$\int_0^1 \left| \int_{\mathbb{T}^2} w_3(x, y) \mathrm{d}x \right|^2 \mathrm{d}y \lesssim \|w\|^2,$$

the map

$$w \mapsto Tw(y) := \int_{\mathbb{T}^2} w_3(x, y) \mathrm{d}x$$

is bounded from L^2 to L^2_y, which implies that H_{curl} is a closed subspace of L^2. Fix $w \in H_{\mathrm{curl}}$ and $\nabla \mathfrak{p} \in H^{\perp}_{\mathrm{curl}}$. According to the definition of $H^{\perp}_{\mathrm{curl}}$,

$$\tilde{\mathfrak{p}} := \mathfrak{p} - \mathfrak{p}_1 y - (1 - y)\mathfrak{p}_0 \in H^1_0,$$

where $\mathfrak{p}_0 := \mathfrak{p}|_{y=0}$ and $\mathfrak{p}_1 := \mathfrak{p}|_{y=1}$ are constants. Integrating by parts and using the definition (1.2) of H_{curl}, one has the orthogonality relation

$$\int_D w \cdot \nabla \mathfrak{p} \mathrm{d}x\mathrm{d}y = \int_D w \cdot \nabla \tilde{\mathfrak{p}} \mathrm{d}x\mathrm{d}y + (\mathfrak{p}_1 - \mathfrak{p}_0) \int_D w_3 \mathrm{d}x\mathrm{d}y = 0.$$

Step II. It remains to construct the decomposition (A.1). Define

$$\mathfrak{p} := \tilde{\mathfrak{p}} + y \int_D f_3 \mathrm{d}x\mathrm{d}y,$$

where $\tilde{\mathfrak{p}}$ solves the Dirichlet–Poisson problem

$$\begin{cases} \Delta \tilde{\mathfrak{p}} = \operatorname{div} f, \\ \tilde{\mathfrak{p}}|_{y=0,1} = 0. \end{cases} \tag{A.3}$$

Then, it follows that $\nabla \mathfrak{p} \in H^{\perp}_{\mathrm{curl}}$ and $w := f - \nabla \mathfrak{p}$ satisfies

$$\operatorname{div} w = \operatorname{div} f - \Delta \mathfrak{p} = \operatorname{div} f - \Delta \tilde{\mathfrak{p}} = 0.$$

To verify the horizontal zero-mean condition for w_3, notice that for any $\varphi = \varphi(y) \in C_0^\infty(0,1)$, one has

$$0 = \int_D \operatorname{div} w \varphi \mathrm{d}x\mathrm{d}y = -\int_D w_3 \varphi' \mathrm{d}x\mathrm{d}y = \int_D \partial_y \left(\int_{\mathbb{T}^2} w_3(x,y)\mathrm{d}x \right) \varphi(y)\mathrm{d}y,$$

which implies

$$\int_{\mathbb{T}^2} w_3(x,y)\mathrm{d}x = c, \qquad a.e.\ y \in (0,1)$$

for some constant c. On the other hand, by the Newton–Leibniz formula and the boundary condition $(\mathrm{A}.2)_2$,

$$0 = \int_D (f_3 - \partial_y \mathfrak{p})\mathrm{d}x\mathrm{d}y = \int_D w_3 \mathrm{d}x\mathrm{d}y = c.$$

Therefore, $w := f - \nabla \mathfrak{p} \in H_{\mathrm{curl}}$ and the proof is complete. □

Let $H^{-1,q}$ be defined in (1.17). The following estimate plays an important role in the a priori estimate.

Corollary A.2. *Let $q \in (1,\infty)$ and $m \geq 0$ be an integer. For any smooth vector field f, it follows that*

$$\|\Pi f\|_{H^{m,q}} \lesssim_{m,q} \|f\|_{H^{m,q}},$$

and

$$\|\Pi f\|_{H^{-1,q}} \lesssim_q \left(\|f\|_{H^{-1,q}} + \left| \int_D f_3 \mathrm{d}x\mathrm{d}y \right| \right). \tag{A.4}$$

Proof. **Step I.** We first address the case $m = 0$. Let $\tilde{\mathfrak{p}}$ be the solution of (A.3). By applying odd extension on $\tilde{\mathfrak{p}}$, f_1, f_2 and even extension on f_3, periodizing the resulting equation, and using the Mikhlin multiplier theorem, one has

$$\|\nabla \tilde{\mathfrak{p}}\|_q \lesssim_q \|f\|_q. \tag{A.5}$$

This, together with the decomposition

$$f = \Pi f + \nabla \tilde{\mathfrak{p}} + \left(\int_D f_3 \mathrm{d}x\mathrm{d}y \right) e_3, \tag{A.6}$$

implies that

$$\|\Pi f\|_q \leq \|f\|_q + \|\nabla \tilde{\mathfrak{p}}\|_q + \|f_3\|_{L^1} \lesssim_q \|f\|_q.$$

Step II. The higher regularity can be obtained through a standard argument. For simplicity, we only discuss the case $m = 1$. Taking ∂_{x_j} on both sides of (A.1), it follows that

$$\partial_{x_j} \Pi f = \partial_{x_j} f - \nabla \partial_{x_j} \mathfrak{p},$$

where $\partial_{x_j} \mathfrak{p}$ satisfies the Dirichlet–Poisson problem

$$\begin{cases} \Delta \partial_{x_j} \mathfrak{p} = \operatorname{div} \partial_{x_j} f, \\ \partial_{x_j} \mathfrak{p}|_{y=0} = \partial_{x_j} \mathfrak{p}|_{y=1} = 0. \end{cases}$$

By the elliptic estimate (A.5), one has

$$\|\partial_{x_j}\Pi f\|_q \lesssim_q \|f\|_{H^{1,q}} + \|\nabla\partial_{x_j}\mathfrak{p}\|_q \lesssim_q \|f\|_{H^{1,q}}.$$

Notice that

$$\|\nabla\partial_y\mathfrak{p}\|_q \le \|\nabla_x\partial_y\mathfrak{p}\|_q + \|\partial_y^2\mathfrak{p}\|_q \lesssim \|f\|_{H^{1,q}} + \|\Delta_x\mathfrak{p}\|_q + \|\operatorname{div} f\|_q \lesssim_q \|f\|_{H^{1,q}},$$

it follows that

$$\|\nabla\Pi f\|_q \le \|\nabla_x\Pi f\|_q + \|\partial_y\Pi f\|_q \lesssim_q \|f\|_{H^{1,q}}.$$

Step Ⅲ. We turn to (A.4). Using (A.6) again, one has

$$\|\Pi f\|_{H^{-1,q}} \le \|f\|_{H^{-1,q}} + \|\nabla\tilde{\mathfrak{p}}\|_{H^{-1,q}} + \left|\int_D f_3 \mathrm{d}x\mathrm{d}y\right|. \tag{A.7}$$

For any $\psi \in L^{q'}$, consider the following Dirichlet–Poisson problem

$$\begin{cases} -\Delta\phi = \psi, \\ \phi|_{y=0,1} = 0. \end{cases} \tag{A.8}$$

By duality, it follows that

$$\|\tilde{\mathfrak{p}}\|_q = \sup_{\|\psi\|_{L^{q'}}=1} |\langle\tilde{\mathfrak{p}}, \psi\rangle|,$$

where, by integrating by parts and the standard elliptic regularity theory for (A.8), one has

$$|\langle\tilde{\mathfrak{p}}, \psi\rangle| = |\langle\tilde{\mathfrak{p}}, \Delta\phi\rangle| = |\langle\operatorname{div} f, \phi\rangle| = |\langle f, \nabla\phi\rangle| \le \|f\|_{H^{-1,q}}\|\nabla\phi\|_{H^{1,q'}} \lesssim_q \|f\|_{H^{-1,q}}.$$

Hence,

$$\|\tilde{\mathfrak{p}}\|_q \lesssim_q \|f\|_{H^{-1,q}}. \tag{A.9}$$

On the other hand, using the boundary conditions of $\tilde{\mathfrak{p}}$, one has

$$\|\nabla\tilde{\mathfrak{p}}\|_{H^{-1,q}} = \sup_{\|\varphi\|_{H^{1,q'}}=1} |\langle\nabla\tilde{\mathfrak{p}}, \varphi\rangle| \le \|\tilde{\mathfrak{p}}\|_q \|\operatorname{div}\varphi\|_{L^{q'}} \lesssim \|\tilde{\mathfrak{p}}\|_q.$$

This, together with (A.7) and (A.9), implies (A.4). □

Next, we turn to the basic properties of the Biot–Savart operator curl^{-1}.

Proposition A.3. *Let $w \in H_{\mathrm{curl}}$. Then, there is a unique solution u of the div-curl system*

$$\begin{cases} \operatorname{curl} u = w, \qquad \operatorname{div} u = 0, \\ u_3|_{y=0,1} = \int_D u_1 \mathrm{d}x = \int_D u_2 \mathrm{d}x = 0. \end{cases} \tag{A.10}$$

Moreover, it follows that

$$\|u\|_{H^1} \lesssim \|w\|. \tag{A.11}$$

We need the following representation of curl-free vector fields.

Lemma A.4. *For any vector field $z \in L^2$ satisfying $\operatorname{curl} z = 0$, there are $\varphi \in H^1$ and $c_h \in \mathbb{R}^2$ such that*

$$z = \nabla\varphi + (c_h, 0).$$

Proof. Consider the tangential Fourier expansion

$$z = \sum_{k\in\mathbb{Z}^2} \hat{z}(k,y)e^{2\pi ik\cdot x}.$$

Since z is curl-free, then for any $k \in \mathbb{Z}^2$,

$$\begin{cases} \partial_y \hat{z}_2(k,y) - 2\pi i k_2 \hat{z}_3(k,y) = 0, \\ \partial_y \hat{z}_1(k,y) - 2\pi i k_1 \hat{z}_3(k,y) = 0, \\ k_1 \hat{z}_2(k,y) - k_2 \hat{z}_1(k,y) = 0. \end{cases} \tag{A.12}$$

For any $k \neq 0$, by using $(\text{A.12})_3$, one has

$$\hat{z}_h(k,y) = 2\pi i k \phi(k,y) \tag{A.13}$$

for some scalar function ϕ, which, together with $(\text{A.12})_1$, yields

$$\hat{z}_3(k,y) = \partial_y \phi(k,y).$$

Therefore,

$$\hat{z}(k,y)e^{2\pi ik\cdot x} = \nabla(\phi(k,y)e^{2\pi ik\cdot x}). \tag{A.14}$$

For $k = 0$, one has

$$0 = \operatorname{curl} \hat{z}(0,y) = (-\partial_y \hat{z}_2(0,y), \partial_y \hat{z}_1(0,y), 0),$$

which implies $\hat{z}_h(0,y) \equiv c_h$ for some $c_h \in \mathbb{R}^2$. Let

$$\varphi := \sum_{k\neq 0} \phi(k,y)e^{2\pi ik\cdot x} + \int_0^y \hat{z}_3(0,y')\mathrm{d}y',$$

then it follows that

$$z = (\hat{z}_h, 0) + \hat{z}_3 e_3 + \nabla\left(\sum_{k\neq 0} \phi(k,y)e^{2\pi ik\cdot x}\right) = (c_h, 0) + \nabla\varphi.$$

Since $\nabla\varphi \in L^2$, by assuming $\int_D \varphi \mathrm{d}x\mathrm{d}y = 0$ if necessary, from the Poincaré inequality, one has $\varphi \in H^1$. □

Now we prove Proposition A.3.

Proof of Proposition A.3. **Step I.** We start with an auxiliary problem. Define

$$V := \left\{ u \in H^1 \,\middle|\, \operatorname{div} u = 0, u_3|_{y=0,1} = \int_D u_1 \mathrm{d}x = \int_D u_2 \mathrm{d}x = 0 \right\} \tag{A.15}$$

and the bilinear form

$$a(u,v) := \int_D \nabla u \cdot \nabla v \mathrm{d}x\mathrm{d}y, \qquad u, v \in V.$$

For any $v \in V$, since v_1, v_2 have a zero mean and v_3 vanishes on the boundary, by applying the Poincaré inequality, one has

$$a(v,v) = \int_D \|\nabla v\|^2 \mathrm{d}x\mathrm{d}y \geq c\|v\|^2.$$

It is clear to have the boundedness of $a(u,v)$. Therefore, by applying the Lax–Milgram theorem, for any $w \in H_{\mathrm{curl}}$, there exists a unique element $u \in V$ satisfying (A.11) and

$$a(u,v) = \int_D w \cdot \operatorname{curl} v \mathrm{d}x\mathrm{d}y, \qquad \forall v \in V.$$

This, together with the identity

$$\int_D \nabla u \cdot \nabla v \mathrm{d}x\mathrm{d}y = \int_D \operatorname{curl} u \cdot \operatorname{curl} v \mathrm{d}x\mathrm{d}y,$$

yields

$$\int_D (\operatorname{curl} u - w) \cdot \operatorname{curl} v \mathrm{d}x\mathrm{d}y = 0, \qquad \forall v \in V. \tag{A.16}$$

Step Ⅱ. Let $z := \operatorname{curl} u - w$. It follows that

$$\operatorname{div} z = \operatorname{div} \operatorname{curl} u - \operatorname{div} w = 0.$$

Moreover, we claim that z is curl-free. Indeed, for any vector field $\psi \in C_0^\infty$, by taking

$$v = \mathrm{P}\psi - \left(\int_D (\mathrm{P}\psi)_1 \mathrm{d}x, \int_D (\mathrm{P}\psi)_2 \mathrm{d}x, 0 \right) \in V$$

in (A.16), where P is the usual Leray projection, one has

$$0 = \int_D z \cdot \operatorname{curl} \psi \mathrm{d}x\mathrm{d}y = - \int_D \operatorname{curl} z \cdot \psi \mathrm{d}x\mathrm{d}y.$$

This implies $\operatorname{curl} z = 0$ as claimed.

Step Ⅲ. As z is curl-free and divergence-free, by Lemma A.4,

$$z = \nabla \varphi + (c_h, 0) \tag{A.17}$$

for some harmonic function φ and constant vector field $c_h := (c_1, c_2)$. We claim that

$$\varphi|_{y=0} \equiv C_0, \qquad \varphi|_{y=1} \equiv C_1 \tag{A.18}$$

for some constants C_0, C_1. To see this, let $\psi \in C^\infty(\mathbb{T}^2)$ and $\vartheta \in C^\infty(0,1)$. By plugging (A.17) and

$$v = \vartheta(y)(-\partial_{x_2}\psi(x), \partial_{x_1}\psi(x), 0)$$

into (A.16) and integrating by parts, it follows that

$$\begin{aligned} 0 &= \int_D \nabla \varphi \cdot \operatorname{curl} v \mathrm{d}x\mathrm{d}y + \int_D c_h \cdot (\operatorname{curl} v)_h \mathrm{d}x\mathrm{d}y \\ &= \int_D \nabla \varphi \cdot \operatorname{curl} v \mathrm{d}x\mathrm{d}y - \int_D \vartheta'(c_h \cdot \nabla_x)\psi \mathrm{d}x\mathrm{d}y = \int_D \nabla \varphi \cdot \operatorname{curl} v \mathrm{d}x\mathrm{d}y. \end{aligned}$$

This, together with integrating by parts again and the fact

$$\operatorname{div} \operatorname{curl} v = 0, \qquad (\operatorname{curl} v)_3 = \vartheta(y)\Delta_x \psi,$$

gives

$$0 = \vartheta(1) \int_{\mathbb{T}^2} \varphi|_{y=1} \Delta_x \psi \mathrm{d}x - \vartheta(0) \int_{\mathbb{T}^2} \varphi|_{y=0} \Delta_x \psi \mathrm{d}x.$$

Since ϑ is arbitrary, one may take $\vartheta(1) = 0$ and $\vartheta(0) = 1$, to get

$$\int_{\mathbb{T}^2} \varphi|_{y=0} \Delta_x \psi \mathrm{d}x = 0$$

for any $\psi \in C^\infty(\mathbb{T}^2)$, which implies $\varphi|_{y=0}$ is a constant and thus (A.18) holds as claimed.

Now note that $z-(c_h,0)=\nabla\varphi$, where φ is a harmonic function satisfying (A.18) and z fulfills $\operatorname{div} z=0$ and $\int_{\mathbb{T}^2} z_3 \mathrm{d}x=0$, then an application of Proposition A.1 yields

$$z\equiv(c_h,0).$$

This, together with taking

$$v=(c_2(y-1/2),-c_1(y-1/2),0)$$

in (A.16), implies

$$c_1^2+c_2^2=0,$$

and $z\equiv 0$. As a result, $\operatorname{curl} u=w$ and u is a solution of (A.10).

Step IV. It remains to show the uniqueness. Suppose that u is a solution of (A.10) with $w=0$. Then, one has

$$\int_D \|\nabla u\|^2 \mathrm{d}x\mathrm{d}y=\int_D \|\operatorname{curl} u\|^2 \mathrm{d}x\mathrm{d}y=0,$$

which implies that u is a constant vector field. This, combined with $(\text{A.10})_2$, gives $u\equiv 0$. □

Define the Biot–Savart operator

$$\operatorname{curl}^{-1}: H_{\operatorname{curl}}\to V,\qquad w\mapsto u, \tag{A.19}$$

where u is the unique solution of (A.10), and V is defined in (A.15).

Corollary A.5. *Let $q\in(1,\infty)$ and $m\geq -1$ be an integer. For any $w\in H_{\operatorname{curl}}\cap C^\infty$, it follows that*

$$\|\operatorname{curl}^{-1} w\|_{H^{m,q}}\lesssim_{m,q}\|w\|_{H^{m-1,q}}. \tag{A.20}$$

Proof. **Step I.** We begin with the case $m=1$. Let $u=\operatorname{curl}^{-1} w$ be the solution of (A.10). Applying even extension on u_1,u_2,w_3 and odd extension on w_1,w_2,u_3, periodizing the resulting equation, and using the boundedness of the Riesz transform, it follows that

$$\|\nabla u\|_q\lesssim_q\|w\|_q.$$

This, combined with the Poincaré inequality, leads to

$$\|u\|_{H^{1,q}}\lesssim_q\|w\|_q.$$

Step II. We use a duality argument to address the case $m=0$. For any vector field $\psi\in L^{q'}$, consider the following Neumann–Poisson problem

$$\begin{cases}-\Delta\phi=\psi-\left(\int_D\psi_1\mathrm{d}x,\int_D\psi_2\mathrm{d}x,0\right)=:\psi-\psi_{h,D},\\ \partial_y\phi_1|_{y=0,1}=\partial_y\phi_2|_{y=0,1}=\phi_3|_{y=0,1}=0.\end{cases} \tag{A.21}$$

The above problem is solvable, since $(\psi-\psi_{h,D})_h$ has zero mean. By duality and $(\text{A.10})_2$, it follows that

$$\|u\|_q=\sup_{\|\psi\|_{q'}=1}\left|\int_D u\cdot\psi\mathrm{d}x\mathrm{d}y\right|=\sup_{\|\psi\|_{q'}=1}\left|\int_D u\cdot(\psi-\psi_{h,D})\mathrm{d}x\mathrm{d}y\right|,$$

where, by applying (A.21), the identity

$$-\Delta\phi=\operatorname{curl}(\operatorname{curl}\phi)-\nabla(\operatorname{div}\phi),$$

and the fact that $n \times u$ is orthogonal to $\operatorname{curl}\phi$ on ∂D, one has

$$\begin{aligned}\left|\int_D u\cdot(\psi-\psi_{h,D})\mathrm{d}x\mathrm{d}y\right| &= |\langle u,\Delta\phi\rangle| = \left|\int_D u\cdot\operatorname{curl}(\operatorname{curl}\phi)\mathrm{d}x\mathrm{d}y - \int_D u\cdot\nabla(\operatorname{div}\phi)\mathrm{d}x\mathrm{d}y\right| \\ &= \left|\int_D \operatorname{curl} u\cdot\operatorname{curl}\phi\mathrm{d}x\mathrm{d}y - \int_{\{y=0,1\}}(n\times u)\cdot\operatorname{curl}\phi\mathrm{d}x\right| \\ &= \left|\int_D w\cdot\operatorname{curl}\phi\mathrm{d}x\mathrm{d}y\right| \le \|w\|_{H^{-1,q}}\|\phi\|_{H^{2,q'}}.\end{aligned}$$

This, combined with the standard elliptic estimate

$$\|\phi\|_{H^{2,q'}} \lesssim_q \|\psi-\psi_{h,D}\|_{q'} \lesssim_q \|\psi\|_{q'},$$

implies (A.20) with $m=0$. The case $m=-1$ can be obtained in the same way.

Step III. As in the proof of Corollary A.2, we only discuss the case $m=2$. Taking ∂_{x_j} on both sides of (A.10), it follows that

$$\begin{cases}\operatorname{curl}\partial_{x_j}u = \partial_{x_j}w, \qquad \operatorname{div}\partial_{x_j}u = 0,\\ \partial_{x_j}u_3|_{y=0,1} = \int_D \partial_{x_j}u_1\mathrm{d}x = \int_D \partial_{x_j}u_2\mathrm{d}x = 0.\end{cases}$$

Applying (A.20) with $m=1$, one has

$$\|\partial_{x_j}u\|_{H^{1,q}} \lesssim \|\partial_{x_j}w\|_{L^q} \lesssim \|w\|_{H^{1,q}},$$

which, combined with the fact that

$$\partial_y^2 u_1 = \partial_y w_2 + \partial_y\partial_{x_1}u_3, \quad \partial_y^2 u_2 = -\partial_y w_1 + \partial_y\partial_{x_2}u_3, \quad \partial_y^2 u_3 = -\partial_y(\partial_{x_1}u_1+\partial_{x_2}u_2),$$

gives

$$\|u\|_{H^{2,q}} \lesssim \|w\|_{H^{1,q}}$$

as desired. □

B Basic properties of the boundary correction operator $\mathfrak{J}$

In this section, we collect some basic properties of the boundary correction operator $\mathfrak{J}$ defined in (2.3). First, we prove the invertibility of $\mathfrak{J}$. Define the corrected vorticity class

$$\widetilde{H}_{\mathrm{curl}} := \left\{\tilde{w}\in L^2 \,\middle|\, \operatorname{div}\tilde{w} = \alpha(1-2y)\tilde{w}_3,\ \int_{\mathbb{T}^2}\tilde{w}_3(x,y)\mathrm{d}x = 0 \text{ a.e. } y\in(0,1)\right\} \tag{B.1}$$

endowed with the usual L^2-norm. Here, the condition on the divergence of $\tilde{w}$ is understood in the sense of distributions.

Proposition B.1. *Let $s\ge 0$, $q\in(1,\infty)$. Then, $\mathfrak{J} = I - \mathcal{B}$ is a topological isomorphism from $H_{\mathrm{curl}}\cap H^{s,q}$ to $\widetilde{H}_{\mathrm{curl}}\cap H^{s,q}$.*

Proof. **Step I.** Let $w\in H_{\mathrm{curl}}$ and $\tilde{w} = \mathfrak{J}w$. Notice that

$$\operatorname{div}\mathcal{B}w = -\alpha(1-2y)w_3, \qquad \tilde{w}_3 = w_3,$$

then one has $\operatorname{div}\tilde{w} = \alpha(1-2y)\tilde{w}_3$ and $\int_{\mathbb{T}^2}\tilde{w}_3\mathrm{d}x = 0$ for almost every $y\in(0,1)$, and thus $\tilde{w}\in\widetilde{H}_{\mathrm{curl}}$. Together with Corollary A.5, this implies

$$\mathfrak{J}\in\mathcal{L}(H_{\mathrm{curl}}\cap H^{s,q};\widetilde{H}_{\mathrm{curl}}\cap H^{s,q})$$

for any $s\ge 0$, $q\in(1,\infty)$.

Step II. We show that $\mathfrak{J}$ can be expressed as a compact perturbation of an isomorphism. Indeed, for $w \in H_{\mathrm{curl}}$, define $\Phi[w]$ as the solution of the Neumann problem

$$\begin{cases} \Delta\Phi[w] - \alpha(1-2y)\partial_y\Phi[w] = \alpha(1-2y)w_3, \\ \partial_y\Phi[w]|_{y=0,1} = 0. \end{cases} \tag{B.2}$$

Since $\int_{\mathbb{T}^2} w_3 \mathrm{d}x = 0$ for almost every $y \in (0,1)$, the solvability condition for the above problem holds:

$$\int_D \alpha(1-2y)w_3 e^{-\alpha(y-y^2)} \mathrm{d}x\mathrm{d}y = 0.$$

Let

$$\mathcal{T}w := w + \nabla\Phi[w].$$

Notice that

$$\mathrm{div}(\mathcal{T}w) = \Delta\Phi[w] = \alpha(1-2y)(w_3 + \partial_y\Phi[w]) = \alpha(1-2y)(\mathcal{T}w)_3,$$

and

$$\int_{\mathbb{T}^2} (\mathcal{T}w)_3 \mathrm{d}x = \partial_y \int_{\mathbb{T}^2} \Phi[w] \mathrm{d}x.$$

Since $\overline{\Phi[w]} := \int_{\mathbb{T}^2} \Phi[w]\mathrm{d}x$ satisfies

$$\partial_y^2\overline{\Phi[w]} - \alpha(1-2y)\partial_y\overline{\Phi[w]} = 0, \qquad \partial_y\overline{\Phi[w]}|_{y=0,1} = 0,$$

one has $\partial_y\overline{\Phi[w]} \equiv 0$ and thus $\mathcal{T}w \in \widetilde{H}_{\mathrm{curl}}$. Moreover, the standard elliptic estimate for (B.2) gives

$$\|\nabla\Phi[w]\|_{H^{s+1,q}} \lesssim_{\alpha,s,q} \|w\|_{H^{s,q}}. \tag{B.3}$$

Therefore, $\mathcal{T}$ is a bounded linear operator from $H_{\mathrm{curl}} \cap H^{s,q}$ to $\widetilde{H}_{\mathrm{curl}} \cap H^{s,q}$. We claim that $\mathcal{T}$ is indeed an isomorphism. Indeed, for any $\tilde{w} \in \widetilde{H}_{\mathrm{curl}} \cap H^{s,q}$, define $\Psi[\tilde{w}]$ as the solution of the following Neumann–Poisson problem

$$\begin{cases} \Delta\Psi[\tilde{w}] = \alpha(1-2y)\tilde{w}_3, \\ \partial_y\Psi[\tilde{w}]|_{y=0,1} = 0. \end{cases} \tag{B.4}$$

Let

$$w := \tilde{w} - \nabla\Psi[\tilde{w}].$$

Then, one has $\mathrm{div}\, w = 0$ and $\int_{\mathbb{T}^2} w_3 \mathrm{d}x = 0$ for almost every $y \in (0,1)$, which implies $w \in H_{\mathrm{curl}}$. Since $\Psi[\tilde{w}]$ satisfies

$$\Delta\Psi[\tilde{w}] - \alpha(1-2y)\partial_y\Psi[\tilde{w}] = \alpha(1-2y)w_3, \qquad \partial_y\Psi[\tilde{w}]|_{y=0,1} = 0,$$

by the unique solvability of (B.2), one has

$$\nabla\Psi[\tilde{w}] = \nabla\Phi[w].$$

This yields

$$\tilde{w} = w + \nabla\Phi[w] = \mathcal{T}w,$$

and thus $\mathcal{T}$ is surjective. It remains to show $\mathcal{T}$ is injective. To see this, suppose that

$$w + \nabla\Phi[w] = 0.$$

Then, $\Phi[w]$ satisfies

$$\Delta\Phi[w]=0,\qquad \partial_y\Phi[w]|_{y=0,1}=0,$$

which gives $\nabla\Phi[w]=0$ and thus $w=0$. Consequently, $\mathcal{T}$ is an isomorphism.

Now notice that

$$\mathcal{T}w-\mathfrak{J}w=\nabla\Phi[w]+\mathcal{B}w,$$

where, by (B.3), Corollary A.5, the standard interpolation theory, and the compact Sobolev embedding theorem, the linear operator $w\mapsto\nabla\Phi[w]+\mathcal{B}w$ is compact. Hence, $\mathfrak{J}$ is a compact perturbation of an isomorphism, and is thus a Fredholm operator of index zero.

Step Ⅲ. By the Fredholm alternative theorem, it remains to show $\mathfrak{J}$ is injective. To this end, suppose that $w=\mathcal{B}w$. Then,

$$\begin{cases}\alpha(1-2y)u_1=\partial_y u_1-\partial_{x_1}u_3,\\ \alpha(1-2y)u_2=\partial_y u_2-\partial_{x_2}u_3,\\ 0=\partial_{x_2}u_1-\partial_{x_1}u_2,\\ \operatorname{div}u=0,\\ \int_D u_1\mathrm{d}x\mathrm{d}y=\int_D u_2\mathrm{d}x\mathrm{d}y=u_3|_{y=0,1}=0,\end{cases}\tag{B.5}$$

where $u:=\operatorname{curl}^{-1}w$. Applying ∂_{x_1} to $(\text{B.5})_1$ and ∂_{x_2} to $(\text{B.5})_2$ respectively, summing the resulting equalities, and using $(\text{B.5})_4$, one has

$$\Delta u_3-\alpha(1-2y)\partial_y u_3=0,\qquad u_3|_{y=0,1}=0,$$

which gives

$$u_3\equiv 0.$$

This, together with $(\text{B.5})_3$, implies that for each fixed $y\in(0,1)$, the two-dimensional vector field $u_h(x,y)$ is divergence-free and curl-free. Therefore,

$$u_h(x,y)=c(y),$$

for some vector field $c(y)$. Combining this with $(\text{B.5})_1$ and $(\text{B.5})_2$, one has

$$u_h=Ce^{\alpha(y-y^2)}.$$

Plugging the above result into $(\text{B.5})_5$ yields $C=0$. Thus, $u\equiv 0$ and $\mathfrak{J}$ is injective. □

Next, we address some commutator estimates.

Proposition B.2. *The following assertions hold.*

1. *For any $m\in\mathbb{N}^2$, $[\mathcal{B},\partial_x^m]=0$.*
2. *For any $w\in H_{\mathrm{curl}}\cap C^\infty$ and $q\in(1,\infty)$,*

$$\|[\mathcal{B},\partial_y]w\|_q\lesssim_{\alpha,q}\|w\|_q,\qquad \|\mathcal{B}\partial_y w\|_q\lesssim_{\alpha,q}\|w\|_q.\tag{B.6}$$

Moreover, if w satisfies the Navier boundary conditions $(1.1)_3$ and $(1.1)_4$, it follows that

$$\|[\mathcal{B},\partial_y^2]w\|_q\lesssim_{\alpha,q}\|w\|_q.\tag{B.7}$$

Proof. **Step I.** The first assertion is ensured by the fact $[\partial_x^m,\operatorname{curl}^{-1}]=0$ and the definition of $\mathcal{B}$. The estimate (B.6) is established in the following way. By applying Corollary A.5, it follows that

$$\begin{aligned}\|[\mathcal{B},\partial_y]w\|_q&=\|\alpha(1-2y)(\operatorname{curl}^{-1}\partial_y w)_h-\alpha\partial_y[(1-2y)(\operatorname{curl}^{-1}w)_h]\|_q\\&\lesssim_{\alpha,q}\|[\operatorname{curl}^{-1},\partial_y]w\|_q+\|w\|_q.\end{aligned}\tag{B.8}$$

Notice that $z := [\mathrm{curl}^{-1}, \partial_y]w$ satisfies the div-curl system

$$\begin{cases} \mathrm{div}\, z = \mathrm{div}(\mathrm{curl}^{-1}\, \partial_y w - \partial_y\, \mathrm{curl}^{-1}\, w) = 0, \\ \mathrm{curl}\, z = \mathrm{curl}(\mathrm{curl}^{-1}\, \partial_y w - \partial_y\, \mathrm{curl}^{-1}\, w) = 0, \end{cases}$$

supplemented with the boundary conditions

$$z_3|_{y=0,1} = -\partial_y(\mathrm{curl}^{-1}\, w)_3|_{y=0,1} = \mathrm{div}_h(\mathrm{curl}^{-1}\, w)_h|_{y=0,1}. \tag{B.9}$$

Moreover, by applying Lemma A.4, it follows

$$z = \nabla\phi + (c_1, c_2, 0) \tag{B.10}$$

for some harmonic function ϕ and constants c_1, c_2. We turn to the estimate of $\nabla\phi$. From (B.9) and (B.10), one gets

$$\begin{cases} \Delta\phi = 0, \\ \partial_y\phi|_{y=0,1} = \mathrm{div}_h(\mathrm{curl}^{-1}\, w)_h|_{y=0,1}. \end{cases}$$

Without loss of generality, one may assume $\int_D \phi \mathrm{d}x\mathrm{d}y = 0$. Then, by the standard elliptic estimate, it follows

$$\|\nabla\phi\|_q \lesssim_q \|(\mathrm{curl}^{-1}\, w)_h\|_{W^{1-\frac{1}{q},q}(\partial D)} \lesssim_q \|w\|_q. \tag{B.11}$$

It remains to bound c_1, c_2. For $j = 1, 2$, by the mean-zero condition required for the horizontal components in the definition (A.19) of curl^{-1}, it follows that

$$\begin{aligned} |c_j| &= \left| \int_D z_j \mathrm{d}x\mathrm{d}y \right| = \left| \int_D \partial_y(\mathrm{curl}^{-1}\, w)_j \mathrm{d}x\mathrm{d}y \right| \\ &\le \left| \int_{\mathbb{T}^2} (\mathrm{curl}^{-1}\, w)_j|_{y=0} \mathrm{d}x \right| + \left| \int_{\mathbb{T}^2} (\mathrm{curl}^{-1}\, w)_j|_{y=1} \mathrm{d}x \right| \\ &\lesssim \|\,\mathrm{curl}^{-1}\, w\|_{L^q(\partial D)} \lesssim_q \|w\|_q, \end{aligned}$$

which, together with (B.10) and (B.11), yields

$$\|[\mathrm{curl}^{-1}, \partial_y]w\|_q = \|z\|_q \lesssim_q \|w\|_q. \tag{B.12}$$

Plugging the above estimate into (B.8), we obtain the first estimate in (B.6), which, together with Corollary A.5, ensures the second estimate therein.

Step Ⅱ. We turn to the proof of (B.7). Proceeding as above, one has

$$\|[\mathcal{B}, \partial_y^2]w\|_q \lesssim_\alpha \|[\mathrm{curl}^{-1}, \partial_y^2]w\|_q + \|\partial_y\, \mathrm{curl}^{-1}\, w\|_q \lesssim_{\alpha,q} \|[\mathrm{curl}^{-1}, \partial_y^2]w\|_q + \|w\|_q,$$

where $\tilde{z} := [\mathrm{curl}^{-1}, \partial_y^2]w$ satisfies

$$\begin{cases} \mathrm{div}\, \tilde{z} = \mathrm{div}(\mathrm{curl}^{-1}\, \partial_y^2 w - \partial_y^2\, \mathrm{curl}^{-1}\, w) = 0, \\ \mathrm{curl}\, \tilde{z} = \mathrm{curl}(\mathrm{curl}^{-1}\, \partial_y^2 w - \partial_y^2\, \mathrm{curl}^{-1}\, w) = 0. \end{cases}$$

By using the Navier boundary conditions in the velocity formulation, one has the boundary conditions

$$\tilde{z}_3|_{y=0} = -\partial_y^2(\mathrm{curl}^{-1}\, w)_3|_{y=0} = \mathrm{div}_h\, \partial_y(\mathrm{curl}^{-1}\, w)_h|_{y=0} = \alpha\, \mathrm{div}_h(\mathrm{curl}^{-1}\, w)_h|_{y=0},$$

and

$$\tilde{z}_3|_{y=1} = -\alpha\, \mathrm{div}_h(\mathrm{curl}^{-1}\, w)_h|_{y=1}.$$

Therefore, by repeating the derivation of (B.12), one gets

$$\|[\mathrm{curl}^{-1}, \partial_y^2]w\|_q = \|\tilde{z}\|_q \lesssim_{\alpha,q} \|w\|_q,$$

and thus (B.7) holds. □

Corollary B.3. *The following assertions hold.*

1. *For any* $m \in \mathbb{N}^2$, $[\mathfrak{J}^{-1}, \partial_x^m] = 0$.
2. *For any* $\tilde{w} \in \widetilde{H}_{\mathrm{curl}} \cap C^\infty$ *satisfying the mixed Dirichlet–Robin boundary condition* $(2.5)_4$, $w := \mathfrak{J}^{-1}\tilde{w}$, *and* $q \in (1, \infty)$, *it follows that*

$$\|[\mathfrak{J}^{-1}, \partial_y]\tilde{w}\|_q + \|[\mathfrak{J}^{-1}, \partial_y^2]\tilde{w}\|_q \lesssim_{\alpha,q} \|\tilde{w}\|_q, \tag{B.13}$$

and

$$\|\partial_y w\|_q \lesssim_{\alpha,q} \|\partial_y \tilde{w}\|_q + \|\tilde{w}\|_q, \qquad \|\partial_y \tilde{w}\|_q \lesssim_{\alpha,q} \|\partial_y w\|_q + \|w\|_q. \tag{B.14}$$

Proof. This follows from the operator identity

$$\mathfrak{J}^{-1}\mathcal{L} - \mathcal{L}\mathfrak{J}^{-1} = \mathfrak{J}^{-1}[\mathcal{B}, \mathcal{L}]\mathfrak{J}^{-1}$$

and Propositions B.1 and B.2. □

Finally, we gather some basic properties of the projection operator $\tilde{\Pi} : L^2 \to \widetilde{H}_{\mathrm{curl}}$ onto the corrected vorticity class $\widetilde{H}_{\mathrm{curl}}$. Define

$$\widetilde{H}^{\perp}_{\mathrm{curl}} := \{\nabla \mathfrak{p} + \alpha(1-2y)\mathfrak{p}e_3 \big|\ \mathfrak{p} \in H^1,\ \mathfrak{p}|_{y=0,1} \text{ are constants}\}.$$

Proposition B.4. *The following decomposition holds*

$$L^2 = \widetilde{H}_{\mathrm{curl}} \oplus \widetilde{H}^{\perp}_{\mathrm{curl}}.$$

For any vector field $f \in L^2$,

$$\tilde{\Pi} f = f - \nabla \mathfrak{p} - \alpha(1-2y)pe_3, \tag{B.15}$$

where $\mathfrak{p}$ *solves the Dirichlet problem*

$$\begin{cases} \Delta \mathfrak{p} - (2\alpha + \alpha^2(1-2y)^2)\mathfrak{p} = \operatorname{div} f - \alpha(1-2y)f_3, \\ \mathfrak{p}|_{y=0} = 0, \qquad \mathfrak{p}|_{y=1} = \int_D e^{\alpha(y-y^2)} f_3 \mathrm{d}x\mathrm{d}y. \end{cases} \tag{B.16}$$

Proof. The orthogonality of subspaces $\widetilde{H}_{\mathrm{curl}}$ and $\widetilde{H}^{\perp}_{\mathrm{curl}}$ can be verified as in the proof of Proposition A.1, while the divergence constraint of $\tilde{w} := f - \nabla \mathfrak{p} - \alpha(1-2y)pe_3$ is obtained by direct computations. It remains to check the horizontal zero-mean condition for $\tilde{w}_3$. Indeed, by integrating with respect to $x \in \mathbb{T}^2$ on both sides of $\operatorname{div} \tilde{w} = \alpha(1-2y)\tilde{w}_3$, one has

$$\partial_y \tilde{w}_{=,3} = \alpha(1-2y)\tilde{w}_{=,3},$$

where $\tilde{w}_{=,3}(y) := \int_{\mathbb{T}^2} \tilde{w}_3(x,y)\mathrm{d}x$. Therefore,

$$\tilde{w}_{=,3} = Ce^{\alpha(y-y^2)}.$$

On the other hand, by integrating with respect to $x \in \mathbb{T}^2$ on both sides of the third component of (B.15), one has

$$\tilde{w}_{=,3} = f_{=,3} - \partial_y p_= - \alpha(1-2y)p_=,$$

which gives

$$\frac{\mathrm{d}}{\mathrm{d}y}(p_= e^{\alpha(y-y^2)}) = e^{\alpha(y-y^2)}(f_{=,3} - \tilde{w}_{=,3}).$$

Together with $(B.16)_2$, this yields

$$\int_D e^{\alpha(y-y^2)} f_3 \mathrm{d}x\mathrm{d}y = \int_0^1 e^{\alpha(y-y^2)} f_{=,3} \mathrm{d}y - C\int_0^1 e^{2\alpha(y-y^2)} \mathrm{d}y.$$

Consequently, the horizontal zero-mean condition holds for $\tilde{w}_3$. □

The following result can be proved in the same way as Corollary A.2, using standard elliptic regularity theory. We therefore omit the proof.

Proposition B.5. *Let $q \in (1,\infty)$. Then, for any smooth vector field f, it follows that*

$$\|\tilde{\Pi} f\|_q \lesssim_{\alpha,q} \|f\|_q, \qquad \|\tilde{\Pi} f\|_{H^{1,q}} \lesssim_{\alpha,q} \|f\|_{H^{1,q}}. \tag{B.17}$$

C Proof of Corollary 4.4

In this section, we prove Corollary 4.4. Although the argument is standard, we include the details in order to keep track of the dependence on κ in the a priori estimates.

Step I. Let w be a solution of (4.49) with $f = 0$, and $\tilde{w} := \mathfrak{J} w$. By applying Propositions B.1, B.2, and Corollary B.3, one has

$$\|\tilde{w}\|_{m,0} \sim_\alpha \|w\|_{m,0}, \qquad \|\nabla_x \tilde{w}\|_{m,0} \sim_\alpha \|\nabla_x w\|_{m,0}. \tag{C.1}$$

Since ∇_x commutes with the harmonic extension operator $\mathcal{H}$ defined in (3.65), together with Proposition 4.3, this implies

$$\|\tilde{w}(t)\|_{m,0}^2 \le C_\alpha e^{-c_\alpha \nu_\kappa t} \|\tilde{w}_{\mathrm{in}}\|_{m,0}^2 \tag{C.2}$$

for any positive integer m. Moreover, by repeating the derivation of (4.46), it follows

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\tilde{w}\|_{m,0}^2 + \nu_\kappa \|\nabla \tilde{w}\|_{m,0}^2 \lesssim_\alpha \nu_\kappa \|\tilde{w}\|_{m,0}^2, \tag{C.3}$$

which, together with (C.2), gives

$$\|\tilde{w}(t)\|_{m,0}^2 + \nu_\kappa \int_0^t \|\nabla \tilde{w}\|_{m,0}^2 \mathrm{d}s \lesssim_\alpha \|w_{\mathrm{in}}\|_{m,0}^2.$$

Therefore, by using (B.14), (C.1), and (C.2), one gets

$$\begin{aligned}\sup_{t\le T} \|w(t)\|_{m,0}^2 + \nu_\kappa \int_0^T (\|\nabla w\|_{m,0}^2 + \|w\|_{m,0}^2)\mathrm{d}s &\lesssim_\alpha \|w_{\mathrm{in}}\|_{m,0}^2 + \nu_\kappa \int_0^T \|\tilde{w}\|_{m,0}^2 \mathrm{d}s \\ &\lesssim_\alpha \|w_{\mathrm{in}}\|_{m,0}^2. \end{aligned} \tag{C.4}$$

Step II. In this step, we derive a smoothing estimate for $\tilde{w}$. By applying Propositions B.1, B.2, and (3.73), one has

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(t\|\tilde{w}\|_{1,0}^2) &= \|\tilde{w}\|_{1,0}^2 + 2t\langle \tilde{w}, \nu_\kappa \Delta \tilde{w} - \nu_\kappa [\mathcal{B}, \partial_y^2] w + \beta_\kappa \mathfrak{J} \nabla \mathcal{H}[n_3 w_3]\rangle_{1,0} \\ &\le \|\tilde{w}\|_{1,0}^2 - 2t\nu_\kappa (\|\nabla \tilde{w}\|_{1,0}^2 + \alpha \|\tilde{w}_3\|_{H^1(\partial D)}^2) \\ &\quad + t\nu_\kappa C_\alpha \|\tilde{w}\|_{1,0} (\|\tilde{w}\|_{1,0} + \|\nabla \tilde{w}\|_{1,0}) \\ &\le \|\tilde{w}\|_{1,0}^2 + t\nu_\kappa C_\alpha \|\tilde{w}\|_{1,0}^2. \end{aligned}$$

Consequently, for $t \in (0, \nu_\kappa^{-1})$, it follows

$$t\|\tilde{w}\|_{1,0}^2 \lesssim_\alpha \int_0^t (\|\nabla_x \tilde{w}\|^2 + \|\tilde{w}\|^2)\mathrm{d}s \lesssim_\alpha \frac{1}{\nu_\kappa} \|\tilde{w}_{\mathrm{in}}\|^2.$$

As $\partial_x^k \tilde{w}$ solves (4.45) with the initial datum $\partial_x^k \tilde{w}_{\mathrm{in}}$, for any positive integer m, one has

$$\|\tilde{w}(t)\|_{m,0}^2 \lesssim_\alpha \frac{1}{\nu_\kappa t} \|\tilde{w}_{\mathrm{in}}\|_{m-1,0}^2, \qquad t \in (0, \nu_\kappa^{-1}). \tag{C.5}$$

Combining this with (C.2), one obtains

$$\|\tilde{w}(t)\|_{m,0}^2 \lesssim_\alpha e^{-c_\alpha \nu_\kappa (t - \nu_\kappa^{-1})} \|\tilde{w}(\nu_\kappa^{-1})\|_{m,0}^2 \lesssim_\alpha e^{-c_\alpha \nu_\kappa t} \|\tilde{w}_{\mathrm{in}}\|_{m-1,0}^2, \qquad t \ge \nu_\kappa^{-1}. \tag{C.6}$$

Step III. Let $\hat{w}$ be a solution of (4.49) with zero initial data. The Duhamel formula gives

$$\hat{w}(t) = \int_0^t e^{(t-s)\mathfrak{L}_{\text{eff}}} f(s)\mathrm{d}s.$$

Applying (C.5) and (C.6) combined with (C.1), one has

$$\|e^{(t-s)\mathfrak{L}_{\text{eff}}} f(s)\|_{m,0} \lesssim_\alpha \left(\frac{\mathbb{I}_{\{t-s<\nu_\kappa^{-1}\}}}{\sqrt{\nu_\kappa(t-s)}} + \mathbb{I}_{\{t-s\geq\nu_\kappa^{-1}\}} e^{-c_\alpha \nu_\kappa (t-s)} \right) \|f(s)\|_{m-1,0},$$

which, together with the Young inequality, implies

$$\begin{aligned}\int_0^T \|\hat{w}(t)\|_{m,0}^2 \mathrm{d}t &\lesssim_\alpha \left(\int_0^{\nu_\kappa^{-1}} \frac{1}{\sqrt{\nu_\kappa t}} \mathrm{d}t + \int_{\nu_\kappa^{-1}}^\infty e^{-c_\alpha \nu_\kappa t} \mathrm{d}t \right)^2 \int_0^T \|f(t)\|_{m-1,0}^2 \mathrm{d}t \\ &\lesssim_\alpha \frac{1}{\nu_\kappa^2} \int_0^T \|f(t)\|_{m-1,0}^2 \mathrm{d}t. \end{aligned} \tag{C.7}$$

Now let $\tilde{\hat{w}} := \mathfrak{J}\hat{w}$. Since $\tilde{\hat{w}}$ solves (4.45) with zero initial data and an additional forcing f, by repeating the derivation of (C.3), one has

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\tilde{\hat{w}}\|_{m,0}^2 + \nu_\kappa \|\nabla \tilde{\hat{w}}\|_{m,0}^2 \lesssim_\alpha \nu_\kappa \|\tilde{\hat{w}}\|_{m,0}^2 + \frac{1}{\nu_\kappa}\|f\|_{m-1,0}^2,$$

which, together with (C.7), yields

$$\|\tilde{\hat{w}}(t)\|_{m,0}^2 + \nu_\kappa \int_0^t \|\nabla \tilde{\hat{w}}\|_{m,0}^2 \mathrm{d}s \lesssim_\alpha \frac{1}{\nu_\kappa} \int_0^t \|f\|_{m-1,0}^2 \mathrm{d}s. \tag{C.8}$$

Therefore, by using (B.14), (C.1), and (C.7), one gets

$$\sup_{t\leq T} \|\hat{w}(t)\|_{m,0}^2 + \nu_\kappa \int_0^T (\|\nabla \hat{w}\|_{m,0}^2 + \|\hat{w}\|_{m,0}^2) \mathrm{d}s \lesssim_\alpha \frac{1}{\nu_\kappa} \int_0^T \|f\|_{m-1,0}^2 \mathrm{d}t.$$

This, together with (C.4) and the principle of superposition, completes the proof.

Data availability

Data sharing is not applicable to this article, as no datasets were generated or analysed during the current study.

Conflict of interest

The author declares no conflict of interest.

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